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Moment map equations in Kähler geometry and gauge theory

Author: Sergio Martín Nieto

Supervisor: Óscar García Prada

Tutor UCM: Marco Castrillón López

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Resumen

Presentamos un estudio de la correspondencia de Hitchin-Kobayashi para fibrados principales sobre superficies de Riemann compactas con grupo de estructura conexo. Damos un esquema de la prueba de la correspondencia, y presentamos la interpretación en términos de una aplicación momento con cierto nivel de detalle para el caso más general de variedades de Kähler compactas, incluyendo una prueba rigurosa del teorema de Marsden-Weinstein para acciones Hamiltonianas de grupos de Lie de Banach sobre variedades de Banach. También damos una prueba de la correspondencia de Chern-Singer, y vemos como la correspondencia de Hitchin-Kobayashi puede verse como una versión infinito-dimensional del teorema de Kempf-Ness. Por último, estudiamos la correspondencia para fibrados principales con grupo de estructura no conexo, y vemos, sin muchos detalles, como esta puede reducirse a la correspondencia para fibrados torcidos con grupo de estructura conexo.

Palabras clave: Correspondencia de Hitchin-Kobayashi, grupos de Lie reductivos, fibrados principales, correspondencia de Chern-Singer.

Abstract

We present an account of the Hitchin-Kobayashi correspondence for principal bundles over compact Riemann surfaces with connected structure group. We sketch a proof of the correspondence, and present the moment map picture associated with some level of detail for the general case of compact Kähler manifolds, including a rigorous proof of the Marsden-Weinstein theorem for Hamiltonian actions of Banach-Lie groups on Banach-manifolds. We also give a proof of the Chern-Singer correspondence, and we see how the Hitchin-Kobayashi correspondence can be seen as an infinite-dimensional version of the Kempf-Ness theorem. Finally, we study the correspondence for principal bundles with non-connected structure group, and we sketch how it can be reduced to a correspondence for twisted bundles with connected structure group.

Keywords: Hitchin-Kobayashi correspondence, Reductive Lie groups, Principal Bundles, Chern-Singer correspondence.

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Introduction

In 1965, M.S. Narasimhan and C.S. Seshadri proved the so-called Narasimhan-Seshadri theorem [43]. In their original formulation, their result provided a relation between the notion of stability for holomorphic vector bundles over a compact Riemann surface, which had been introduced by D. Mumford in 1963 in the context of Geometric Invariant Theory, and representations of the fundamental group of the surface. Briefly, stability of vector bundles is defined by assigning a numerical invariant to each bundle, called its slope. A vector bundle over a Riemann surface is stable if its slope is greater than the slope of any of its proper holomorphic subbundles. In the 1970s and 1980s, the works by M. Atiyah, R. Bott, S. Kobayashi, S. Donaldson, N. Hitchin and others showed that the result by Narasimhan and Seshadri could be translated to the language of differential geometry, and more specifically, in terms of Yang-Mills connections, notions arising from aspects of differential geometry and theoretical physics that were being studied at the time. More precisely, Donaldson gave a new proof of the theorem by Narasimhan and Seshadri in 1983 [14]. He proved that an indecomposable holomorphic vector bundle over a compact Riemann surface $\mathcal{E} \to X$ is stable if and only if it has a hermitian fiber metric h, whose Chern connection ∇ satisfies the equation $\star F_\nabla = -\mathrm{i}\, c \cdot \mathrm{id}_\mathcal{E}$ for a fixed constant c that depends on the topology of the bundle and the surface. This is called the *constant central curvature* condition. If the condition of being indecomposable is omitted, one gets that a hermitian holomorphic vector bundle is polystable if and only if it admits a hermitian fiber metric whose Chern connection satisfies the constant central curvature condition. The relation with Yang-Mills theory can be seen even more directly if one realizes that the theorem by Donaldson is equivalent to the following statement: a smooth complex vector bundle $E \to X$ over a compact Riemann surface endowed with a hermitian fiber metric has a *Hermite-Einstein-Yang-Mills connection*, i.e. a connection satisfying $\star F_\nabla = -\mathrm{i}\, c \cdot \mathrm{id}_E$ if and only if it admits a holomorphic structure making the bundle polystable.

In 1980, Kobayashi generalized the notion of stability to holomorphic vector bundles over higher-dimensional compact Kähler manifolds and Kobayashi and Hitchin conjectured independently that the theorem of Narasimhan and Seshadri should hold for compact Kähler surfaces (hence the name, *Hitchin-Kobayashi correspondence*) with a generalized notion of stability and a generalized equation.

In 1975, the result by Narasimhan and Seshadri was generalized by Ramanathan for principal bundles over a compact Riemann surface with compact connected reductive complex structure groups [45]. Among the contributions of Ramanathan to the discipline, we should highlight the fact that he gave a new notion of stability. This was generalized to complex projective manifolds by Ramanathan and Subramanian [56]. From the beginning of the century onwards, several generalizations of the correspondence have been proven. Namely, I. Mundet i Riera gave a generalization to Kähler fibrations [46], and O. García Prada and P.B. Gothen gave a generalization of the correspondence for Higgs bundles [16].

The remaining piece of the puzzle was given by M. Atiyah and R. Bott, who published a ground-breaking article [4] in 1983 that laid the foundations of the moment map picture. Basically, they showed that the moduli space of Hermite-Einstein connections could be described by means of a symplectic reduction for an appropriate moment map on the set of connections, which is acted on by the gauge group of the bundle. Phrased that way, the Narasimhan-Seshadri theorem and, more generally, the Hitchin-Kobayashi correspondence, can be thought of as infinite-dimensional versions of the Kempf-Ness

theorem.

The aim of this work is to present an account of the Hitchin-Kobayashi correspondence for principal bundles over compact Riemann surfaces, although some sections deal with the more general case of compact Kähler manifolds. We explain the moment map picture and the sense in which it can be seen as an infinite-dimensional version of the Kempf-Ness theorem. Detailed proofs of these results are beyond the scope of this work. We have followed an exposition close to that of [46] and [16], but making the corresponding simplifications (which basically consist of ignoring any extra structure). When studying moduli spaces for these structures, rigorous constructions require working with Sobolev completions of several spaces, something that is often dismissed in the literature by omitting the technical details. We have tried to specify why Sobolev completions are needed to make everything rigorous and how to work with them. Our second objective is to take the first steps towards a generalization of the correspondence for principal bundles with non-connected reductive structure groups. Again, this requires both a new Hermite-Einstein-Yang-Mills equation and a new notion of stability, but more importantly, it relies on the work by García Prada, G. Barajas, Gothen and Mundet i Riera in [5] on twisted equivariant bundles over coverings. The key ideas and results concerning this generalization have been taken from a draft of a yet unpublished article that my supervisor kindly shared with me [8].

As a brief outline, in chapter 1 we address the finite-dimensional version of symplectic reduction and the Kempf-Ness theorem. The former is generalized to Banach manifolds in chapter 2. Chapter 3 introduces connections in the context of principal bundles, and treats the relation between connections and complex structures: the so-called Chern or Chern-Singer correspondence. Finally, chapters 4 and 5 build on all the previous work and introduce the hermitian Yang-Mills equation and the Hitchin-Kobayashi correspondence respectively.

I have tried to make this work as self-contained as possible, an ultimately impossible task, given the sheer amount of concepts that play a role in the Hitchin-Kobayashi correspondence. For the convenience of the reader, and for my own convenience when citing results, I have included three appendices that give short introductions to some of the prerequisites needed for the main body of the text. Appendix A is a short introduction to the theory of complex manifolds and vector bundles. It should be read before the main body of the text. Appendix B contains some notions of the theory of Lie groups which may not be covered in a first course on the subject, as well as some tools from [5] which set the basic framework needed to work with principal bundles with non-connected structure group. It should be read before chapter 5 although some of its results are used earlier in the text. Finally, appendix C gives a short introduction to Banach manifolds. It mostly contains relevant definitions, and statements that show that Banach manifolds are sufficiently well-behaved to still satisfy many of the properties of usual finite-dimensional smooth manifolds (with the exception, of course, of the existence of charts of the form $(x^1, \dots x^n)$). It should be read before chapter 2. With my deepest sorrow, and in an effort to keep the length of the document manageable, I have omitted many proofs and only sketched others. However, whenever a proof has been omitted, I have tried to give a reference in the literature where it can be found. On the other hand, I have tried to pay special attention to detail in the proofs that I have included. This is a matter of personal taste, but I hope it will be helpful to future readers.

I would like to thank my supervisor, Óscar García Prada for his guidance, and Diego Ruiz Cases for his useful observations and enriching discussions, which made section 3.8 possible.

Chapter 1

Finite-dimensional symplectic reduction and the Kempf-Ness theorem

The aim of this chapter is to provide a very brief account of the Kempf-Ness theorem. The motivation behind including this chapter is the fact that the Hitchin-Kobayashi correspondence, treated in chapter 5 can be seen as an infinite-dimensional version of this theorem. It is important to remark that all the results of chapter 5 can be stated and proved without ever mentioning moment maps. However, the framework of moment maps and symplectic reduction proves to be very useful, because, together with the results we are about to expose in this chapter, it motivates the correspondence, and hints at many correct results.

In this chapter we will assume basic knowledge of symplectic geometry, Lie theory, and elementary projective algebraic geometry. We will recall some basic concepts, but most results here will be stated without proof. Good references on these topics are [54], [22] and [21]. To fix some notation, we will use e_G for the unit element of a group G.

1.1 Symplectic geometry

Recall that a symplectic manifold is a (necessarily even-dimensional) smooth manifold M endowed with a symplectic form ω, i.e. a closed two-form such that $\omega_p \colon T_pM \times T_pM \to \mathbb{R}$ is skew-symmetric and non-degenerate. A symplectomorphism between symplectic manifolds is a diffeomorphism $\phi \colon (M_1, \omega_1) \to (M_2, \omega_2)$ satisfying $\phi^* \omega_2 = \omega_1$.

Example 1.1. A non-trivial example of a symplectic manifold is the complex projective space $\mathbb{CP}^n$. One way to endow it with a symplectic structure is theorem 1.4 below. Another option is to do it directly by defining a symplectic form. This is called the *Fubini-Study form* (see [33, Ex. 8.14]). Consider the holomorphic coordinates of $\mathbb{CP}^n$ given by the affine open covering $\{U_i\}_{i=0}^n$, where $U_i = \{[z_0 : z_1 : \ldots : z_n] : z_i \neq 0\}$. Recall that we have holomorphic coordinates

$$\varphi_i \colon U_i \to \mathbb{C}^n, [z_0 : \ldots : z_i : \ldots z_n] \mapsto \left(\frac{z_0}{z_i}, \ldots \frac{\hat{z}_i}{z_i}, \ldots \frac{z_n}{z_i} \right),$$

where the hat means that the corresponding term is missing. Writing w^j for the projection of φ_i onto the j-th coordinate, the coordinate expression of the Fubini-Study form in U_i is

$$\omega_{\mathrm{FS}} = \sum_{j,k=1}^{n} \left(\frac{\delta_{jk} \cdot (1 + ||w||^2) - \bar{w}^j \cdot w^k}{(1 + ||w||^2)^2} \right) \cdot \mathrm{d}w^j \wedge \mathrm{d}\bar{w}^k. \tag{1.1}$$

In particular, for any embedded submanifold $X \subset \mathbb{CP}^n$, the pullback through the inclusion $\imath \colon X \hookrightarrow \mathbb{CP}^n$ induces a closed two-form $\imath^* \omega \in \Omega^2_{\mathbb{R}}(X)$ which turns out to be symplectic.

The latter is not true in general: not every embedded submanifold of a symplectic manifold is symplectic, but it is true in this case because $\mathbb{CP}^n$ is actually Kähler (see appendix A).

1.1.1 Symplectic reduction

Symplectic reduction is a technique whose origin lies in the formalization of mechanics in the context of symplectic geometry. Briefly, when a Lie group acts on a symplectic manifold, and the action is *Hamiltonian* (see definition 1.3 below), then, symplectic reduction provides a way to obtain, from (M,ω), another symplectic manifold whose dimension is $\dim M - 2\cdot\dim G$. That is, we obtain a symplectic manifold whose dimension is smaller than that of M.

Let G be a Lie group, and M be a smooth manifold, and suppose that G acts on M on the left (right actions are handled analogously). We say that the action is *smooth* if the map $\phi\colon G\times M\to M,(g,m)\mapsto g\cdot m$ is smooth. Equivalently, the action is smooth if for every $g\in G$, the map $\phi_g\colon M\to M, m\mapsto g\cdot m$ is smooth. Moreover, we say that the action is *proper* if the map $G\times M\to M\times M,(g,m)\mapsto(g\cdot m,m)$ is a proper map. As we will work extensively with actions of Lie groups on smooth manifolds, let us fix some notation. When a Lie group acts on a smooth manifold, we will denote, unless otherwise stated, the action map by $\phi(g,m)$ or simply $g\cdot m$ for left actions (for right actions we will use $p\cdot m$). We will also denote $\phi_g\colon M\to M, m\mapsto\phi(g,m)$ for $g\in G$, and $\phi^{(m)}\colon G\to M, g\mapsto\phi(g,m)$ for $m\in M$. Both of them are smooth under the assumption that the action is smooth. The following result is standard in the theory of smooth manifolds, but we will use it extensively.

Theorem 1.2 (Quotient manifold theorem). Let G be a Lie group acting smoothly, freely and properly on a smooth manifold M. Then, the orbit space M/G is a topological manifold of dimension equal to $\dim M-\dim G$. Moreover, it admits a unique smooth structure such that the topological quotient map $q\colon M\to M/G$ is a surjective smooth submersion.

Proof. See [34, Th. 21.10]. □

Hamiltonian actions are those admitting moment maps. Let us recall some facts from Lie group theory that are needed to define moment maps. Let G be a Lie group with Lie algebra $\mathfrak{g}$. Recall that the *adjoint representation* is the Lie group representation $G\to\mathrm{GL}(\mathfrak{g})$ given by $\mathrm{Ad}_g:=(C_g)_*$, where $C_g\colon G\to G, h\mapsto ghg^{-1}$ is the conjugation map. That is, for each $\xi\in\mathfrak{g}$ $\mathrm{Ad}_g\xi\in\mathfrak{g}$ is the unique vector field C_g-related to ξ. If we identify the Lie algebra with T_eG, then $\mathrm{Ad}_g=\mathrm{d}_eC_g$. On the other hand, let

$$\begin{array}{rccc}(\cdot,\cdot)\colon & \mathfrak{g}^*\times\mathfrak{g} & \longrightarrow & \mathbb{R}\\ & (\varphi,\xi) & \mapsto & \varphi(\xi)\end{array} \tag{1.2}$$

be the pairing induced by the dual (as a vector space) of the Lie algebra. We define the *coadjoint representation* $\mathrm{Ad}^*\colon G\to\mathrm{GL}(\mathfrak{g}^*), g\mapsto\mathrm{Ad}^*_g$, where Ad^*_g is given by

$$(\mathrm{Ad}^*_g(\varphi),\xi):=(\varphi,\mathrm{Ad}_{g^{-1}}(\xi)).$$

The definition using $\mathrm{Ad}_{g^{-1}}$ instead of Ad_g ensures that we actually get a representation.

Finally, let us introduce some notation:

1. Consider a Lie group G that acts smoothly on a smooth manifold M on the left. Let $\xi \in \mathfrak{g}$. We define the *infinitesimal generator of the one-parameter subgroup* $\exp(t\xi)$, also called the *fundamental vector field* generated by $\xi \in \mathfrak{g}$ as the vector field
$$X_{\xi,m} := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \exp(t\xi)\cdot m = \mathrm{d}_e\phi^{(m)}(\xi). \tag{1.3}$$
For a right action the definition is analogous. We also say that the vector $X_{\xi,m}$ is the infinitesimal action of ξ on m.
2. Let $\mu\colon M \to \mathfrak{g}^*$ be a smooth map. For each $\xi \in \mathfrak{g}$, we set $\mu^\xi\colon M \to \mathbb{R}, m \mapsto (\mu(m), \xi)$.
3. When a Lie group G acts on a symplectic manifold (M,ω), we say that the action is symplectic, or that G *acts by symplectomorphisms*, if ϕ_g is a symplectomorphism for every $g \in G$.

Definition 1.3. Let (M,ω) be a symplectic manifold, and let G be a Lie group that acts smoothly, freely and properly by symplectomorphisms on M. The action is said to be *Hamiltonian* if there exists a smooth map, called *moment map* $\mu\colon M \to \mathfrak{g}^*$ satisfying:

1. $\mathrm{d}\mu^\xi = i_{X_\xi}\omega$ for all $\xi \in \mathfrak{g}$,
2. μ is equivariant with respect to the action and coadjoint representation. That is, for every $g \in G$, we have $\mu \circ \phi_g = \mathrm{Ad}_g^* \circ \mu$.

$$\begin{array}{ccc} M & \xrightarrow{\phi_g} & M \\ {\scriptstyle\mu}\downarrow & & \downarrow{\scriptstyle\mu} \\ \mathfrak{g}^* & \xrightarrow{\mathrm{Ad}_g^*} & \mathfrak{g}^* \end{array}$$

The quadruple (M,ω,G,μ) is called a *Hamiltonian space* or *Hamiltonian G-space*.

The first condition of the definition is equivalent to $(\mathrm{d}_m\mu(v),\xi) = \omega_p(X_{\xi,m}, v)$ for all $m \in M$, $v \in T_mM$ and $\xi \in \mathfrak{g}$. Moment maps may or may not exist for a symplectic action. However, if they exist, they are unique up to a $\mathfrak{g}^*$ constant factor.

We are finally ready to state the main result in the theory of finite-dimensional symplectic reduction.

Theorem 1.4 (Marsden-Weinstein). Let (M,ω,G,μ) be a Hamiltonian G-space. Then, $M_{\mathrm{red}} := \mu^{-1}(0)/G$ is a symplectic manifold, whose symplectic form ω_{red} satisfies
$$i^*\omega = q^*\omega_{\mathrm{red}},$$
where $i\colon \mu^{-1}(0) \hookrightarrow M$ is the inclusion, and $q\colon \mu^{-1}(0) \to \mu^{-1}(0)/G$ the quotient map.

We will not give a detailed proof here because chapter 2 gives a proof of the theorem for Banach manifolds, which can be easily adapted (in fact, simplified) to the finite-dimensional case. A direct proof for the finite-dimensional case can be found in [**54**, Part IX], however, it is worth making some comments on the result. Firstly, $\mu^{-1}(0)$ is stable under the G-action. That is, if $m \in \mu^{-1}(0)$, then $G\cdot m \in \mu^{-1}(0)$. Indeed,
$$\mu(g\cdot m) = \mathrm{Ad}_g^* \circ \mu(m) = \mathrm{Ad}_g^*(0) = 0. \tag{1.4}$$
Moreover, $0 \in \mathfrak{g}^*$ is a regular value of the smooth map μ, which shows that $\mu^{-1}(0) \subset M$ is an embedded submanifold. It can be shown that $\dim \mu^{-1}(0) = \dim M - \dim G$, and $T_m\ \mu^{-1}(0) = \ker(\mathrm{d}_m\mu) \subset T_mM$. What is more, $G\cdot m$ is an embedded submanifold of $\mu^{-1}(0)$ for all $m \in \mu^{-1}(0)$, and the tangent spaces of all the manifolds involved relate as follows:
$$T_{q(m)}M_{\mathrm{red}} \cong T_m\ \mu^{-1}(0)/T_m(G\cdot m).$$

1.2 Geometric invariant theory

The Kempf-Ness theorem relates GIT (geometric invariant theory) to symplectic geometry; two branches of geometry that seem completely unrelated at first glance, but end up being intertwined. GIT is the theory developed by David Mumford in his book [39] which basically provides the framework needed to define quotients in algebraic geometry. We will not follow his approach, because he develops his theory in the language of schemes, and that is a level of generality far beyond what we need. A good reference for GIT that requires less abstraction is [22]. We will follow this source, and especially [57]. Our objective for this section is to give a brief account of projective GIT theory that requires minimal previous knowledge.

As the discussion in [22] illustrates, orbit spaces do not behave well in algebraic geometry, and one must look for an alternative notion of quotient. This is done as explained below. First, recall from basic projective algebraic geometry that we define a projective variety as the zero set of a family of homogeneous polynomials, i.e. $X = V(F_1, \dots F_r) \subset \mathbb{CP}^n$ where $F_i \in \mathbb{C}[X_0, \dots X_n]$ is homogeneous. If we ask X to be irreducible, then $I(X) := \{F \in \mathbb{C}[X_0, \dots X_n] : F(x) = 0 \ \ \forall\, x \in X\}$ is prime. Then, we define the *homogeneous coordinate ring*

$$R(X) := \mathbb{C}[X_0, \dots X_n]/I(X),$$

which is a graded ring whose grading is given by degree. Every finitely generated graded $\mathbb{C}$-algebra that can be generated by elements of degree one gives rise to a projective variety as follows: first, choose such a set of generators $r_0, \dots r_n \in R$, and consider the $\mathbb{C}$-algebra homomorphism given by $\varphi\colon \mathbb{C}[X_0, \dots X_n] \to R, X_i \to r_i$. Let $I := \ker\varphi$. Then, by the first isomorphism theorem we have $R \cong \mathbb{C}[X_0, \dots X_n]/I$, and we define $\mathrm{Proj}(R) := V(I) = \{x \in \mathbb{CP}^n : F(x) = 0 \ \ \forall\, F \in I\}$. This is a projective variety. The choice of generators is irrelevant as different choices yield isomorphic varieties.

Consider now a complex reductive group G (we refer the reader to appendix B for the relevant notions), and suppose that G acts on a projective variety X through a representation $\rho\colon G \to \mathrm{GL}(n+1, \mathbb{C})$. Formally, the action is given by $g \cdot [v] := [\rho(g)(v)]$ where $[\cdot]$ denotes the equivalence class in complex projective space, and in a way such that $[v] \in X$ implies $g \cdot [v] \in X$. Equivalently, G acts through a representation in the affine cone associated to X. We call these actions *linear*. This induces an action of G on $\mathbb{C}[X_0, \dots X_n]$ given by

$$G \times \mathbb{C}[X_0, \dots X_n] \longrightarrow \mathbb{C}[X_0, \dots X_n], \ \ g \cdot (F(X)) \to F(\rho(g^{-1}) \cdot (X)).$$

By a theorem of Nagata [42], the ring $R(X)^G := \{F \in \mathbb{C}[X_0, \dots X_n] : g \cdot F = F\}$ is finitely generated, and one may show that there is an algebra

$$\tilde{R}(X)^G = \bigoplus_{k \geq 0} R(X)^G_{dk}$$

for some $d \geq 0$ that is generated by elements of degree d. Changing the grading to make degree d the new degree one, $\tilde{R}(X)^G$ is generated by degree one elements. Here, $R(X)^G_k$ is the degree k piece defined by the grading.

Definition 1.5 (Projective GIT quotient). Let G be a complex reductive group that acts linearly on a projective variety $X \subset \mathbb{CP}^n$. The GIT quotient is the projective variety $X/\!\!/G := \mathrm{Proj}\left(\tilde{R}(X)^G\right)$.

The double bar notation is used to emphasize that this is generally not a usual quotient in the sense of equivalence classes for an action. Right now it is quite unclear why

we choose to name this a quotient, or what the geometric meaning of this set is, so we will try to answer those questions. To answer the first one, choose $F_0, \dots F_n$ degree one generators of $\tilde{R}(X)^G$, and define a "map"

$$\varphi \colon X \dashrightarrow \mathbb{CP}^n, \ x \mapsto [F_0(x) : \dots : F_n(x)]. \tag{1.5}$$

This is well-defined only at points $x \in X$ that do not annihilate all generators, i.e. if there is at least one element in $\tilde{R}(X)^G$ that does not vanish on x. The fact that it is well-defined follows from the fact that all the polynomials have the same degree. Observe that the image of φ is precisely $X/\!\!/G$. This follows from the definition of Proj and the fact that

$$\mathrm{Proj}\left(\tilde{R}(X)^G\right) = \{x \in \mathbb{CP}^n : Q(x) = 0 \ \ \forall\, Q \ \text{ such that } \ Q(F_0, \dots F_n) = 0\}\,.$$

Definition 1.6. Let G be a complex reductive group that acts linearly on a complex projective variety $X \subset \mathbb{CP}^n$.

1. A point $x \in X$ is *semistable* if there is $F \in \tilde{R}(X)^G$ such that $F(x) \neq 0$.
2. A point $x \in X$ is *stable* if the orbit $G \cdot x$ is closed in X^{ss} and the stabilizer G_x is finite.
3. A point $x \in X$ is *polystable* if the orbit $G \cdot x$ is closed in X^{ss}.

Otherwise, we say that x is *unstable*. The sets of stable, semistable and polystable points are denoted, respectively, X^s, X^{ss} and X^{ps}.

Remark 1.7.

1. The quotient $X/\!\!/G$ parametrizes orbits for which there is at least one non-vanishing element of $\tilde{R}(X)^G$. That is, φ is well-defined in X^{ss}.
2. φ identifies orbits that cannot be distinguished by elements of $\tilde{R}(X)^G$. Observe in (1.5) that φ maps points in the same orbit to the same point, but it may map points in different orbits to the same orbit as well.
3. X^s and X^{ss} are open subsets of X.
4. The quotient $X/\!\!/G$ is the quotient (in the sense of equivalence classes) of X^{ss} by the equivalence relation that identifies x_1 and x_2 if and only if $\overline{G \cdot x_1} \cap \overline{G \cdot x_2} \cap X^{ss} \neq \varnothing$. More rigorously, the latter is equivalent to $\varphi(x_1) = \varphi(x_2)$, see [22, Th. 4.11].
5. The restriction of the action to X^s yields what is called a *geometric quotient*, i.e. a quotient that is indeed a set of equivalence classes.

Finally, [22, Lemma 4.16] states that for every $x \in X^{ss}$, the set $\overline{G \cdot x}$ contains a unique polystable orbit. This result, together with point 4 above implies that the quotient $X/\!\!/G$ can be thought of as parametrizing polystable orbits. That is, there is a bijection $X/\!\!/G \leftrightarrow X^{ps}/G$, the latter being the usual set of equivalence classes. This uses the fact that the preimage of any point through φ contains a unique closed orbit [22, Cor. 2.39].

1.2.1 The Kempf-Ness theorem

We refer the reader to appendix B for the basics of complex Lie groups and complexifications of real compact Lie groups. As we recall in the appendix, complex reductive Lie

groups are characterized precisely by the fact that they are complexifications of compact real Lie groups.

We have defined a projective variety $X \subset \mathbb{CP}^n$ as the zero set of a collection of homogeneous polynomials. Observe that $\mathbb{CP}^n$ is also a complex manifold (in the sense of complex differential geometry), and we may ask if projective varieties can be equipped with a complex smooth structure such that they become embedded complex submanifolds of $\mathbb{CP}^n$. It turns out that for this to be true, we have to impose some extra conditions on the polynomials defining X. If $X = V(I)$, where $I = (F_1, \dots F_r)$ is a radical ideal, and $\mathrm{Jac}(F_1, \dots F_r)(z)$ has constant rank for every $z \in X$, then it is a complex manifold. Observe that this condition is well-defined and does not depend on the representative we choose for z. Indeed, choosing another representative λz yields a matrix obtained from the Jacobian by multiplying each row by a power of λ, which does not change the rank. We may then show that if $s = \mathrm{rank}(\mathrm{Jac}(F_1, \dots F_r))$, and $d := n - s$, then X is a complex manifold of complex dimension $\dim_{\mathbb{C}}(X) = d$ (see [41, Prop. 5.42]).

Suppose now that $\mathrm{U}(n+1)$ acts on $\mathbb{CP}^n$ through

$$\mathrm{U}(n+1) \times \mathbb{CP}^n \to \mathbb{CP}^n, (A, [z]) \mapsto [A \cdot z].$$

One can then show that, with respect to the Fubini-Study form, the action is symplectic, and in fact Hamiltonian. The moment map is given by $\tilde{\mu} \colon \mathbb{CP}^n \to \mathfrak{u}(n+1)^*, [z] \mapsto \tilde{\mu}([z])$ and

$$(\tilde{\mu}([z]), A) := \frac{\mathrm{Tr}(z^* \cdot A \cdot z)}{2\mathrm{i} \cdot (z^* \cdot z)} \qquad \text{for} \qquad A \in \mathfrak{u}(n+1), \tag{1.6}$$

where z^* is the transpose conjugate [22, Ex. 8.10].

More generally, consider a reductive complex group $G = K^{\mathbb{C}}$ for some maximal compact subgroup (those are not unique but are conjugate to each other). Suppose that G acts on a smooth projective variety $X \subset \mathbb{CP}^n$ via a representation $\rho \colon G \to \mathrm{GL}(n+1, \mathbb{C})$. Restricting the action, there is also an action of K on $\mathbb{CP}^n$. We want to relate the GIT quotient with the symplectic quotient given by the Marsden-Weinstein theorem. Thus, we take a smooth complex variety $X \subset \mathbb{CP}^n$, which is also a complex manifold, and therefore a symplectic manifold (the symplectic form is the pullback of the Fubini-Study form, see the comments in example 1.1). On the one hand, we may form the GIT quotient $X /\!\!/ G$, and on the other hand, if the restricted action of K was symplectic, we could consider the symplectic quotient. Using Weyl's unitary trick, we may choose coordinates appropriately, such that the representation is unitary $\rho \colon K \to \mathrm{U}(n+1)$ (see [21, Th. 12.3.9]). In that case, the action of G on both X and $\mathbb{CP}^n$ is symplectic, and actually Hamiltonian. A moment map may be given as follows: consider $\mathrm{d}_{e_K}\rho \colon \mathfrak{k} \to \mathfrak{u}(n+1)$. The dual map is $(\mathrm{d}_{e_K}\rho)^* \colon \mathfrak{u}(n+1)^* \to \mathfrak{k}^*$. Hence, we can define $\mu := (\mathrm{d}_{e_K}\rho)^* \circ \tilde{\mu}$, where $\tilde{\mu}$ is given in (1.6). Explicitly,

$$(\mu([z]), B) := \frac{\mathrm{Tr}(z^* \cdot \mathrm{d}_{e_K}\rho(B) \cdot z)}{2\mathrm{i}\ (z^* \cdot z)}.$$

Now that both the symplectic and the GIT quotients make sense, their relation is given by the Kempf-Ness theorem.

Theorem 1.8 (Kempf-Ness). Let $G = K^{\mathbb{C}}$ be a complex reductive group acting linearly on a smooth complex projective variety, $X \subset \mathbb{CP}^n$. Suppose that one maximal compact subgroup K is connected (hence all of them are) and acts symplectically on X, and let $\mu\colon X \to \mathfrak{k}^*$ be the associated moment map. Then

1. $G \cdot \mu^{-1}(0) = X^{ps}$.
2. If $x \in X$ is polystable, then its orbit $G \cdot x$ meets $\mu^{-1}(0)$ in a single K-orbit, i.e. $G \cdot x \cap \mu^{-1}(0)$ is contained in a single K-orbit.
3. $x \in X$ is semistable if and only if its orbit's closure $\overline{G \cdot x}$ meets $\mu^{-1}(0)$.

Proof. See [22, Th. 10.3] □

Corollary 1.9. Under the previous hypothesis, the inclusion $\mu^{-1}(0) \subset X^{ps}$ induces a homeomorphism

$$\mu^{-1}(0)/K \to X/\!\!/G.$$

To be more explicit, the previous corollary should be interpreted as follows: $X/\!\!/G$ is in bijection with X^{ps}/G. X^{ps} carries the subspace topology it inherits from $\mathbb{CP}^n$ and the quotient is endowed with the usual quotient topology. It is with respect to this topology that we have a homeomorphism.

Proof. Observe that $\mu^{-1}(0) \subset G \cdot \mu^{-1}(0) \subset X^{ps}$ by part 1 of theorem 1.8. Moreover, $\mu^{-1}(0) \subset X$ is a closed subset of a compact set, hence compact, and so is $\mu^{-1}(0)/K$. Furthermore, the composition $\mu^{-1}(0) \hookrightarrow X^{ps} \to X^{ps}/G$ is continuous, and it is the map sending each $x \in \mu^{-1}(0)$ to the corresponding G-orbit. By point 2 of theorem 1.8, for each $x \in X^{ps}$, $G \cdot x \cap \mu^{-1}(0)$ in a single K-orbit, so the composition factors through the quotient $\mu^{-1}(0)/K$. By points 1 and 2 the map is bijective. We thus have a continuous bijection from a compact space to a Hausdorff topological space. It is therefore a homeomorphism. The fact that $X^{ps}/G \leftrightarrow X/\!\!/G$ is not obvious, but it is a consequence of the way that GIT is constructed. □

Chapter 2

Infinite-dimensional symplectic reduction

The aim of this chapter is to give a Banach manifold version of symplectic reduction (theorem 1.4), for the action of a Banach-Lie group on a Banach manifold. We refer the reader to appendix C for the relevant notions concerning infinite-dimensional Banach manifolds. The chapter starts with the (straightforward) generalization of symplectic manifolds to infinite dimensions. Once this is done, we can give the definition of moment map, and state the reduction theorem for Banach manifolds. As we shall see, the proof does not really change much from the classical theorem, aside from the fact that we no longer have a quotient manifold theorem, and that has to be addressed by requiring additional properties to moment maps. However, if we assume that we can define quotients, and they are Banach manifolds, the proof is quite similar to the finite-dimensional case. In fact, the original article by Marsden and Weinstein [36] gives a proof of theorem 1.4 in the context of Banach manifolds. A very thorough exposition of the theorem can be found in [18]. These results will be used in chapter 4, as they provide a rigorous framework for the moment map picture of the Hitchin-Kobayashi correspondence.

Given a skew-symmetric bilinear form on a vector space Ω, we define $\Omega^\flat \colon V \to V^*, v \mapsto \Omega(v, \cdot)$. The definition of an infinite-dimensional symplectic manifold is quite similar to the finite-dimensional one except for the fact that we no longer ask the symplectic form $\Omega_m^\flat$ to define an isomorphism $T_m\mathcal{M} \cong T_m\mathcal{M}^*$, we just ask it to be injective. Observe that in the finite-dimensional setting this already implies that $\Omega_m^\flat$ is an isomorphism because $T_m\mathcal{M}$ and its dual are vector spaces of the same finite dimension.

Definition 2.1 (Banach symplectic manifold). Let $\mathcal{M}$ be a Banach manifold, we say that a pair $(\mathcal{M}, \Omega)$ is a *Banach symplectic manifold* if Ω is a closed two-form satisfying:

1. For each $m \in \mathcal{M}$, $\Omega_m \colon T_m\mathcal{M} \times T_m\mathcal{M} \to \mathbb{R}$ is continuous.
2. For each $m \in \mathcal{M}$, Ω_m is non-degenerate, that is, if $u \in T_m\mathcal{M}$ and $\Omega_m(u, v) = 0$ for all $v \in T_m\mathcal{M}$, then $u = 0$.

We say that Ω is a symplectic form.

The notions of symplectomorphism, Hamiltonian action of a Banach-Lie group on a Banach manifold and moment map are defined just as in the finite-dimensional case (see section 1.1.1). Observe that given a Banach-Lie group G and $\xi \in \mathrm{Lie}(G)$, the infinitesimal action, i.e. the vector field X_ξ can be defined just as in (1.3) because Banach-Lie groups have an exponential map. The first major difference with respect to the finite setting is the fact that we do not have a direct generalization of the quotient manifold theorem. What ends up working is to ask more conditions to the moment map. Before dwelling

on the problem of the lack of a quotient manifold theorem, let us consider the quotient $\mu^{-1}(0)/G$. The fact that μ is G-equivariant means that $\mu^{-1}(0)$ is invariant under the G-action (same justification as (1.4)). Hence, G acts on $\mu^{-1}(0)$, and the *reduced space* $\mathcal{M}_{\text{red}} := \mu^{-1}(0)/G$ is well-defined as a topological space. In the classical theorem, one could show that 0 was always a regular value of μ, and this implied that $\mu^{-1}(0)$ was an embedded submanifold of the original manifold. Since we may not have that condition, we have to impose it on μ. This motivates the following definition.

Definition 2.2. Let G be a Banach-Lie group acting symplectically on a Banach manifold $\mathcal{M}$, and let $\mu\colon \mathcal{M} \to \mathfrak{g}^*$ be a moment map for the action.

1. We say that $0 \in \text{Lie}(G)^*$ is a *weakly regular value* of μ if $\mu^{-1}(0)$ is an embedded submanifold of $\mathcal{M}$ and $T_m\ \mu^{-1}(0) = \ker \mathrm{d}_m\mu$ for each $m \in \mu^{-1}(0)$.
2. We say that μ *satisfies the slice condition* if the action of G on $\mathcal{M}$ is free, and for every $m \in \mu^{-1}(0)$ there is a submanifold $S_m \subset \mu^{-1}(0)$, called a *slice* such that
$$T_m\ \mu^{-1}(0) = T_mS_m \oplus T_m(G \cdot m).$$

Aside from the motivation we just gave for the definition of a weakly regular value, the condition $T_m\ \mu^{-1}(0) = \ker \mathrm{d}_m\mu$ is always satisfied for a moment map in the classical sense. Moreover, if μ is a submersion at 0, then 0 is a weakly regular value (see [1, Th. 3.5.4]), but not every weakly regular value is also regular. It remains to somehow motivate the slice condition. This is precisely what will be the substitute of the quotient manifold theorem. Indeed, [1, Sec. 3.5] gives a detailed exposition of the theory of quotient Banach manifolds. Under the hypothesis of the existence of a slice, we may use it to get local coordinates around every point (see the proof of [1, Th. 3.5.25 and Lemma 3.5.26]).

Moreover, observe that if $\phi\colon G \times \mathcal{M} \to \mathcal{M}$ is the action, then $\phi^{(m)}\colon G \to \mathcal{M}$ is an embedding for each $m \in \mathcal{M}$, and if $m \in \mu^{-1}(0)$, then $G \cdot m \subset \mu^{-1}(0)$ is an embedded submanifold. If the slice condition is satisfied, we have $T_m\ \mu^{-1}(0) = T_mS_m \oplus T_m(G \cdot m)$, hence $T_m\ \mu^{-1}(0)/T_m(G \cdot m) \cong T_mS_m$. Moreover, for each sufficiently small slice [1, Th. 3.5.25 and Lemma 3.5.26] state that the map $\varphi\colon S_m \to \mathcal{M}_{\text{red}}, p \mapsto \pi(p)$ ($\pi\colon \mu^{-1}(0) \to \mathcal{M}_{\text{red}}$ being the quotient map) is a diffeomorphism onto its image, therefore $T_mS_m \cong T_{\pi(m)}\mathcal{M}_{\text{red}}$. Putting everything together, we get:

$$T_{\pi(m)}\mathcal{M}_{\text{red}} \cong T_m\ \mu^{-1}(0)/T_m(G \cdot m), \tag{2.1}$$

where the isomorphism is given by the first isomorphism theorem. Indeed, $\ker \mathrm{d}_m\pi = T_m(G \cdot m)$, hence we get an isomorphism $\overline{\mathrm{d}_m\pi}\colon T_m\ \mu^{-1}(0)/T_m(G \cdot m) \to T_{\pi(m)}\mathcal{M}_{\text{red}}$.

Observe as well that the fact that $\phi^{(m)}$ is an embedding means that $T_m(G \cdot m) = \mathrm{d}_e\phi^{(m)}(\text{Lie}(G))$, thus, by equation (1.3)

$$T_m(G \cdot m) = \{X_{\xi,m} : \xi \in \text{Lie}(G)\}. \tag{2.2}$$

Remark 2.3. Recall that the Hausdorff condition is something imposed to manifolds to avoid pathological situations, but it is not implied by the condition of being locally euclidean. When the action is proper, we have a topological group acting continuously and properly on a topological space. It is a standard result (see for example [34, Prop. 21.4]) that under those assumptions the quotient space is Hausdorff.

We can now state the reduction theorem for Banach manifolds.

Theorem 2.4 (Banach symplectic reduction). Let $\mathcal{M}$ be a symplectic manifold endowed with a symplectic form Ω, and a smooth, symplectic, left action of a Banach-Lie group G. Suppose that $\mu\colon \mathcal{M} \to \mathrm{Lie}(G)^*$ is a moment map for which 0 is a weakly regular value, and satisfying the slice condition. Then, $\mathcal{M}_{\mathrm{red}} := \mu^{-1}(0)/G$ has the structure of a symplectic (possibly non-Hausdorff) Banach manifold, which is endowed with a symplectic form Ω_{red} and satisfying

$$\pi^*\Omega_{\mathrm{red}} = i^*\Omega$$

where $i\colon \mu^{-1}(0) \hookrightarrow \mathcal{M}$ is the inclusion and $\pi\colon \mu^{-1}(0) \to \mathcal{M}_{\mathrm{red}}$ the natural projection. Moreover, if the restricted action of G on $\mu^{-1}(0)$ is proper, then $\mathcal{M}_{\mathrm{red}}$ is Hausdorff.

Before giving the proof, let us state and prove some lemmas we will need. Recall that, given a skew-symmetric bilinear map $\Omega\colon \mathbf{E} \times \mathbf{E} \to \mathbb{R}$ such that $\Omega(u, v) = 0$ for all $v \in \mathbf{E}$ implies $u = 0$, (that is, $\ker \Omega^\flat = \{0\}$) we say that it is non-degenerate.

Lemma 2.5. Let $(\mathbf{E}, ||\cdot||_{\mathbf{E}})$ be a Banach space and let $\Omega\colon \mathbf{E} \times \mathbf{E} \to \mathbb{R}$ be a skew-symmetric, non-degenerate, continuous bilinear map. Consider, for each $v \in \mathbf{E}$ the maps $p_v\colon \mathbf{E} \to \mathbb{R}, u \mapsto |\Omega(u, v)|$. Then, $\{p_v\}_{v\in\mathbf{E}}$ is a family of seminorms with respect to which $\mathbf{E}$ becomes a Hausdorff locally convex TVS. Moreover, given a linear map $\alpha\colon \mathbf{E} \to \mathbb{R}$ continuous with respect to the locally convex topology, there exists $v \in \mathbf{E}$ such that $\alpha(u) = \Omega(u, v)$ for all $u \in \mathbf{E}$.

Proof. The fact that $\{p_v\}_{v\in\mathbf{E}}$ is a family of seminorms on $\mathbf{E}$ follows immediately from the fact that Ω is bilinear. Moreover, non-degeneracy of Ω implies that if $p_v(u) = 0$ for all $v \in \mathbf{E}$, then $u = 0$. Endow $\mathbf{E}$ with the initial topology with respect to the natural projections $q_i\colon \mathbf{E} \to \mathbf{E}/p_i^{-1}(0)$. This turns $\mathbf{E}$ into a Hausdorff TVS.

For the second statement, let $\alpha\colon \mathbf{E} \to \mathbb{R}$ be continuous with respect to the TVS topology we just defined. It is a general fact of locally convex vector spaces that a linear map $\alpha\colon \mathbf{E} \to \mathbb{R}$ is continuous if and only if there are seminorms $\{p_{i_1}, \dots p_{i_n}\}$ and a constant $C > 0$ such that $|\alpha(v)| \leq C \cdot \max_{1\leq j\leq n} \{p_{i_j}(v)\}$ (see [29, Lemma 1.4.2 and Lemma 3.4.2]). Back to our setting, this means that there exist $v_1, \dots v_n \in \mathbf{E}$ and a constant $C > 0$ such that

$$|\alpha(v)| \leq C \cdot \max_{1\leq j\leq n} \{p_{v_j}(v)\} = C \cdot \max_{1\leq j\leq n} \{\, |\Omega(v_j, v)| \,\}.$$

Observe that α vanishes on $F = \bigcap_{j=1}^n \ker \Omega^\flat(v_j)$. Moreover, since each $\Omega^\flat(v_j)$ is continuous in the norm topology, F is closed with respect to the norm-topology and its codimension (the dimension of the quotient $\mathbf{E}/F$) is at most n. Choose any algebraic complement W of F in $\mathbf{E}$. That is, as vector spaces, $W \oplus F = \mathbf{E}$. Since $W \cong \mathbf{E}/F$, it must be finite dimensional, hence closed in $\mathbf{E}$ with respect to the Banach topology. This means that W^* is also finite-dimensional, and spanned by $\Omega^\flat(v_1)|_W, \dots, \Omega^\flat(v_n)|_W$. Then $\alpha|_W \in W^*$ can be written as $\alpha|_W = \sum_{i=1}^n \lambda_i \cdot \Omega^\flat(v_i)|_W$ for some real coefficients λ_i. Since in F both sides are identically 0, this linear combination can be extended to all of $\mathbf{E}$, thus

$$\alpha = \sum_{i=1}^n \lambda_i \cdot \Omega^\flat(v_i) = \Omega^\flat\left(\sum_{i=1}^n \lambda_i \cdot v_i\right) = \Omega^\flat(v),$$

where we defined $v = \sum_{i=1}^n \lambda_i \cdot v_i$. This proves the result. □

Lemma 2.6. Let $(\mathbf{E}, ||\cdot||_{\mathbf{E}})$ be a Banach space, and let $\Omega\colon \mathbf{E}\times\mathbf{E}\to\mathbb{R}$ be a skew-symmetric, non-degenerate, continuous bilinear map. For any closed subspace $Y\subset\mathbf{E}$, define the Ω-*orthogonal complement* of Y by

$$Y^{\Omega} := \{v\in\mathbf{E} : \Omega(u,v)=0 \text{ for all } u\in Y\}.$$

Then, $\left(Y^{\Omega}\right)^{\Omega} = Y$.

Proof. The inclusion $Y\subset (Y^{\Omega})^{\Omega}$ follows directly from the definition. For the reverse inclusion, let $v\notin Y$. By a standard consequence of the Hahn-Banach theorem for locally convex TVS, there exists $\alpha\colon E\to\mathbb{R}$ (continuous with respect to the locally convex topology) such that $\alpha|_Y\equiv 0$ and $\alpha(v)\neq 0$. By the previous lemma, there is an element $z\in E$ such that $\alpha(u)=\Omega(u,z)$. This means that $\Omega(v,z)=\alpha(v)\neq 0$. However, $z\in Y^{\Omega}$ because for every $y\in Y$, $0=\alpha(y)=\Omega(y,z)=0$. Therefore, $v\notin (Y^{\Omega})^{\Omega}$. □

Lemma 2.7. Let $\mathcal{M}$ and $\mathcal{N}$ be Banach manifolds, and let $f\colon\mathcal{M}\to\mathcal{N}$ be a surjective smooth submersion. Let $\alpha\in\Omega^k_{\mathbb{R}}(\mathcal{N})$. If $f^*\alpha=0$, then $\alpha=0$.

Proof. $\alpha=0$ if and only if it vanishes when applied to any set of tangent vectors at any point. Let $n\in N$ and $m\in M$ such that $f(m)=n$. Choose any vectors $v_1,\dots v_k\in T_n\mathcal{N}$, and $w_1,\dots w_k\in T_m\mathcal{M}$ such that $\mathrm{d}_m f(w_j)=v_j$. Then:

$$\alpha_n(v_1,\dots v_k)=\alpha_{f(m)}(\mathrm{d}_m f(w_1),\dots \mathrm{d}_m f(w_k))=(f^*\alpha)_m(w_1,\dots w_k)=0.$$

□

Lemma 2.8. Let μ be a moment map on a Banach manifold $\mathcal{M}$ for the left smooth action of a Banach-Lie group G. Then, for each $m\in\mu^{-1}(0)$, we have $(T_m\,\mu^{-1}(0))^{\Omega_m} = T_m(G\cdot m)$ and $T_m\,\mu^{-1}(0)=(T_m(G\cdot m))^{\Omega_m}$.

Proof. By lemma 2.6 it suffices to check the second identity. Let $u\in T_m\mathcal{M}$ be such that $\Omega_m(X_{\xi,m},u)=0$ $X_{\xi,m}\in T_m(G\cdot m)$ (see (2.2)), and let us see that $u\in T_m\,\mu^{-1}(0)$. Observe that

$$\Omega(X_{\xi,m},u)=\left(i_{X_\xi}\Omega\right)_m(u)=\mathrm{d}_m\mu^{\xi}(u)=(\mathrm{d}_m\mu(u),\xi)$$

the latter being zero for all $\xi\in\mathrm{Lie}(G)$ if and only if $u\in\ker\mathrm{d}_m\mu = T_m\,\mu^{-1}(0)$. □

Proof of theorem 2.4. We are now ready to prove the Banach manifold version of the symplectic reduction theorem. First, observe that the fact that $\mathcal{M}_{\mathrm{red}}$ is a Banach manifold (Hausdorff if G acts properly on $\mu^{-1}(0)$) is taken care of by the slice condition, so we just need to concern ourselves with obtaining the symplectic form.

The requirement that π be a smooth surjective submersion defines Ω_{red} uniquely. Firstly, by (2.1) there is an isomorphism $T_{\pi(m)}\mathcal{M}_{\mathrm{red}}\cong T_m(\mu^{-1}(0))/T_m(G\cdot m)$ for all $m\in\mu^{-1}(0)$. Let us denote equivalence classes using brackets $[\,\cdot\,]$. Define, pointwise

$$(\Omega_{\mathrm{red}})_{\pi(m)}([u],[v]) := \Omega_m(u,v) \qquad u,v\in T_m\,\mu^{-1}(0).$$

Let us check that the previous definition does not depend on the representatives chosen. To this end, let $[v]=[v']$ and $[u]=[u']$, where all vectors are elements of $T_m\,\mu^{-1}(0)$. Explicitly, by (2.2), this means that there exist $\xi,\zeta\in\mathrm{Lie}(G)$ such that $u'=u+X_{\xi,m}$ and

$v' = v + X_{\zeta,m}$. Then

$$\begin{aligned}\Omega_m(u', v') &= \Omega_m(u + X_{\xi,m}, v + X_{\zeta,m}) = \Omega_m(u, v) + \Omega_m(u, X_{\zeta,m}) + \Omega_m(X_{\xi,m}, v) + \Omega_m(X_{\xi,m}, X_{\zeta,m}) \\ &= \Omega_m(u, v),\end{aligned}$$

where we used that $\Omega_m(X_{\xi,m}, X_{\zeta,m}) = \Omega_m(u, X_{\zeta,m}) = \Omega_p(X_{\xi,m}, v) = 0$ because, by lemma 2.8 $T_m\ \mu^{-1}(0)$ is the Ω_m-orthogonal complement of $T_m(G \cdot m)$. Observe that $T_m(G \cdot m) \subset T_m\ \mu^{-1}(0) = (T_m(G \cdot m))^{\Omega_m}$ so elements of $T_m(G \cdot m)$ are Ω_m-orthogonal to each other. Let us now see that the definition is also independent of the point m chosen to define Ω_m. Suppose that $\pi(m) = \pi(m')$. Since π is the natural projection induced by the action $\phi\colon G \times \mu^{-1}(0) \to \mu^{-1}(0)$, there is some $g \in G$ such that $\phi_g(m) = m'$. Now, since $\pi(m) = \pi(m')$, we obviously have $T_{\pi(m)}\mathcal{M}_{\text{red}} = T_{\pi(m')}\mathcal{M}_{\text{red}}$, but then, the isomorphism (2.1) applies for both m and m'. We get that given $u \in T_m\ \mu^{-1}(0)$ and $u' \in T_{m'}\ \mu^{-1}(0)$, $[u]$ and $[u']$ correspond to the same element in $T_{\pi(m)}\mathcal{M}_{\text{red}}$ if and only if $\mathrm{d}_m\pi(u) = \mathrm{d}_{m'}\pi(u')$ where $m' = \phi_g(m)$ (see the comments below equation (2.1)). Then, since the action is symplectic

$$\begin{aligned}(\Omega_{\text{red}})_{\pi(m')}\big([\mathrm{d}_m\phi_g(u)], [\mathrm{d}_m\phi_g(v)]\big) &= \Omega_{m'}(\mathrm{d}_m\phi_g(u), \mathrm{d}_m\phi_g(v)) = \Omega_{\phi_g(m)}(\mathrm{d}_m\phi(u), \mathrm{d}_m\phi(v)) \\ &= \left(\phi_g^*\Omega\right)_m(u, v) = \Omega_m(u, v) = (\Omega_{\text{red}})_{\pi(m)}([u], [v]),\end{aligned}$$

where we used the fact that $\mathrm{d}_{m'}\pi\left(\mathrm{d}_m\phi_g(u)\right) = \mathrm{d}_m(\pi \circ \phi_g)(u) = \mathrm{d}_m\pi(u)$ to deduce that $[u]$ and $[\mathrm{d}_m\phi_g(u)]$ represent the same element in $T_{\pi(m)}\mathcal{M}_{\text{red}}$.

This means that Ω_{red} is well-defined. It is smooth because the definition implies that $\pi^*\Omega_{\text{red}} = i^*\Omega$, and the right-hand side is smooth. Moreover, it is closed because d commutes with pullbacks

$$\pi^*(\mathrm{d}\Omega_{\text{red}}) = \mathrm{d}\left(\pi^*\Omega_{\text{red}}\right) = \mathrm{d}\left(i^*\Omega\right) = i^*\left(\mathrm{d}\Omega\right) = 0,$$

and π is a surjective smooth submersion, so we may apply lemma 2.7. It remains to show that Ω is non-degenerate. This is a linear issue, so choose $m \in \mu^{-1}(0)$, and suppose that there is some $[u] \in T_{\pi(m)}\mathcal{M}_{\text{red}}$ such that for every $[v] \in T_{\pi(m)}\mathcal{M}$ we have $(\Omega_{\text{red}})_{\pi(m)}([u], [v]) = 0$, let us see that $[u] = [0]$. The definition of Ω_{red} allows us to translate this problem to one on $T_m\ \mu^{-1}(0)$. That is, given $u \in T_m\mu^{-1}(0)$ such that for every $v \in T_m\ \mu^{-1}(0)$ we have $\Omega_m(u, v) = 0$, we have to show that $u \in T_m(G \cdot m)$. Let us apply lemma 2.5. Take $T_m\mathcal{M}$ as the Banach space (the linear isomorphisms of lemma C.3 can be used to endow $T_m\mathcal{M}$ with different but equivalent norms), and take $T_m(G \cdot m)$ as the closed subspace. Now, lemma 2.6 implies that

$$\begin{aligned}T_m(G \cdot m) &= \left((T_m(G \cdot m))^{\Omega_m}\right)^{\Omega_m} = (T_m\ \mu^{-1}(0))^{\Omega_m} \\ &= \left\{u \in T_m\mathcal{M} : \Omega_m(u, v) = 0 \quad \text{for all} \quad v \in T_m\ \mu^{-1}(0)\right\},\end{aligned}$$

which is what we wanted to show. □

Chapter 3

Principal bundles

The aim of this chapter is to introduce basic concepts regarding principal bundles. As we did for vector bundles, we will describe simultaneously principal bundles in the smooth and holomorphic categories. Hence, we recover the notation $\mathcal{S}$ introduced in appendix A to mean either smooth or holomorphic. The expression *$\mathcal{S}$-Lie group* means a smooth or holomorphic Lie group, and an *$\mathcal{S}$-action* means either a smooth or holomorphic action (that is, the map $G \times M \to M, (g, m) \mapsto g \cdot m$ is an $\mathcal{S}$-morphism). A good reference for principal G-bundles is [48, Ch. 1].

We will also describe connections in principal bundles. Since principal bundles do not have linear fibers, connections will no longer be differential operators as happens in vector bundles (see definition A.10), but we will see that they can be understood from a very geometric viewpoint. There is also a concept of curvature, and this can be seen as a section of an associated vector bundle, which means that differential operators may be applied to the curvature of a connection. This is, as we will see in chapter 4, essential to write down the hermitian Yang-Mills equation. The chapter ends by introducing the Chern-Singer correspondence, which shows how reductions of the structure group parametrize complex structures on smooth principal bundles.

3.1 Definition and first properties

The spirit of the definition of a principal bundle is quite similar to that of vector bundles: we attach a Lie group to each point of the manifold in a way such that the resulting manifold becomes locally trivial. Before stating the definition, recall that an action ϕ of a group G on a set S is free if $\phi_g(s) = s$ for some $s \in S$ implies that $g = e$ (that is, only the group identity can have fixed points).

Definition 3.1 (Principal G-bundle). A *principal G-bundle*, or *principal bundle with structure group G* is a 4-tuple (P, G, M, π), where P and M are $\mathcal{S}$-manifolds, G is an $\mathcal{S}$-Lie group, and $\pi \colon P \to M$ is an $\mathcal{S}$-morphism satisfying

1. There is a free, $\mathcal{S}$, right action of G on P.
2. $\pi^{-1}(\pi(p)) = \{p \cdot g : g \in G\} = p \cdot G$.
3. For every $m \in M$ there are an open neighborhood $U \subset M$ of m and an $\mathcal{S}$-isomorphism, called a *local trivialization* $\Phi \colon \pi^{-1}(U) \to U \times G$, where $\Phi(p) = (\pi(p), s_U(p))$, and $s_U \colon \pi^{-1}(U) \to G$ commutes with the G-action, that is, $s_U(p \cdot g) = s_U(p) \cdot g$.

We say that M is the *base space*, and P is the *total space*. Observe that holomorphic principal bundles require G and M to be complex. If there is no room for confusion,

we will denote a principal G-bundle by P, or P_G. Observe that π must be a surjective submersion. Moreover, local triviality implies that $\dim P(M,G) = \dim G + \dim M$. Since principal bundles are equipped with a natural projection to the base manifold, we can define *global sections* (resp. *local sections*) as $\mathcal{S}$-morphisms $\sigma\colon M \to P$ (resp. $\sigma\colon U \to P$ where $U \subset M$ is open) satisfying $\pi \circ \sigma = \mathrm{id}_M$ (resp. $\pi \circ \sigma = \mathrm{id}_U$).

Remark 3.2. If $P(M,G)$ is a principal G-bundle, then the action of G on P is proper, and $P/G \cong M$. Some references, such as [27], define principal G-bundles by replacing part 2 of definition 3.1 by the condition $P/G \cong M$.

Definition 3.3 (Principal bundle homomorphism). Let (P_1, M_1, G_1, π_1) and (P_2, M_2, G_2, π_2) be two $\mathcal{S}$-principal bundles with respective structure groups G_1 and G_2. A *principal bundle homomorphism* is a pair of mappings (ϑ, λ), such that $\vartheta\colon P_1 \to P_2$ is an $\mathcal{S}$-morphism, $\lambda\colon G_1 \to G_2$ is an $\mathcal{S}$-Lie group homomorphism, and $\vartheta(p \cdot g) = \vartheta(p) \cdot \lambda(g)$ for all $p \in P_1, g \in G_1$.

We will say that a principal bundle homomorphism is an *isomorphism* if ϑ is an $\mathcal{S}$-isomorphism, and λ is a Lie group isomorphism.

In the situation described in the previous definition, ϑ induces a smooth map on the base manifolds M_1 and M_2. Indeed, ϑ must send fibers of π_1 to fibers of π_2, and therefore, we may define $\vartheta'\colon M_1 \to M_2$ by $\vartheta'(\pi_1(p)) := \pi_2(\vartheta(p))$ (which is an $\mathcal{S}$-morphism by local triviality). This makes the following diagram commutative.

$$\begin{array}{ccc} P_1 & \xrightarrow{\vartheta} & P_2 \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle \pi_2} \\ M_1 & \xrightarrow{\vartheta'} & M_2 \end{array}$$

Observe that $\vartheta'(\pi_1(p \cdot g)) = \pi_2(\vartheta(p \cdot g)) = \pi_2(\vartheta(p) \cdot \lambda(g)) = \pi_2(\vartheta(p))$ so it is well-defined and independent of the representative chosen. It is smooth because π_1 is a surjective smooth submersion [34, Th. 4.29], and $\vartheta' \circ \pi_1 = \pi_2 \circ \vartheta$, the right-hand side being smooth.

Given an $\mathcal{S}$-principal bundle $P(M,G)$, another $\mathcal{S}$-manifold, and an $\mathcal{S}$-morphism $\varphi\colon N \to M$, we can define the *pullback principal bundle*

$$\varphi^* P := \{(n,p) \in N \times P : \varphi(n) = \pi(p)\} \subset N \times P. \tag{3.1}$$

Endowed with the action given by $(n,p)\cdot g = (n, p\cdot g)$ and whose projection is the restriction of the projection $N \times P \to N$, it becomes a principal G-bundle [48, Remark 1.1.9].

Finally, sometimes vector bundles and fiber bundles are not enough; sometimes we need general manifolds to be the fibers of a bundle. This motivates the definition of a general fiber bundle, which generalizes vector and principal bundles.

Definition 3.4 (Fiber bundle). A *fiber bundle* is a 4-tuple (E, M, F, π), where M, E and F are $\mathcal{S}$-manifolds, $\pi\colon E \to M$ is a surjective $\mathcal{S}$-morphism, and for every $m \in M$ there is an open neighborhood $U \subset M$, and an $\mathcal{S}$-isomorphism $\Phi\colon \pi^{-1}(U) \to U \times F$ called a *local trivialization*, such that $\pi_1 \circ \Phi = \pi_U$.

Observe that we use the term *local trivialization* for all the types of bundles we have discussed. This shall lead to no confusion, as it will be clear from the context the type of bundle we are dealing with.

3.1.1 Associated bundles

Given an $\mathcal{S}$-principal G-bundle, and another $\mathcal{S}$-manifold F which is acted upon from the left by G by $\mathcal{S}$-morphisms, we may construct another fiber bundle called an *associated bundle* to the original principal G-bundle.

Firstly, observe that if $\phi\colon F\times G\to F$ is the left action originally defined, we may always define a right action as follows

$$\begin{array}{rccc}\tilde{\phi}\colon & F\times G & \longrightarrow & F\\ & (p,\xi) & \mapsto & \phi_{g^{-1}}(\xi).\end{array} \tag{3.2}$$

Thus, we can define a right action on the usual cartesian product $P\times F$ by

$$\begin{array}{ccc}(P\times F)\times G & \longrightarrow & P\times F\\ ((p,\xi),g) & \mapsto & (p\cdot g, g^{-1}\cdot\xi).\end{array}$$

If G acts freely and by $\mathcal{S}$-morphisms on F, then it does so on $P\times F$. Certainly, the action is component-wise free, therefore free. Moreover, by remark 3.2, the action of G on P is proper. Furthermore, the product of a proper action with any other action is again proper (see [47, Remark 6.3.9]). Having that in mind, we define $P\times_G F:=(P\times F)/G$. That is, we define $P\times_G F$ to be the set of equivalence classes for the equivalence relation $(p_1,\xi_1)\sim(p_2,\xi_2)$ if and only if $(p_1\cdot g, g^{-1}\cdot\xi_1)=(p_2,\xi_2)$ for some $g\in G$. Let us denote by $[p,\xi]$ the equivalence class of a pair (p,ξ). We endow it with the quotient topology, and by the quotient manifold theorem (theorem 1.2 or theorem B.3 depending on the category we are working in), $P\times_G F$ has a unique $\mathcal{S}$-manifold structure that makes the usual quotient map $q\colon P\times F\to P\times_G F$ into an $\mathcal{S}$-morphism and a submersion. Define the projection $\pi_G\colon P\times_G F\to M, [p,\xi]\mapsto\pi(p)$. It is well-defined because π is invariant by the action of G. Moreover, the following diagram commutes

$$\begin{array}{ccc}P\times F & & \\ {\scriptstyle q}\downarrow & \searrow^{\pi\circ\pi_P} & \\ P\times_G F & \xrightarrow[\pi_G]{} & M.\end{array}$$

Since $\pi\circ\pi_P$ is an $\mathcal{S}$-morphism, the characteristic property of surjective smooth submersions implies that π_G is an $\mathcal{S}$-morphism (see [34, Th. 4.29] and [33, Cor. 2.6] for the smooth and complex categories respectively). The only thing left to check is that this structure admits local trivializations, a claim for which we refer the reader to [48, Sec. 1.2]. We sometimes denote $P(F):=P\times_G F$ if the group is clear from the context.

Remark 3.5. The natural projection $q\colon P\times F\to P\times_G F$ induces a natural $\mathcal{S}$-isomorphism between the fibers of π_G and F. Indeed, fix $p\in P$, and define the map $q_p\colon F\to P\times_G F, f\mapsto[p,f]$. The image is clearly contained in $\pi_G^{-1}(\pi(p))$, and q_p is equivariant:

$$q_{p\cdot g}(f)=[p\cdot g,f]=[p\cdot g,g^{-1}\cdot g\cdot f]=[p,g\cdot f]=q_p(g\cdot f),$$

thus $q_{p\cdot g}=q_p\circ\phi_g$. If we choose an element of the fiber $\pi_G^{-1}(\pi(p))$, it will be $[p\cdot g,f]=q_p(g\cdot f)$, so each q_p is a bijection. In fact, it is an $\mathcal{S}$-isomorphism.

Let us now discuss the case of the adjoint bundle. Let $P(M,G)$ be a principal G-bundle. Take F to be the Lie algebra $\mathfrak{g}$. This is clearly a smooth manifold because it is a finite-dimensional vector space. Moreover, $\mathrm{Ad}\colon G\to\mathrm{GL}(\mathfrak{g})$ is a left action of G on $\mathfrak{g}$. We can thus define the *adjoint bundle* $\mathrm{ad}(P):=P\times_G\mathfrak{g}$. This case is special because we actually get a vector bundle by defining $\lambda\cdot[p,\xi]+\mu\cdot[p,\zeta]:=[p,\lambda\xi+\mu\zeta]$. Observe that the fibers are (non-canonically) isomorphic to $\mathfrak{g}$ through $\mathfrak{g}\to(P\times_G\mathfrak{g})_m, \xi\mapsto[p,\xi]$ with $\pi(p)=m$. This actually works for any vector space V on which there is a representation of G: $P\times_G V$ is a vector bundle.

Example 3.6. Let us comment on another example of an associated bundle that we will use: the *group adjoint bundle*. Let $P(M,G)$ be a principal G-bundle. Consider $F = G$ in the notation at the beginning of the section, where G acts on itself from the left by conjugation $g \cdot h := ghg^{-1}$. We then define $\mathrm{Ad}(P) := P \times_G G$. Observe that the fibers of $\mathrm{Ad}(P)$ have a natural group structure. The product is defined by $[p,g] \cdot [p,h] := [p, g \cdot h]$, and one readily sees that it is well-defined. In general, $\mathrm{Ad}(P)$ is a fiber bundle over M.

3.2 Theory of connections

In this section we will work in the smooth category. The last section of the chapter dives into the relation between connections and complex structures on smooth principal bundles. As we will see, there are several equivalent definitions of a connection on a principal G-bundle.

Since every principal bundle has a natural Lie group action, every $\xi \in \mathfrak{g}$ defines the *fundamental vector field* $X_\xi \in \mathfrak{X}(P)$ through

$$X_{\xi,p} := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} p \cdot \exp(t\xi) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \phi^{(p)}(\exp(t\xi)) = \left(\phi^{(p)} \circ \gamma\right)'(0) = \mathrm{d}_e\phi^{(p)}(\xi), \tag{3.3}$$

where γ is the curve given by $\gamma(t) = \exp(t\xi)$, and we used [34, Cor. 3.25]. Observe that these vector fields are always tangent to the orbits of the action (because the curves $\exp(tX) \cdot p$ are contained in said orbits). The fact that the assignment $p \mapsto X_{\xi,p}$ defines a smooth vector field is readily checked (see, for example [60, Prop. 27.11]). The natural projection $\pi\colon P \to M$ defines, for each $p \in P$, a *vertical subspace* given by $V_pP := \ker \mathrm{d}_p\pi$. Its elements are called *vertical vectors*, and a vector field that is always vertical is called a *vertical vector field*. Observe that Killing vector fields are always vertical. Indeed,

$$\mathrm{d}_p\pi(X_{\xi,p}) = \mathrm{d}_e\left(\pi \circ \phi^{(p)}\right)(\xi) = 0$$

because $\pi \circ \phi^{(p)}$ is constant. In fact, $\mathrm{d}_e\phi^{(p)}\colon \mathfrak{g} \to V_pP$ is a vector space isomorphism (see [48, Lemma 1.3.1] or [60, Prop. 27.18]). Moreover, the union of all vertical subspaces defines a subbundle, called the *vertical subbundle* of TP, and $VP \cong P \times \mathfrak{g}$, so VP is a trivial bundle. The isomorphism is given by $(p,\xi) \mapsto X_{\xi,p}$.

Definition 3.7 (Connection on a principal G-bundle). Let $P(M,G)$ be a principal G-bundle. A *connection* on P is a smooth distribution HP such that

1. $T_pP = V_pP \oplus H_pP$ for every $p \in P$.
2. $H_{\phi_g(p)}P = \mathrm{d}_p\phi_g(H_pP)$ for every $p \in P, g \in G$. We say that the distribution is G-invariant.

We call H_pP the *horizontal subspace* at p.

The direct sum decomposition $T_pP = H_pP \oplus V_pP$ on each tangent space implies that every vector $w \in T_pP$ can be uniquely decomposed as a vertical and a horizontal vector $w = w^H + w^V$. In fact, given a vector field $X \in \mathfrak{X}(P)$, the fiberwise decomposition defines two smooth vector fields $X^V \in \Gamma(P; VP)$ and $X^H \in \Gamma(P; HP)$ which are called the *vertical* and *horizontal components* of X respectively [60, Prop. 28.3]. Observe that both components depend on the connection. To sum up, the main idea of the definition is that a connection is a *choice* of a complementary subspace to the vertical subspace V_pP because, in general, there is no canonical way of making this choice.

Remark 3.8.

1. The vertical-horizontal decomposition is G-invariant. More precisely, if $w = w^H + w^V \in T_pP$, then the vertical and horizontal parts of $\mathrm{d}_p\phi_g(w) \in T_{p\cdot g}P$ are $\mathrm{d}_p\phi_g\left(w^V\right)$ and $\mathrm{d}_p\phi_g\left(w^H\right)$ respectively. Indeed, $\mathrm{d}_p\phi_g(w) = \mathrm{d}_p\phi_g\left(w^H\right) + \mathrm{d}_p\phi_g\left(w^V\right)$. The latter is vertical because $\mathrm{d}_{p\cdot g}\pi\left(\mathrm{d}_p\phi_g\left(w^V\right)\right) = \mathrm{d}_p\pi(w^V) = 0$, and the former is horizontal because the distribution is G-invariant. The result follows from the uniqueness of the decomposition.
2. The projection $\mathrm{d}_p\pi$ defines a vector space isomorphism between H_pP and $T_{\pi(p)}M$. This follows from the fact that $\mathrm{d}_p\pi$ is surjective and $\ker \mathrm{d}_p\pi = V_pP$, so by the first isomorphism theorem for vector spaces $T_pP/V_pP \cong H_pP \cong T_{\pi(p)}M$.

Moreover, connections allow us to speak about horizontal lifts. Let $P(M,G)$ be a principal bundle endowed with a connection, which we see as a horizontal subbundle. Let $Y \in \mathfrak{X}(M)$. A *horizontal lift* is another vector field Z that is horizontal such that $\mathrm{d}_p\pi(Z_p) = Y_{\pi(p)}$ for each $p \in P$. As we see in the following lemma, horizontal lifts always exist.

Lemma 3.9 (Existence of horizontal lifts). Let $P(M,G)$ be a principal bundle, and let $Y \in \mathfrak{X}(M)$. Then, Y admits a unique horizontal lift. Moreover, if $Y^{\mathbf{HL}}$ is the lift, it satisfies $(\phi_g)_* Y^{\mathbf{HL}} = Y^{\mathbf{HL}}$.

Proof. By remark 3.8, the only option is to define $Y^{\mathbf{HL}}_p := (\mathrm{d}_p\pi|_{H_pP})^{-1}(Y_{\pi(p)})$. Since smoothness is a local issue, up to choosing a trivialization, we can assume that P is trivial. Hence, $TP \cong TM \oplus TG$. Then, we choose any smooth vector field $\hat{Y} \in \mathfrak{X}(P)$ satisfying $\mathrm{d}_{p,g}\pi(\hat{Y}_{p,g}) = Y_p$, for example, $(p,g) \mapsto (Y_p, 0) = \hat{Y}_p$. By uniqueness of the vertical-horizontal decomposition, $\hat{Y}^H = Y^{\mathbf{HL}}$, and it follows that the horizontal lift is smooth. Invariance under the action is clear from the fact that

$$\mathrm{d}_p\phi_g|_{H_pP} \circ \left(\mathrm{d}_p\pi|_{H_pP}\right)^{-1} = \left(\mathrm{d}_{p\cdot g}\pi|_{H_{p\cdot g}P}\right)^{-1},$$

so $\mathrm{d}_p\phi_g\left(Y^{\mathbf{HL}}_p\right) = \mathrm{d}_p\phi_g\left(\left(\mathrm{d}_p\pi|_{H_pP}\right)^{-1}\left(Y_{\pi(p)}\right)\right) = \left(\mathrm{d}_{p\cdot g}\pi|_{H_{p\cdot g}P}\right)^{-1}(Y_p) = Y^{\mathbf{HL}}_{p\cdot g}$. □

We will write $Y^{\mathbf{HL}}$ for the horizontal lift of a vector field Y. It is not explicit in the notation, but it clearly depends on the connection.

Definition 3.7 is quite intuitive, but impractical to work with. There is a less intuitive, but more practical definition in terms of a $\mathfrak{g}$-valued one-form. Recall that, given a vector space V. By a V-valued p-form we mean a p-form with values in the trivial vector bundle $M \times V$, that is, a section of the vector bundle $(M \times V) \otimes \bigwedge^p T^*M$ (see section A.5.1). Observe that a general V-valued p-form may be written as a sum

$$\alpha = v_1 \otimes \alpha_1 + v_2 \otimes \alpha_2 + \dots v_r \otimes \alpha_r \tag{3.4}$$

where $v_i \in V$, and $\omega_j \in \Omega^p_{\mathbb{R}}(M)$. If V is a complex vector space and the forms are taken to be complex, everything works the same. We will write $\Omega^p_{\mathbb{R}}(M;V)$ for the set of real V-valued p-forms.

Definition 3.10 (Connection 1-form). Let $P(M,G)$ be a principal G-bundle, and let HP be a connection. Define the *connection 1-form* by

$$A_p(w) = \left(\mathrm{d}_e\phi^{(p)}\right)^{-1}\left(w^V\right) \qquad w \in T_pP.$$

Observe that we are using A instead of a greek letter to denote the one-form. This notation is not universal. The following theorem shows that A is smooth, and some properties that characterize it.

Theorem 3.11. Let $P(M,G)$ be a principal G-bundle, and consider a connection whose connection one-form is A. Then, A is a smooth one-form, and

1. $A(X_\xi) = \xi$ for every $\xi \in \mathfrak{g}$.
2. $(\phi_g)^* A = \mathrm{Ad}_{g^{-1}} \circ A$. We say that A is *G-equivariant*.

Moreover, if $A \in \Omega^1_{\mathbb{R}}(P;\mathfrak{g})$ satisfies the previous conditions, then there is a unique connection in P whose connection one-form is A.

Proof. See [60, Prop. 28.5]. □

Intuitively, a connection projects a vector into its vertical component, which can be identified with an element of the Lie algebra. Moreover, one may show that every principal bundle admits a connection. This is done through a partition of unity argument [48, Prop. 1.3.7]. The horizontal subbundle determined by a connection whose one-form is A will be denoted $H^A P$.

3.3 Curvature of a connection

First and foremost, when we deal with vector-valued p-forms, we can extend the exterior derivative as follows. If V is a real vector space, and $\beta \in \Omega^p_{\mathbb{R}}(M;V)$, we have seen that we may express β in the following manner (we may suppose that $\{v_1, \dots v_r\}$ is a basis of V) $\beta = \beta^j \otimes v_j$, where $\beta^j \in \Omega^p_{\mathbb{R}}(M)$. We can simply define

$$\mathrm{d}\beta := \mathrm{d}\beta^j \otimes v_j \in \Omega^{p+1}_{\mathbb{R}}(M;V),$$

and one readily checks that the definition does not depend on the basis chosen.

Definition 3.12 (Covariant exterior derivative). Let $P(M,G)$ be a principal G-bundle, and let V be a finite dimensional vector space. Given $\alpha \in \Omega^p_{\mathbb{R}}(M;V)$, we define its *covariant exterior derivative with respect to a connection* A as the V-valued $(p+1)$-form given by

$$\mathrm{D}_A\alpha(X_0, X_1, \dots X_p) := \mathrm{d}\alpha\left(X_0^H, X_1^H, \dots X_p^H\right), \qquad X_1, \dots X_p \in \mathfrak{X}(P).$$

Given $\alpha \in \Omega^p_{\mathbb{R}}(M;V)$, it is common to denote by α^H the V-valued p-form

$$(\alpha)^H(X_1, \dots X_p) := \alpha\left(X_1^H, \dots X_p^H\right).$$

Therefore, we may also write $\mathrm{D}_A\alpha = (\mathrm{d}\alpha)^H$. The symbol A in the notation is intended to reference the connection one-form. This is because the covariant derivative depends on the connection (because the horizontal components of the vectors depend on the connection). We are now ready to define the curvature of a connection.

Definition 3.13 (Curvature of a connection). Let $P(M,G)$ be a principal G-bundle endowed with a connection with connection one-form A. We define the curvature of this connection as $\mathrm{D}_A A \in \Omega^2_{\mathbb{R}}(P;\mathfrak{g})$.

Given $\alpha \in \Omega^k_{\mathbb{R}}(M;\mathfrak{g})$ and $\beta \in \Omega^\ell_{\mathbb{R}}(M;\mathfrak{g})$, we define $[\beta \wedge \alpha] \in \Omega^{k+\ell}_{\mathbb{R}}(M;\mathfrak{g})$ by

$$[\beta \wedge \alpha](X_1, \dots X_{k+\ell}) := \frac{1}{k! \cdot \ell!} \sum_{\sigma \in S_{k+\ell}} \operatorname{sgn}(\sigma) \cdot [\beta(X_{\sigma(1)}, \dots X_{\sigma(\ell)}), \alpha(X_{\sigma(\ell+1)}, \dots X_{\sigma(\ell+k)})] \in \mathfrak{g},$$

Equivalently, we may define $[\cdot \wedge \cdot]$ by linear extension of $[(\beta \otimes \xi) \wedge (\alpha \otimes \zeta)] := (\beta \wedge \alpha) \otimes [\xi, \zeta]$ for α, β usual differential forms and $\xi, \zeta \in \mathfrak{g}$ (see [19, Sec. 5.2]). Then, the curvature may be expressed directly in terms of the connection one-form.

Theorem 3.14 (Structure equation). Let $P(M,G)$ be a principal G-bundle endowed with a connection, and let A be the corresponding connection one-form. Then,

$$\mathrm{D}_A A = \mathrm{d}A + \frac{1}{2} \cdot [A \wedge A].$$

Proof. The proof is rather lengthy, see [27, Th. 5.2] or [55, Th. 11.2]. The strategy is to consider separately the cases where $\mathrm{D}_A A$ acts on two vertical vectors, two horizontal vectors, and a horizontal and a vertical vector. Then, one concludes using the fact that both sides are linear. □

We devote the rest of the section to showing that the curvature of a connection on a principal G-bundle can be viewed in a canonical way as a $\mathrm{ad}(P)$-valued two-form, that is, a section of a vector bundle. Let us start with a definition. For now, we keep the discussion general, and we define horizontal differential forms for any finite-dimensional representation of the Lie group.

Definition 3.15. Let $P(M,G)$ be a principal G-bundle with right action $\phi\colon P \times G \to P$, and consider a finite-dimensional representation $\sigma\colon G \to \mathrm{Aut}(V)$ for some finite-dimensional vector space V. Given $\alpha \in \Omega^k_{\mathbb{R}}(P;V)$ we say that it is *horizontal (of type σ)* if it satisfies:

1. For every vertical vector field X, $i_X \alpha = 0$.
2. $(\phi_g)^* \alpha = \sigma(g^{-1}) \circ \alpha$ for every $g \in G$.

We denote the space of all horizontal k-forms with respect to σ by $\Omega^k_{\sigma,\mathrm{hor}}(P;V)$.

The space of connections (more precisely, connection one-forms) on a principal bundle, which we denote $\mathscr{A}(P_G)$, is an affine space modelled on the space of horizontal one-forms with respect to the adjoint representation, i.e. $\Omega^1_{\mathrm{Ad},\mathrm{hor}}(P;\mathfrak{g})$. Indeed, let A_1 and A_2 be the connection one-forms of two connections on a principal G-bundle, and let $A := A_1 - A_2$. Being the difference of two $\mathfrak{g}$-valued one-forms, it is a one-form, so we need only check that it is horizontal. The fact that $(\phi_g)^* A = \mathrm{Ad}_{g^{-1}} \circ A$ follows from the second statement of theorem 3.11. Now, the vertical bundle is spanned by vector fields X_ξ for $\xi \in \mathfrak{g}$ and $i_{X_\xi} A = A(X_\xi) - A_2(X_\xi) = \xi - \xi = 0$, therefore, $i_X A = 0$ for each vertical vector field X. Conversely, given a connection $A \in \Omega^1_{\mathbb{R}}(P;\mathfrak{g})$, and a horizontal one-form $\alpha \in \Omega^1_{\mathrm{Ad},\mathrm{hor}}(P;\mathfrak{g})$, the sum $A + \alpha$ is a connection one-form. By property 2 of definition 3.15, $A + \alpha$ is G-equivariant. Moreover, given a vertical vector field X_ξ, we have $(A+\alpha)(X_\xi) = A(X_\xi) + \alpha(X_\xi) = A(X_\xi) + i_{X_\xi}\alpha = \xi + 0$.

This discussion implies that if we fix a connection one-form A, then any other connection may be obtained as $A + \alpha$ for $\alpha \in \Omega^1_{\mathrm{Ad},\ \mathrm{hor}}(P;\mathfrak{g})$. It is then natural to ask how the curvature of $A + \alpha$ relates to that of A. From the structure equation, (theorem 3.14) it

follows that

$$\mathrm{D}_{A+\alpha}(A+\alpha) = \mathrm{d}(A+\alpha) + \frac{1}{2}\cdot[(A+\alpha)\wedge(A+\alpha)] = \mathrm{D}_A A + \mathrm{d}\alpha + [A\wedge\alpha] + \frac{1}{2}\cdot[\alpha\wedge\alpha].$$

Moreover, for $\alpha \in \Omega^p_{\mathrm{Ad},\ \mathrm{hor}}(P;\mathfrak{g})$, the following relation holds $\mathrm{D}_A\alpha = \mathrm{d}\alpha + [A\wedge\alpha]$ (see [48, Remark 1.4.12]). Taking this identity into account, the previous equation reads

$$\mathrm{D}_{A+\alpha}(A+\alpha) = \mathrm{D}_A A + \mathrm{D}_A\alpha + \frac{1}{2}\cdot[\alpha\wedge\alpha]. \tag{3.5}$$

A straightforward calculation shows that $\mathrm{D}_A A$ is always horizontal (see [48, Page 40]), that is, $\mathrm{D}_A A \in \Omega^2_{\mathrm{Ad},\ \mathrm{hor}}(P;\mathfrak{g})$. It is sometimes useful to see the curvature as a section of a vector bundle over the base manifold. To that end, we are going to show now that there is an isomorphism $\Omega^k_{\sigma,\mathrm{hor}}(P;V) \cong \Omega^k_{\mathbb{R}}(M;P\times_G V)$. In particular, this means that there is an isomorphism $\Omega^2_{\mathrm{Ad},\mathrm{hor}}(P;\mathfrak{g}) \cong \Omega^2_{\mathbb{R}}(M;\mathrm{ad}(P))$.

Theorem 3.16. Let $P(M,G)$ be a principal G-bundle, and $\sigma\colon G\to\mathrm{Aut}(V)$ a finite-dimensional representation.

To every $\alpha\in\Omega^k_{\sigma,\mathrm{hor}}(P;V), k\geq 1$ corresponds a unique element $\Upsilon(\alpha)\in\Omega^k_{\mathbb{R}}(M;P\times_G V)$ such that the following diagram commutes

$$\begin{array}{ccc} \bigwedge^k TP & \xrightarrow{\mathrm{pr}\times\alpha} & P\times V \\ {\scriptstyle \Lambda^k\mathrm{d}\pi}\downarrow & & \downarrow{\scriptstyle q} \\ \bigwedge^k TM & \xrightarrow{\Upsilon(\alpha)} & P\times_G V \end{array}$$

Moreover, the assignment $\Upsilon\colon\Omega^k_{\sigma,\mathrm{hor}}(P;V)\to\Omega^k_{\mathbb{R}}(M;P\times_G V), \alpha\mapsto\Upsilon(\alpha)$ is a vector space isomorphism.

Proof. Recall that $P\times_G V = P\times V/\sim$ where $\sim$ denotes the equivalence relation: $(p,v)\sim(p',v')$ if and only if $p'=p\cdot g$ and $v'=\sigma_{g^{-1}}(v)$. It is a vector bundle. To fix notation, set $q\colon P\times V\to P\times_G V$ to be the quotient map, $\mathrm{pr}\colon\bigwedge^k TP\to P$ the usual projection, and $\Lambda^k\mathrm{d}\pi\colon\bigwedge^k TP\to\bigwedge^k TM$ the bundle homomorphism uniquely determined by

$$\Lambda^k\mathrm{d}\pi(w_1\wedge w_2\ldots\wedge w_k) := (\mathrm{d}_p\pi(w_1))\wedge(\mathrm{d}_p\pi(w_2))\wedge\ldots\wedge(\mathrm{d}_p\pi(w_k)) \quad \text{if} \quad w_1,\ldots,w_k\in T_pP.$$

$\Lambda^k\mathrm{d}\pi$ is a surjective smooth submersion. Define as well $q_p\colon V\to P\times_G V$ by $q_p(v):=[p,v]$. Having said that, let $m\in M$, $v_i\in T_mM$, $1\leq i\leq k$, and choose $p\in P$ that projects onto m (that is $\pi(p)=m$). Choose as well $w_i\in T_pP$ such that $\mathrm{d}_p\pi(w_i)=v_i$ for $1\leq i\leq k$. We define

$$\Upsilon(\alpha)_m(v_1,\ldots,v_k) := (q_p\circ\alpha_p)(w_1,\ldots,w_k). \tag{3.6}$$

Once we see that this is independent of the choices of p and vectors $w_1,\ldots w_k$, α will be well-defined as a map, and commutativity of the diagram will be obvious. Recall that

$$\Omega^k_{\mathbb{R}}(M;P\times_G V) = \Gamma\left(M;\left(\bigwedge^k TM\right)^*\otimes P\times_G V\right)$$

and $\left(\bigwedge^k TM\right)^*\otimes P\times_G V$ is identified with the bundle whose fibers are alternating multilinear forms from $T_mM\times\overset{k}{\cdots}\times T_mM\to(P\times_G V)_m$. Hence, $\Upsilon(\alpha)\circ\Lambda^k\mathrm{d}\pi(w_1\wedge\ldots\wedge w_k)$ is just the form $\Upsilon(\alpha)$ acting on $v_1,\ldots,v_r$. As usual, smoothness follows from the characteristic property of surjective smooth submersions [34, Th. 4.29] applied to $\Lambda^k\mathrm{d}\pi$.

To see that (3.6) does not depend on our choices, choose another point p' projecting to m. Thus, we will have $p'=p\cdot g$ for some $g\in G$. Let $w_i'\in T_{p'}P$ such that $\mathrm{d}_{p'}\pi(w_i')=v_i$

for $1 \leq i \leq k$. Then, since $\mathrm{d}_{p'}\pi(w_i' - \mathrm{d}_p\phi_g(w_i)) = v_i - \mathrm{d}_p(\pi \circ \phi_g)(w_i) = v_i - v_i = 0$, there are vertical vectors $u_i \in V_{p'}P$ satisfying $w_i' = \mathrm{d}_p\phi_g(w_i) + u_i$. Thus, we get

$$\begin{aligned} q_{p'} \circ \alpha_{p'}(w_1', \dots w_k') &= q_{p\cdot g} \circ \alpha_{p\cdot g}(\mathrm{d}_p\phi_g(w_1) + u_1, \dots, \mathrm{d}_p\phi_g(w_k) + u_k) \\ &\overset{(1)}{=} q_{p\cdot g} \circ \alpha_{p\cdot g}(\mathrm{d}_p\phi_g(w_1), \dots, \mathrm{d}_p\phi_g(w_k)) \\ &\overset{(2)}{=} q_p \circ \sigma(g) \circ \left((\phi_g)^*\alpha\right)_p(w_1, \dots, w_k) \overset{(3)}{=} q_p \circ \alpha_p(w_1, \dots, w_k). \end{aligned}$$

In (1), we used the fact that, if α is horizontal, then it vanishes when one of its entries is a vertical vector field. (2) uses the fact that $q_{p\cdot g} = q_p \circ \sigma_g$ (see remark 3.5), and (3) follows from the identity $\sigma(g^{-1}) \circ \alpha = (\phi_g)^*\alpha$ for horizontal forms. Observe that the map Υ is clearly linear. For the converse, given $\beta \in \Omega^k_{\mathbb{R}}(M; P \times_G V)$, we define $\hat{\Upsilon}(\beta)$ by $(\hat{\Upsilon}(\beta))_p = q_p^{-1} \circ (\pi^*\beta)_p$. Observe that, looking at (3.6) this is the only reasonable choice. $\hat{\Upsilon}(\beta)$ is horizontal. Indeed, if w^V is vertical, $i_{w^V}\hat{\Upsilon}(\beta) = 0$ because $\mathrm{d}_p\pi\left(w^V\right) = 0$. Moreover,

$$\begin{aligned} \left((\phi_g)^*\hat{\Upsilon}(\beta)\right)_p(w_1, \dots w_k) &= \left(\hat{\Upsilon}(\beta)\right)_{p\cdot g}(\mathrm{d}_p\phi_g(w_1), \dots \mathrm{d}_p\phi_g(w_k)) \\ &= q_{p\cdot g}^{-1} \circ \beta_{\pi(p)}\left(\mathrm{d}_{p\cdot g}\pi(\mathrm{d}_p\phi_g(w_1)), \dots \mathrm{d}_{p\cdot g}\pi(\mathrm{d}_p\phi_g(w_k))\right) \\ &= \sigma(g^{-1}) \circ q_p^{-1} \circ \beta_{\pi(p)}(\mathrm{d}_p\pi(w_1), \dots \mathrm{d}_p\pi(w_k)). \end{aligned}$$

See (2) for the last equality. The map $\hat{\Upsilon} \colon \Omega^k_{\mathbb{R}}(M; P \times_G V) \to \Omega^k_{\sigma,\mathrm{hor}}(P; V)$ is also linear. One readily verifies that the two compositions of Υ and $\hat{\Upsilon}$ yield the corresponding identities. □

Remark 3.17.

1. From the fact that $\mathscr{A}(P)$ is an affine space modelled on $\Omega^1_{\mathrm{Ad},\ \mathrm{hor}}(P; \mathfrak{g})$ and theorem 3.16, it follows that $\mathscr{A}(P)$ can be seen as an affine space modelled on $\Omega^1_{\mathbb{R}}(M; \mathrm{ad}(P))$.

2. When we see the curvature of a connection as an element of $\Omega^2_{\mathbb{R}}(M; P \times_G \mathfrak{g}) \cong \Omega^2_{\mathrm{Ad},\ \mathrm{hor}}(M; \mathrm{ad}(P))$, we will write F_A. More precisely, we define $F_A := \Upsilon(\mathrm{D}_A A)$.

3.4 Connections on associated vector bundles

Let $P(M, G)$ be a principal bundle over a smooth manifold M, and let $\sigma \colon G \to \mathrm{Aut}(V)$ be a finite-dimensional representation. Consider $P(V) := P \times_G V$ the associated vector bundle. Our objective for this section is to show that a connection A on P induces an exterior covariant derivative d_A (in the sense of theorem A.11) on the associated vector bundle $P(V)$.

Definition 3.18. Under the previous hypotheses, let $\alpha \in \Omega^k_{\mathbb{R}}(M; P(V))$, and suppose that the principal G-bundle $P(M, G)$ is endowed with a connection A. The covariant exterior derivative of α induced by A is defined by

$$\mathrm{d}_A\alpha := \Upsilon\left(\mathrm{D}_A(\Upsilon^{-1}(\alpha))\right).$$

As a direct computation shows, if α is horizontal, then so is $\mathrm{D}_A\alpha$, hence the previous definition makes sense. According to the proof given in theorem 3.16, $\mathrm{d}_A\alpha \in \Omega^k_{\mathbb{R}}(M; P(V))$ satisfies

$$(\mathrm{d}_A\alpha)_{\pi(p)}(\mathrm{d}_p\pi(w_0), \dots, \mathrm{d}_p\pi(w_k)) = q_p \circ (\mathrm{D}_A\Upsilon^{-1}(\alpha))_p(w_0, \dots, w_k) \tag{3.7}$$

In particular, this construction defines a family of maps

$$\mathrm{d}_A\colon \Omega^k_{\mathbb{R}}(M;P(V)) \longrightarrow \Omega^{k+1}_{\mathbb{R}}(M;P(V)).$$

These maps are covariant exterior derivatives in the sense of theorem A.11. Observe that, for now, we work with real representations, but most of the topics we covered in section A.6 apply to real vector bundles if we do not complexify the tangent bundle. In particular, d_A defines a connection ∇^A in the sense of definition A.10, that is, $\nabla^A = \mathrm{d}_A$ acting on sections (i.e. zero-forms) satisfies Leibniz's rule $\nabla^A(f\cdot s) = \mathrm{d}f\otimes s + f\cdot\nabla^A s$ [**48**, Prop. 1.5.4]. To make sense of this, we must observe first that theorem 3.16 does not cover the case $k=0$. In that case, we can still relate 0-forms (sections). Indeed, there is a vector space isomorphism [**48**, Prop. 1.2.6]

$$\left\{\hat{\xi}\colon P\to V \text{ such that } \hat{\xi}\circ\phi_g = \sigma_{g^{-1}}\circ\hat{\xi}\right\} \longleftrightarrow \Gamma(M;P\times_G V), \tag{3.8}$$

which assigns to a G-equivariant map $\hat{\xi}$ the section $\boldsymbol{\xi} := \Upsilon(\hat{\xi})$ given by $\boldsymbol{\xi}(m) = [p,\hat{\xi}(p)], m = \pi(p)$ (that is, $\boldsymbol{\xi}\circ\pi(p) = q_p(\hat{\xi}(p))$). If we call the isomorphism Υ, we can extend d_A to act on sections, and it is precisely acting on sections where we have $\mathrm{d}_A = \nabla^A$.

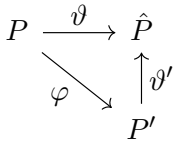
3.5 Reductions of the structure group

Definition 3.19 (Reduction of principal bundles). Let $P_1(M,G_1)$ and $P_2(M,G_2)$ be two $\mathcal{S}$-principal bundles over the same manifold, and $\vartheta\colon P_1\to P_2$ a bundle morphism. If $\lambda\colon G_1\to G_2$ is an embedding, and $\vartheta' = \mathrm{id}_M$, then we say that P_1 is a G_1-*reduction* or *reduction of the structure group* of P_2 from G_2 to G_1.

Under the conditions of the previous definition, P_1 may be identified with its image in P_2 through ϑ. We say that P_1 is an *embedded subbundle*. The definition may be relaxed by requiring only that λ be an immersion. First, observe that ϑ is injective. Indeed, if $\vartheta(p) = \vartheta(p')$, the fact that $\vartheta' = \mathrm{id}_M$ implies that $\pi_1(p) = \pi_1(p')$, thus $p' = p\cdot g$, so $\vartheta(p) = \vartheta(p') = \vartheta(p)\cdot\lambda(g)$ which means $\lambda(g) = e_{G_2}$ because the action is free. Since λ is an embedding, it follows that $g = e_{G_1}$. Finally, working with local trivializations one may show that $\vartheta\colon P_1\to P_2$ is an embedding as well.

When one identifies $P_1\equiv\vartheta(P_1)\subset P_2$, P_1 is a principal G_1-bundle whose G_1 action is given by restriction of the G_2 action on P_2. Moreover, its projection simply is the restriction of the projection of P_2. This means that $VP_1\subset VP_2$, and for each $p\in P_1$, $\mathrm{d}_e\phi^{(p)}|_{\mathfrak{g}_1}\colon \mathfrak{g}_1\to V_pP_1$ defines a vector space isomorphism.

Two reductions $P(M,H)$ and $P'(M,H)$ of the structure group of a principal G-bundle $\hat{P}(M,G)$ are *equivalent* if there is an isomorphism of principal bundles $\varphi\colon P\to P'$ covering the identity such that the diagram is commutative.



Consider now $P(M,G)$ a principal G-bundle, and let $H\subset G$ be a closed subgroup. In that case:

1. The set $G/H := \{gH : g\in G\}$ has a unique manifold structure such that the projection $G\to G/H$ is a smooth embedding. This is a consequence of the quotient manifold theorem. See [**34**, Th. 21.17] for a detailed proof. Moreover, the projection $G\to G/H$ admits local smooth sections.

2. G acts on G/H by $G \times G/H \to G/H, (g, \tilde{g}H) \mapsto g\tilde{g}H$, and this action is smooth with respect to the previous smooth structure (see, for example, the proof of [34, Th. 21.17]).

Taking this into account, we may consider the associated bundle $P \times_G (G/H)$, where G acts on G/H by right multiplication as we just defined. This is a fiber bundle over M with fiber G/H. We will write $P(G/H) := P \times_G (G/H)$. Moreover, there is a diffeomorphism $P \times_G (G/H) \to P/H, [p, gH] \mapsto [p \cdot g]$. The interest of this construction lies in the fact that sections of the bundle $P(G/H)$ parametrize reductions of the structure group of P to H.

Proposition 3.20. Let $P(M, G)$ be a principal bundle, and let $H \subset G$ be a closed subgroup. Then, there is a reduction of the structure group to H if and only if the bundle $P(G/H)$ has a global section. In fact, there is a bijection

$$\Gamma(M; P(G/H)) \longleftrightarrow \{\text{reductions of the structure group of } P(M, G) \text{ to } H\} / \sim$$

where $\sim$ stands for the relation of being equivalent.

Proof. We refer the reader to [27, I Prop. 5.6] for the details. We shall only sketch how the bijection goes. First, one may show that $P \to P/H$ has the structure of a principal H-bundle (see [27, I Prop. 5.5]). Let $\pi_{P/H} \colon P \to P/H$, $\pi \colon P \to M$ and $\hat{\pi} \colon P \times_G G/H \cong P/H \to M$ be the projections corresponding to all the principal bundles involved. Given a section $\sigma \colon M \to P(G/H) \cong P/H$, the reduction is given by

$$P_\sigma = \sigma^* P = \left\{(m, p) \in M \times P : \sigma(m) = \pi_{P/H}(p)\right\}.$$

Since $P \to P/H$ is a principal H-bundle, this endows P_σ with the structure of a principal H-bundle over M. Observe that if $(m, p) \in P_\sigma$, then $\pi(p) = m$. Certainly, form the definition of pullback $\pi_{P/H}(p) = \sigma(m)$. Applying $\hat{\pi}$ to both sides we get $\pi(p) = \hat{\pi}(\pi_{P/H}(p)) = \hat{\pi}(\sigma(m)) = m$. Finally, we consider $\vartheta \colon P_\sigma \to P, (m, p) \mapsto p$. Hence, we may identify P_σ with the set $P' := \left\{p \in P : \pi_{P/H}(p) = \sigma(\pi(p))\right\} \subset P$. Conversely, let $\pi' \colon P' \to M$ be a reduction of the structure group to H, and let $\vartheta \colon P' \to P$ the corresponding embedding. Then, the section is given by $\sigma(m) = \pi_{P/H}(\vartheta(p'))$ for any p' such that $\pi'(p') = m$. □

Remark 3.21. There is a natural identification $P_\sigma \times_H G \cong P$ where H acts on G by left multiplication. Just to be clear, recall that

$$P_\sigma \times_H G := \left\{(m, p, g) \in M \times P \times G : \pi_{P/H}(p) = \sigma(m)\right\} / \sim$$

where $(m, p, g) \sim (m, p \cdot h, h^{-1} \cdot g)$. This has the structure of a principal G-bundle over M, which is isomorphic to P though the map $P_\sigma \times_H G \to P, [m, p, g] \mapsto p \cdot g$. Observe that then we may identify $P_\sigma \equiv P_\sigma \times \{e_G\} \subset P_\sigma \times_H G \cong P$, and see it as an embedded submanifold of P. Note that by $P_\sigma \times \{e_G\}$ we mean the set of equivalence classes, that is $[m, p, e_G]$ such that $\pi_{P/H}(p) = \sigma(m)$. The G-action that turns $P \times_H G$ into a principal G-bundle (equivalently the G-action on P seen under the isomorphism) is given by $[m, p, g] \cdot g' = [m, p, g \cdot g']$. This is clearly independent of the representative. Moreover, P_σ is invariant under the restriction of the action to H. Indeed, $(P_\sigma \times \{e_G\}) \cdot h = P_\sigma \times \{h\} = P_\sigma \cdot h^{-1} \times \{e_G\} = P_\sigma \times \{e_G\}$. Finally, observe that the restriction of the G-action on P to an action of H on P_σ is precisely the H-action that P_σ has as a principal H-bundle over M.

By the discussion above, we will speak about sections of $P(G/H)$ and reductions of the structure group of P to H interchangeably.

Finally, connections on reductions of the structure group induce cconnections on the original bundle.

Proposition 3.22. Let $(\vartheta, \lambda)\colon P_1(M_1, G_1) \to P_2(M_2, G_2)$ be a principal bundle homomorphism between smooth principal bundles. Suppose that the induced map on the base spaces $\vartheta'\colon M_1 \to M_2$ is a diffeomorphism. Let A be a connection on P_1. Then, there is a unique connection $A' = \vartheta(A)$ on P_2 such that $\mathrm{d}\vartheta(H^A P_1) \subset H^{\vartheta(A)} P_2$. Moreover, the connection one-form and curvature two-form are related as follows

$$\vartheta^*(A') = \mathrm{d}\lambda(A) \qquad \text{and} \qquad \vartheta^*(\mathrm{D}_{A'} A') = \mathrm{d}\lambda(\mathrm{D}_A A).$$

Proof. See [27, Chap. II, Prop. 6.1]. The connection $\vartheta(A)$ is defined as follows (we define it in terms of the horizontal subbundle). For each $p_2 \in P_2$ with $m_2 := \pi_2(p_2)$ choose $p_1 \in \pi_1^{-1}\left((\vartheta')^{-1}(m_2)\right) \subset P_1$. Then, $p_2 = \vartheta(p_1) \cdot g$ for some $g \in G$. We set

$$H^{\vartheta(A)}_{p_2} P_2 := \mathrm{d}_{\vartheta(p_2)}\phi_g \circ \mathrm{d}_{p_1}\vartheta \ \left(H^A_{p_1} P_1\right).$$

One then shows that this satisfies the conditions of definition 3.7. □

3.6 Holomorphic principal bundles

We devote this section to studying holomorphic principal bundles specifically. Whenever we have a holomorphic principal G-bundle over a complex manifold M, we will use `mathcal` symbols (for example, $\mathcal{P}$). Then, the corresponding plain letter P will refer to the underlying smooth principal G-bundle. Moreover, whenever we say "holomorphic principal bundle" we will always assume that the structure group is a complex Lie group, and that the base space is a complex manifold (the definition does not make sense otherwise).

Let $\mathcal{P}_G \to M$ be a holomorphic principal bundle, and let A be a connection on the underlying smooth principal bundle, which we see using the connection one-form. For each $p \in P$, we have $A\colon T_p P_G \to \mathfrak{g}$. We may complexify both sides, thus getting $A\colon (T_p P_G)_{\mathbb{C}} \to \mathfrak{g}_{\mathbb{C}}$. Since $\mathfrak{g}$ is already a complex Lie algebra (this is because G is a complex Lie group, see appendix B), we would like this map to take values on $\mathfrak{g}$. To do so, observe that the almost complex structure $J_{\mathfrak{g}}$ induces a direct sum decomposition into $\mathbb{C}$-linear subspaces $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g}^{(1,0)} \oplus \mathfrak{g}^{(0,1)}$. Hence, we have a $\mathbb{C}$-linear projection $\mathfrak{g}_{\mathbb{C}} \to \mathfrak{g}^{(1,0)}$, and the latter is naturally isomorphic to $\mathfrak{g}$ with the $\mathbb{C}$-linear structure it has as a complex Lie algebra. The composition of A with the projection and the isomorphism yields a complex one-form $A \in \Omega^1_{\mathbb{C}}(\mathcal{P}; \mathfrak{g})$.

Definition 3.23 (Almost holomorphic structure). Let $P(M, G)$ be a smooth principal G-bundle, where G is a complex Lie group, and M is a complex manifold. An *almost holomorphic structure* is an almost complex structure on the smooth manifold P such that

1. $\mathrm{d}\pi \circ J = J_M \circ \mathrm{d}\pi$.
2. $J_{p\cdot g}(\mathrm{d}_{(p,g)}\phi(w, X)) = \mathrm{d}_{(p,g)}\phi(J_p(w), (J_G)_g(X))$, for each $p \in P, g \in G$ and $X \in T_g G, B \in T_p P$.

J_M and J_G are the canonical almost complex structures on M and G.

We say that an almost holomorphic structure is integrable if it is integrable as a usual almost complex structure in the sense of Newlander-Nirenberg.

Remark 3.24.

1. The previous definition has a very clear meaning: if J is an integrable almost holomorphic structure, then, by the Newlander-Nirenberg theorem P admits a complex structure turning it into a complex manifold. In that case, since M and G are complex manifolds, theorem A.8 states that the projection and the action of G are holomorphic maps. In that case, one may show that P admits holomorphic local trivializations (see [50, Prop. 2.5.3]).
2. The second condition is equivalent to ask J to satisfy the following

$$J_{p\cdot g}\circ \mathrm{d}_p\phi_g = \mathrm{d}_p\phi_g \circ J_p \tag{3.9}$$
$$J_{p\cdot g}\circ \mathrm{d}_g\phi^{(p)} = \mathrm{d}_g\phi^{(p)} \circ (J_G)_g \tag{3.10}$$

The proof is just a straightforward application of the Leibniz rule [27, I Prop. 1.4]

$$\mathrm{d}_p\phi_g(J_p(w)) + \mathrm{d}_g\phi^{(p)}((J_G)_g(X)) = J_{p\cdot g}(\mathrm{d}_p\phi_g(w)) + J_{p\cdot g}(\mathrm{d}_g\phi^{(p)}(X))$$

followed by setting X or w equal to 0. Moreover, (3.10) is equivalent to

$$J_p(X_{\zeta,p}) = X_{\mathrm{i}\,\zeta,p}, \qquad \zeta\in\mathfrak{g}. \tag{3.11}$$

Let us denote by $\mathscr{J}(P)$ the set of integrable almost holomorphic structures, and $\bar{\mathscr{J}}(P)$ the set of almost holomorphic structures.

Remark 3.25. The curvature $\mathrm{D}_A A$ can be seen as a $\mathfrak{g}$-valued complex two-form applying the same reasoning we used to see the connection one-form as a complex one-form. This is useful because it allows us to decompose $\mathrm{D}_A A$ as $\mathrm{D}_A A^{(2,0)} + \mathrm{D}_A A^{(1,1)} + \mathrm{D}_A A^{(0,2)}$, where $\mathrm{D}_A A^{(1,1)}(X,Y) = \mathrm{D}_A A\left(X_J^{(1,0)}, Y_J^{(0,1)}\right) + \mathrm{D}_A A\left(X_J^{(0,1)}, Y_J^{(1,0)}\right)$, $\mathrm{D}_A A^{(0,2)}(X,Y) = \mathrm{D}_A A\left(X^{(0,1)}\right)$ and similarly for $\mathrm{D}_A A^{(2,0)}$. Here $X_J^{(1,0)}$ is the component of X lying in $T^{(1,0)}P_G$. Moreover, the linear isomorphism $\mathrm{d}_e\phi^{(p)}\colon \mathfrak{g}\to V_pP$ may be complexified. The collection of all vector spaces $(V_pP_G)_{\mathbb{C}}$ produces a smooth complex vector subbundle $V_{\mathbb{C}}P_G$ of $T_{\mathbb{C}}P_G$. We still have $(V_pP_G)_{\mathbb{C}} = \ker \mathrm{d}_p\pi$ (now $\mathrm{d}_p\pi\colon (T_pP_G)_{\mathbb{C}}\to (T_{\pi(p)}M)_{\mathbb{C}}$ is the complexified map). We can now speak of vertical vector fields, and horizontal forms with respect to the adjoint representation. Moreover, we can then generalize theorem (3.16) and define $F_A\in\Omega^2_{\mathbb{C}}(M; P_G\times_G\mathfrak{g})$, which admits an analogous decomposition $F_A = F_A^{(2,0)} + F_A^{(1,1)} + F_A^{(0,2)}$. That is,

$$(F_A)_m(X_m,Y_m) = \left[p, \mathrm{D}_A\left(\hat{X}_p,\hat{Y}_p\right)\right] \qquad \pi(p)=m$$

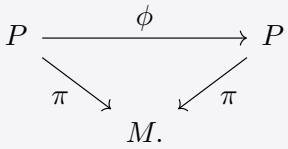
where $\hat{X}$ and $\hat{Y}$ project to X and Y. Since the projection is pseudoholomorphic, it preserves the decomposition of vectors into types $(1,0)$ and $(0,1)$ (we say that it is *type preserving*, see [28, Prop. 2.9]). Hence, $(F_A^{(1,1)})_m(X_m,Y_m) = [p, \mathrm{D}_A^{(1,1)}(\hat{X}_p,\hat{Y}_p)]$ and the same applies to the other components.

3.7 The gauge group of a principal bundle

Definition 3.26. Let $P(M,G)$ be a principal G-bundle.

A *gauge transformation* of P is a G-equivariant diffeomorphism covering the identity on M. The group of all gauge transformations is called the *gauge group*.

$$\mathscr{G} := \{\phi\in \mathrm{Diff}(P) : \phi(p\cdot g) = \phi(p)\cdot g \quad \text{and} \quad \phi\circ\pi = \pi\}.$$



A lot of sources claim that $\mathscr{G}$ is a Hilbert-Lie group (or a Banach-Lie group, it depends on the construction), which is not very accurate. As we are going to see, the way one endows $\mathscr{G}$ with a Hilbert-Lie group structure is rather elaborate, and $\mathscr{G}$ itself *is not* a Hilbert-Lie group: we have to consider a group it is isomorphic to and complete it (in a certain sense we will make precise in section 4.1) to get a Hilbert-Lie group. The construction is not straightforward at all, but we will give an outline. For now, let us see a different way to look at the gauge group.

Proposition 3.27. Let M be a smooth manifold, and $P(M,G)$ a principal G-bundle. Then, $\mathscr{G} \cong \Gamma(M;\mathrm{Ad(P)})$ as groups and $\Gamma(M;\mathrm{Ad}(P))$ is given a group structure through pointwise multiplication.

Proof. We will only give the isomorphism. Checking that it is indeed a group homomorphism is a routine verification, and to check smoothness of the different maps that will appear one uses explicit forms of the local trivializations. Given $\phi \in \mathscr{G}$, $\phi(p) \in \pi^{-1}(p)$, since the action is free there is a unique $\tau(p) \in G$ such that $\phi(p) = p \cdot \tau(p)$. We define a section $\boldsymbol{\tau} \in \Gamma(M;\mathrm{Ad}(P))$ by setting

$$\boldsymbol{\tau}\colon M \to \mathrm{Ad}(P), \pi(p) \mapsto [p, \tau(p)]$$

which is well-defined and smooth. □

Using theorem 3.27 above, we shall identify $\mathscr{G} \cong \Gamma(M;\mathrm{Ad}(P))$. It is the completion of $\Gamma(M;\mathrm{Ad}(P))$ with respect to a Sobolev norm what actually *is* a Hilbert-Lie group. We will see this in more detail in section 4.1

3.8 The Chern-Singer correspondence

In this section, we shall see that, for a principal bundle having a reductive complex Lie group as structure group, we can find a bijection between the set of integrable almost holomorphic structures, $\mathscr{J}(P)$ and the set of connections on a reduction of the principal bundle to a maximal compact subgroup K such that $G = K^{\mathbb{C}}$. This section is mostly based on the thorough exposition given in [**50**]. We fix G and K for the remainder of the section. Recall that we can see the reduction P_K as an embedded submanifold of P_G. Hence, TP_K is a subbundle of TP_G [**34**, Example 10.33-(c)].

Theorem 3.28 (Chern-Singer correspondence, I). The map $\mathsf{CS}\colon \bar{\mathscr{J}}(P_G) \to \mathscr{A}(P_K), J \mapsto A_J$, where A_J denotes the connection corresponding to the distribution $H^J_p P_K := T_pP_K \cap J_p(T_pP_K),\ p \in P_K$ is a bijection.

Proof. Let $J \in \bar{\mathscr{J}}(P_G)$, and let us show first that the assignment $p \mapsto H^J_p P_K$ is a smooth distribution, and that it satisfies properties 1 and 2 of definition 3.7. Note that $H^J_p P_K \cap V_pP_K = \{0\}$. Indeed, an element of the intersection is on the vertical subbundle, hence it is X_ζ for some $\zeta \in \mathfrak{k}$. Then, by (3.10)

$$J_p(X_{\zeta,p}) = J_{p\cdot e}\left(\mathrm{d}_e\phi^{(g)}(\zeta)\right) = \mathrm{d}_e\phi^{(p)}(J_{\mathfrak{g}}(\zeta)) = \mathrm{d}_e\phi^{(g)}(\mathrm{i}\,\zeta) = X_{\mathrm{i}\,\zeta,p}.$$

Moreover, since $X_{\zeta,p} \in H^J_p P_K$, it readily follows that $J(X_{\zeta,p}) \in H^J_p P_K$. Using the fact that $\mathrm{d}_e\phi^{(p)}\colon \mathfrak{g} \to V_pP_G$ is an isomorphism which restricts to an isomorphism $\mathfrak{k} \to V_pP_K$, it

follows that $\zeta \in \mathfrak{k} \cap \mathrm{i}\,\mathfrak{k} = \{0\}$. Finally, since $T_pP_K + J_p(T_pP_K) \subset T_pP_G$, we have

$$\begin{aligned}\dim(H^J_p P_K) &= \dim(T_pP_K \cap J_p(T_pP_K)) = \dim(T_pP_K) + \dim(J_p(T_pP_K)) - \dim(T_pP_K + J_p(T_pP_K)) \\ &\geq \dim_{\mathbb{R}} M + \dim_{\mathbb{R}} K + \dim_{\mathbb{R}} M + \dim_{\mathbb{R}} K - (\dim_{\mathbb{R}} G + \dim_{\mathbb{R}} M) = \dim_{\mathbb{R}} M\end{aligned} \tag{3.12}$$

($\dim_{\mathbb{R}} M$ is the dimension of M as a real manifold, and we used the fact that $\dim_{\mathbb{R}} G = 2 \cdot \dim_{\mathbb{R}} K$, see appendix B). Hence, $\dim(H^J_p P_K \oplus V_p P_K) = \dim H^J_p P_K + \dim K \geq \dim_{\mathbb{R}} M + \dim_{\mathbb{R}} K = \dim T_pP_K$ and the sum covers the whole tangent bundle. Finally, $\mathrm{d}_p\phi_k(H^J_p P_K) \subset H^J_{p\cdot k} P_K$ (we need only check it for $k \in K$) and since the dimensions are equal, we have the equality. Indeed, choose $w \in T_pP_K \cap J_p(T_pK)$. We may write $w = J_p(w')$ for some $w' \in T_pP_K$. Then, by (3.9), we have

$$\mathrm{d}_p\phi_k(J_p(w')) = J_{p\cdot k}(\mathrm{d}_p\phi_k(w')) \in J_{p\cdot k}(T_{p\cdot k}P_K) \cap T_{p\cdot k}P_K$$

where we used that $\mathrm{d}_p\phi_k(w') \in T_{p\cdot k}P_K$ because the action of G restricts to the action of K on P_K. Smoothness is somewhat tedious to prove, the key idea is to use a local frame of TP_K to obtain one of $H^J P_K$ and then apply the local frame criterion for subbundles [**34**, Lemma 10.32]. We refer the reader to [**50**, Th. 3.1.1] for the details.

Let us now construct the inverse map. From proposition 3.22, we know that every connection $A \in \mathscr{A}(P_K)$ induces a connection on the original bundle P_G. Let us still denote by A the induced connection. We define an almost holomorphic structure J^A on P_G as follows. For $p \in P_G$ and $w \in T_pP_G$, the connection A induces a decomposition $w = w^V + w^H$. Then, for some $\zeta \in \mathfrak{g}$, we have $w^V = X_{\zeta,p}$. Define

$$J^A_p(w) := X_{\mathrm{i}\,\zeta,p} + \left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_p. \tag{3.13}$$

Observe that J^A is an almost holomorphic structure. The fact that $(J^A)^2 = -\mathrm{id}$ is clear from the fact that (3.13) already gives us the decomposition of $J^A(w)$ as a sum of a vertical and horizontal vector. Hence,

$$\begin{aligned}J^A_p(J^A_p(w)) &= X_{-\zeta,p} + \left((J_M)_{\pi(p)}\left(\mathrm{d}_p\pi\left(\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_p\right)\right)\right)^{\mathbf{HL}}_p \\ &= -X_{\zeta,p} - \mathrm{d}_p\pi(w)^{\mathbf{HL}}_p = -\left(w^V + w^H\right) = -w.\end{aligned}$$

Now, $\mathrm{d}\pi \circ J^A = J_M \circ \mathrm{d}\pi$ and (3.11) are obvious. To check (3.9), let $w \in T_pP$ and let $w^V = X_{\zeta,p}$. Observe that $\mathrm{d}_p\phi(w)^V = \mathrm{d}_p\phi(w^V + w^H)^V = \mathrm{d}_p\phi_g(w^V)^V$ (cf. remark 3.8). Moreover, since $(\phi_g)_*(X_\xi) = X_{\mathrm{Ad}_{g^{-1}}(\xi)}$ (cf. [**48**, Eq. 1.3.1])

$$\mathrm{d}_p\phi(w^V) = \mathrm{d}_p\phi_g\left(X_{\zeta,p}\right) = X_{\mathrm{Ad}_{g^{-1}}(\zeta),p\cdot g}, \tag{3.14}$$

which is again vertical. Thus,

$$\begin{aligned}J^A_{p\cdot g}(\mathrm{d}_p\phi_g(w)) &= X_{\mathrm{i}\,\mathrm{Ad}_{g^{-1}}(\zeta),p} + \left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_{p\cdot g} \\ &\overset{(1)}{=} \mathrm{d}_p\phi_g(X_{\mathrm{i}\,\zeta,p}) + \mathrm{d}_p\phi_g\left(\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_p\right) \\ &= \mathrm{d}_p\phi_g(J^A_p(w)),\end{aligned}$$

where we used in (1) equation (3.14) but with $\mathrm{i}\,\zeta$ and lemma 3.9.

Finally, observe that these constructions are inverses of each other. Let us see first that $A = A_{J^A}$. This amounts to checking whether $H^A_p P_K$ and $T_pP_K \cap J^A_p(T_pP_K) = H^{J^A}_p P_K$ coincide for each $p \in P_K$. Choose $w \in H^A_p P_K \subset T_pP_K$, then $w^V = 0$, and

$$w = J^A_p\left(-\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}\right) \in J^A_p(H^A_p P_K) \subset J^A_p(T_pP_K).$$

Hence, $w \in T_pP_K \cap J_p^A(T_pP_K)$. Thus, $H_p^AP_K \subset H_p^{J^A}P_K$ and since they have the same dimension (the calculation of equation (3.12) applies to both sides), they are equal.

Conversely, consider $J \in \bar{\mathscr{J}}(P_G)$. Let $A_J := \mathsf{CS}(J)$. Let us see that $J^{A_J} = J$. Both coincide on vertical vectors: ($J_p(X_{\zeta,p}) = X_{\mathrm{i}\,\zeta,p}$ follows from applying (3.10) to ζ). Since both are linear as well, it suffices to check that they coincide when applied to horizontal vectors (with respect to A_J). Hence, if $w \in H_p^AP_K$, we have

$$\mathrm{d}_p\pi(J_p(w)) = (J_M)_{\pi(p)}(\mathrm{d}_p\pi(w)). \tag{3.15}$$

Moreover, $J_p(w)$ is horizontal (observe that $H_p^JP_K = T_pP_K \cap J_p(T_pP_K)$), so considering the horizontal lift of (3.10) we get

$$J^{A_J}(w) = \big((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\big)^{\mathbf{HL}} = J_p(w).$$

□

We say that a connection $A \in \mathscr{A}(P_G)$ is of type $(1,1)$ if its curvature form is of type $(1,1)$. Similarly, $A \in \mathscr{A}^{(1,1)}(P_K)$ if its extension to P_G is of type $(1,1)$.

Theorem 3.29 (Chern-Singer correspondence, II). The map CS restricts to a bijection $\mathsf{CS}|_{\mathscr{J}(P_G)}\colon \mathscr{J}(P_G) \to \mathscr{A}^{(1,1)}(P_K)$.

For the proof of this result we will diverge from [50]. We will instead make use of the following lemma, which is taken from [58]. The proof consists of showing that J^A is integrable if and only if A is of type $(1,1)$, hence we fix A and $J = J^A$ for the remainder of the section.

Lemma 3.30. Let $A \in \mathscr{A}(P_K)$ be a connection, and let $J = \mathsf{CS}^{-1}(A) \in \bar{\mathscr{J}}(P_G)$. Then, J is integrable (that is, $J \in \mathscr{J}(P_G)$) if and only if the two-form

$$\begin{array}{rccc} \mathfrak{f}_J\colon & \Gamma(P_G;T_{\mathbb{C}}P_G)\times\Gamma(P_G;T_{\mathbb{C}}P_G) & \longrightarrow & \mathscr{C}^\infty(P_G;\mathfrak{g}) \\ & (X,Y) & \mapsto & -A\left(\left[X_J^{(0,1)},Y_J^{(0,1)}\right]\right) \end{array} \tag{3.16}$$

vanishes identically.

The connection one-form A in (3.16) should be understood as the extension of A to P_G (as in proposition 3.22), which we still denote A. Observe that $V_{\mathbb{C}}P_G \subset T_{\mathbb{C}}P_G$ is a vector bundle, and the isomorphism $P_G \times \mathfrak{g} \cong VP_G$ extends to the complexification as $\mathfrak{g}_{\mathbb{C}} \to (V_pP_G)_{\mathbb{C}}$, $\xi = \zeta_1 + \mathrm{i}\,\zeta_2 \mapsto X_{\zeta_1,p} + \mathrm{i}\,X_{\zeta_2,p}$, for $\zeta_1,\zeta_2 \in \mathfrak{g}$. We can legitimately speak of vertical vectors in the complexified tangent bundle. We want to be able to speak of horizontal vectors in $T_{\mathbb{C}}P_G$ as well. This is done by considering the complexification of H^AP_G. In other words, we get a decomposition $T_{\mathbb{C}}P_G = H^A_{\mathbb{C}}P_G \oplus V_{\mathbb{C}}P_G$. Moreover, since $\mathrm{d}\phi_g$ satisfies (3.9), we know that ϕ_g is pseudoholomorphic, hence $\mathrm{d}\phi_g$ is type preserving. In particular, $T^{(0,1)}P_G$ is G-invariant.

Proof of lemma 3.30. This is a consequence of the classical Newlander-Nirenberg theorem. Recall that the Newlander-Nirenberg theorem implies that J is integrable if and only if $\left[T^{(0,1)}P_G, T^{(0,1)}P_G\right] \subset T^{(0,1)}P_G$. Thus, we are led to check that the condition $\mathfrak{f}_J \equiv 0$ is equivalent to the condition that for every pair of complex vector fields X,Y, we have $\left[X_J^{(0,1)},Y_J^{(0,1)}\right] \in T^{(0,1)}P_G$. To show this, consider first $T_{\mathbb{C}}M = T^{(1,0)}M \oplus T^{(0,1)}M$ (recall that M is a complex manifold). Since π commutes with J and J_M by definition, it is

pseudoholomorphic, hence type preserving. Consider the pullback bundle $\pi^*(T^{(1,0)}M) \subset \pi^*(T_{\mathbb{C}}M)$ over P_G. Explicitly:

$$\pi^*(T^{(1,0)}M) = \left\{(p,v) \in P_G \times T^{(1,0)}M : \pi(p) = \pi_M(v)\right\} \subset \pi^*(T_{\mathbb{C}}M)$$

where $\pi_M \colon T_{\mathbb{C}}M \to M$ is the usual projection of the (complexified) tangent bundle and the latter is defined analogously. Here, $\pi \colon P_G \to M$ is the projection as a principal bundle. The projection $\pi \colon P_G \to M$ induces a map $\mathrm{d}\pi' \colon T_{\mathbb{C}}P_G \to \pi^*(T_{\mathbb{C}}M)$ defined fiberwise by $(T_pP_G)_{\mathbb{C}} \to (\pi^*(T_{\mathbb{C}}M))_p, v \mapsto (v, \mathrm{d}_p\pi(v))$. We may consider the vector subbundle $\hat{T}^{(1,0)}M := (\mathrm{d}\pi')^{-1}(\pi^*(T^{(1,0)}M)) \subset T_{\mathbb{C}}P_G$. Since π is pseudoholomorphic, it must preserve type, hence $\hat{T}^{(0,1)}M = T^{(0,1)}P_G \oplus V^{(1,0)}P_G$, where the second factor is explained by the fact that vertical vectors annihilate $\mathrm{d}\pi$ even if they are of type $(1,0)$. The fact that M is a complex manifold implies, using the Newlander-Nirenberg theorem, that $[T^{(0,1)}M, T^{(0,1)}M] \subset T^{(0,1)}M$. Let us see that the same holds for $\hat{T}^{(0,1)}M$. Choose $X, Y \in \Gamma(P_G; \hat{T}^{(0,1)}M)$. We want to see that $[X,Y] \in \hat{T}^{(0,1)}M$. This is equivalent to showing that $\mathrm{d}_p\pi([X,Y]_p) \in T^{(0,1)}_{\pi(p)}M$ for each $p \in P_G$. Now, if either X or Y is vertical, then so is $[X,Y]$. Furthermore, if both X and Y are horizontal, then they are horizontal lifts of some vector fields in $T_{\mathbb{C}}M$, call those X', Y'. We thus have $\mathrm{d}_p\pi(X_p) = X'_{\pi(p)} \in T^{(0,1)}_{\pi(p)}M$ and the same applies to Y. We deduce that $\mathrm{d}_p\pi([X,Y]_p) = [X',Y']_{\pi(p)}$, the latter being a vector field in $T^{(0,1)}M$ as we wanted to show (see [34, Prop. 8.30]).

Therefore, we know that $[T^{(0,1)}P_G, T^{(0,1)}P_G] \subset T^{(0,1)}P_G \oplus V^{(1,0)}P_G$. Thus, it suffices to show that the vanishing of $\mathfrak{f}_J$ is equivalent to the condition that for every pair of vector fields X, Y, the projection of $[X_J^{(0,1)}, Y_J^{(0,1)}]$ on $V^{(1,0)}P_G$ is zero. Since $A \in \Omega^1_{\mathbb{C}}(P_G; \mathfrak{g})$ is of type $(1,0)$ (we give a proof below), this projection is precisely $\mathfrak{f}_J(X,Y)$ (up to the isomorphism between $\mathfrak{g}$ and $V_p^{(1,0)}P_G$ given by the maps $\mathrm{d}_e\phi^{(p)}$).

Let us finally check that A is of type $(1,0)$ with respect to the almost holomorphic structure it induces. To do so, pick any vector $w \in (T_pP_G)_{\mathbb{C}}$, we have

$$A\left(w_J^{(0,1)}\right) = \frac{1}{2} \cdot (A(w) + \mathrm{i}\, A(J(w))) = \frac{1}{2} \cdot (A(w) + (\mathrm{i} \cdot \mathrm{i}) \cdot A(w)) = 0$$

where we used the definition of J^A, (3.13), and the fact that A vanishes on $H_{\mathbb{C}}P_G$. □

Lemma 3.31. Let $P(M,G)$ be a principal bundle. Then, for any pair of horizontal vector fields, $X, Y \in \Gamma(P; T_{\mathbb{C}}P)$, we have

$$\mathrm{D}_A A(X,Y) = -A([X,Y]).$$

Proof. First, for a $\mathfrak{g}$-valued one-form and a vector field, we define $X(\omega \otimes \xi) := X(\omega) \cdot \xi$.

$$\mathrm{D}_A A(X,Y) = (\mathrm{d}A)^H(X,Y) = \mathrm{d}A(X,Y) \overset{(2)}{=} X(A(Y)) - Y(A(X)) - A([X,Y]) \overset{(3)}{=} -A([X,Y]).$$

(2) follows from the corresponding identity for usual one-forms (see [34, Prop. 14.29]), and (3) from the fact that $A(X) = 0$ for horizontal vector fields. □

We can now show the equivalence.

Proof of theorem 3.29. By the previous lemma, we have to check that $\mathfrak{f}_J \equiv 0$. If either X or Y is a vertical vector field, then $\mathfrak{f}_J(X,Y) = 0$ (see below), so we may restrict to the case where they are both horizontal. Indeed, for any $\zeta \in \mathfrak{g}^{\mathbb{C}}$ and any vector field Y, the

vector field $\left[X_\zeta, Y_J^{(0,1)}\right]$ is of type $(0,1)$. This follows from the fact that $T^{(0,1)}P_G \subset T_{\mathbb{C}}P_G$ is G-invariant, so (suppose $\zeta \in \mathfrak{g}$, if that is not the case, decompose it as a sum of elements $\mathfrak{g} \oplus \mathrm{i}\,\mathfrak{g}$ and use linearity of the bracket)

$$\left[X_\zeta, Y_J^{(0,1)}\right]_p = \mathcal{L}_{X_\zeta}\left(Y_J^{(0,1)}\right)_p = \lim_{t\to 0} \frac{\mathrm{d}_{p\cdot\exp(t\zeta)}\phi_{\exp(-t\zeta)}\left(Y^{(0,1)}_{J,p\cdot\exp(t\zeta)}\right) - Y^{(0,1)}_{J,p}}{t}$$

where the numerator is the difference between two vectors in $T_p^{(1,0)}P_G$, hence an element of that subspace. We used the fact that the flow generated by a vector field X_ζ is precisely $(t,p) \mapsto p\cdot\exp(t\zeta)$. Invariance under the G-action means that the numerator is a vector in $T^{(0,1)}P_G$. Finally, since A is type $(1,0)$, $\mathfrak{f}_J(X,Y)$ vanishes if either X or Y are vertical. Of course, $\mathrm{D}_A A$ also vanishes when applied to a vertical vector field because it is horizontal.

Once we show that $\mathfrak{f}_J$ restricted to horizontal vector fields coincides with $\mathrm{D}_A^{(0,2)}A$ we will be finished (recall that the curvature is a horizontal two-form). Hence, from proposition A.6 it follows that for all $Z, Y \in \Gamma(P_G; H_{\mathbb{C}}P_G)$ (using $Z_J^{(1,0)} = 1/2\cdot(Z + \mathrm{i}\,JZ)$, see proposition A.6)

$$\begin{aligned} -A\left(\left[Z_J^{(0,1)}, Y_J^{(0,1)}\right]\right) &= -\frac{1}{4}\cdot A\left([Z+\mathrm{i}\,JZ, Y+\mathrm{i}\,JY]\right) \\ &= -\frac{1}{4}\cdot A([Z,Y]) + \frac{1}{4}\cdot A([JZ,JY]) - \frac{\mathrm{i}}{4}\cdot A([Z,JY]) - \frac{\mathrm{i}}{4}\cdot A([JZ,Y]). \end{aligned} \tag{3.17}$$

Similarly, for all $Z, Y \in \Gamma(P_G; H_{\mathbb{C}}P_G)$

$$\begin{aligned} \mathrm{D}_A^{(0,2)}(Z,Y) = \mathrm{D}_A\left(Z_J^{(0,1)}, Y_J^{(0,1)}\right) &= \frac{1}{4}\cdot \mathrm{D}_A(Z+\mathrm{i}\,JZ, Y+\mathrm{i}\,JY) \\ &= \frac{1}{4}\cdot \mathrm{D}_A(Z,Y) - \frac{1}{4}\cdot \mathrm{D}_A(JZ,JY) + \frac{\mathrm{i}}{4}\cdot \mathrm{D}_A(Z,JY) + \frac{\mathrm{i}}{4}\cdot \mathrm{D}_A(JZ,Y). \end{aligned} \tag{3.18}$$

Finally, by lemma 3.31 for horizontal vector fields, $\mathrm{D}_A([X,Y]) = -A([X,Y])$, which applied to each component of (3.17) and (3.18) when the vector fields are horizontal yields the desired result. We have used the fact that $\mathrm{D}_A A^{(0,2)} = 0$ if and only if $F_A^{(0,2)} = 0$. □

Given an integrable almost holomorphic structure $J \in \mathscr{J}(P_G)$, the connection $A := \mathsf{CS}^{-1}(J)$ on P_K is called the associated *Chern connection*.

Remark 3.32. If $M = X$ is a Riemann surface, then every connection A is of type $(1,1)$. Indeed, a Riemann surface only needs a single holomorphic coordinate, so for every two-form, the terms $\mathrm{d}z \wedge \mathrm{d}z$ and $\mathrm{d}\bar{z} \wedge \mathrm{d}\bar{z}$ would vanish in any local expression, leaving only terms of type $\mathrm{d}z \wedge \mathrm{d}\bar{z}$.

Finally, we will comment without proof how quotients by the gauge group play a role in the correspondence. We refer to [**50**] and references therein for further details. Again, we do not intend to give these spaces further structure for now (that is, we do not study moduli spaces), so we will not complete any space with respect to any Sobolev norm. Let $\mathscr{G}^{\mathbb{C}} = \Gamma(M; P_G \times_G G)$ and $\mathscr{G} = \Gamma(M; P_K \times_K K)$ be the gauge groups associated to both bundles (see example 3.6 and theorem 3.27). Observe that the fibers of $P_G \times_G G$ are isomorphic to $G = K^{\mathbb{C}}$, so it seems reasonable to say that $\Gamma(M; P_G \times_G G)$ is a complexification of $\Gamma(P_K \times_K K)$, hence the notation.

We may see $\mathscr{G}$ as a subgroup of $\mathscr{G}^{\mathbb{C}}$. To that end, we observe that $P_G \times_G G \cong P_K \times_K G$, where K acts on G by the adjoint conjugation instead of inverse left multiplication in

the latter. Then, $\mathrm{Ad}(P_K) \subset \mathrm{Ad}(P_G)$, and sections of $P_K \times_K K$ can be naturally seen as sections of $P_G \times_G G$, hence $\mathscr{G}$ is a subgroup of $\mathscr{G}^{\mathbb{C}}$. Moreover, there is a right action of $\mathscr{G}$ on $\mathscr{A}(P_K)$ given by pullback $\mathscr{G} \times \mathscr{A}(P_K) \to \mathscr{A}(P_K), (A, \phi) \mapsto (\phi^{-1})^* A$. A straightforward computation shows that the pullback of a connection one-form is a connection one-form [7, Th. 3.2.5]. For an explicit expression, suppose that $\phi(p) = p \cdot \tau(p)$ (cf. 3.27). Then, (see [7, Th. 3.2.14])

$$((\phi^{-1})^* A)_p = \mathrm{Ad}_{\tau(p)} \circ A_p + \mathrm{d}_{\tau(p)^{-1}} L_{\tau(p)} \circ \mathrm{d}_p \tau^{-1}. \tag{3.19}$$

The previous expression is sometimes written as $\phi^* A = \mathrm{Ad}_{\tau^{-1}} \circ A + \mathrm{d}\tau^{-1}\tau$. The second summand can also be written as $(\tau^{-1})^*\theta$, where θ is the *Maurer-Cartan* form. Hence, the action is equivalently written as

$$(\phi^{-1})^* A = \mathrm{Ad}_{\tau^{-1}}(A) + (\tau^{-1})^*\theta. \tag{3.20}$$

This action leaves $\mathscr{A}^{(1,1)}(P_K)$ invariant. On the other hand, there is a right action of $\mathscr{G}^{\mathbb{C}}$ on $\bar{\mathscr{J}}(P_G)$ given by $\bar{\mathscr{J}}(P_G) \times \mathscr{G}^{\mathbb{C}} \to \bar{\mathscr{J}}(P_G), (J, \varphi) \mapsto \varphi(J)$, where

$$\varphi(J) := (\mathrm{d}\varphi)^{-1} \circ J \circ \mathrm{d}\varphi.$$

This action leaves $\mathscr{J}(P_G)$ invariant. Moreover, via the Chern-Singer correspondence (version I, theorem 3.28), we get an action of $\mathscr{G}^{\mathbb{C}}$ on $\mathscr{A}(P_K)$ given by $(A, \varphi) \mapsto A_{\varphi(J_A)}$, which extends the original $\mathscr{G}$-action.

Now, two almost holomorphic integrable structures in the same $\mathscr{G}^{\mathbb{C}}$-orbit induce isomorphic complex principal bundle structures on P_G (this is a direct consequence of theorem A.8 because then φ is a biholomorphism). Thus, the quotient $\mathscr{J}(P_G)/\mathscr{G}^{\mathbb{C}}$ can be seen as the set of essentially different holomorphic structures on the bundle P_G. Now, the Chern-Singer correspondence (version II, theorem 3.29) establishes that there is a bijection

$$\frac{\mathscr{J}(P_G)}{\mathscr{G}^{\mathbb{C}}} \leftrightarrow \frac{\mathscr{A}^{(1,1)}(P_K)}{\mathscr{G}^{\mathbb{C}}}, \tag{3.21}$$

the left-hand side being the set of complex principal bundle structures on P_G.

Chapter 4

The hermitian Yang-Mills equation

The aim of this chapter is to study the hermitian Yang-Mills equation for connections on principal bundles over compact Kähler manifolds having compact connected structure groups. At the end of the chapter, we study the problems the equation has and how it should be modified when the structure group is not connected. Let K be a compact Lie group, and $P_K \to M$ be a principal K-bundle. The *hermitian Yang-Mills equation* is an equation *for a connection* A on P_K. The equation is given by

$$\Lambda_\omega F_A = -\mathrm{i}\, c \tag{4.1}$$

for some element $c \in \mathrm{i}\,\mathfrak{z}_K$ where $\mathfrak{z}_K = \mathfrak{z}(\mathfrak{k})$ is the center of the Lie algebra of K.

Remark 4.1. Observe that $\Lambda_\omega F_A = -\mathrm{i}\, c$ should be understood as an equality between smooth sections of the bundle $P_K \times_K \mathfrak{k}$. Let us ignore, for the time being, the factor i. The right-hand side then corresponds to a constant section $m \mapsto [p, c]$ for $\pi(p) = m$. One should check that this section is well-defined. Indeed, choose another representative $[p \cdot k, \mathrm{Ad}_{k^{-1}}(c)]$. Since $c \in \mathfrak{z}_K$ *and* K *is connected*, $\mathrm{Ad}_k(c) = c$ for each $k \in K$. It is very important to remark that this only holds for connected Lie groups. A quick proof is as follows:

$$\mathrm{Ad}_k(c) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} L_k \circ R_{k^{-1}} \circ \exp(tc) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} k \exp(tc) k^{-1} = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} k k^{-1} \exp(tc) = c.$$

This uses the fact that the exponential of a compact connected Lie group is surjective [**21**, Cor. 12.2.3] and the Baker-Campbell-Hausdorff formula.

Moreover, if equation (5.1) is satisfied, then the constant c is determined uniquely by the topology of the bundle. Uniqueness is not difficult to verify. Let $\langle \cdot, \cdot \rangle$ be an Ad-invariant positive definite inner product on $\mathfrak{k}$ (the opposite of the Killing form for instance). Hence, let $\{\zeta_1, \ldots, \zeta_r\}$ be an orthonormal basis. $-\mathrm{i}\, c \in \mathfrak{k}$, hence $-\mathrm{i}\, c = \sum_{j=1}^r \langle -\mathrm{i}\, c, \zeta_j \rangle \cdot \zeta_j$. Moreover, $\langle \zeta_j, -\mathrm{i}\, c \rangle = \langle \zeta_j, \Lambda_\omega F_A \rangle$ where this is an equality between smooth real-valued maps. Integrating over X with respect to the volume form ω, we get

$$\int_X \langle \zeta_j, -\mathrm{i}\, c \rangle \cdot \omega = \langle \zeta_j, -\mathrm{i}\, c \rangle \cdot \mathrm{Vol}(X) = \int_X \langle \zeta_j, \Lambda_\omega F_A \rangle \cdot \omega = \int_X \langle \zeta_j, F_A \rangle$$

Using lemma A.18 in the last identity and extending the inner product to differential forms the usual way. Then,

$$-\mathrm{i}\, c = \sum_{j=1}^r \left[\frac{1}{\mathrm{Vol}(X)} \cdot \int_X \langle \zeta_j, F_A \rangle \right] \cdot \zeta_j. \tag{4.2}$$

Hence, we have shown that the Chern connection may only satisfy (5.1) for a single value of c. For a proof of the fact that (4.2) depends only on the topology of the manifold and the bundle, see [**17**, Prop. 3.4] or [**2**, Remark 1.2] and the references therein.

We will fix the Ad-invariant positive definite inner product for the whole section. This induces a bilinear form on the space of sections of $\mathrm{ad}(P_K)$. Indeed, given two sections $\boldsymbol{\xi}, \boldsymbol{\zeta} \in \Gamma(X; \mathrm{ad}(P_K))$, we define $\langle \boldsymbol{\xi}(x), \boldsymbol{\zeta}(x) \rangle := \langle \xi, \zeta \rangle$ where $\boldsymbol{\xi}(x) = [p, \xi], \boldsymbol{\zeta}(x) = [p, \zeta]$ and $\pi(p) = x$. This is well-defined because taking another pair of representatives $[p \cdot g, \mathrm{Ad}_{g^{-1}}\xi]$ and $[p \cdot g, \mathrm{Ad}_{g^{-1}}\zeta]$, Ad-invariance implies that $\langle \mathrm{Ad}_{g^{-1}}\xi, \mathrm{Ad}_{g^{-1}}\zeta \rangle = \langle \xi, \zeta \rangle$. Moreover, this extends to $\mathrm{ad}(P_G)$-valued differential forms. More precisely, we define

$$\langle \cdot \wedge \cdot \rangle \colon \quad \Omega^p_{\mathbb{R}}(X; \mathrm{ad}(P_K)) \times \Omega^q_{\mathbb{R}}(X; \mathrm{ad}(P_K)) \quad \longrightarrow \quad \Omega^{p+q}_{\mathbb{R}}(X)$$

by extension of $\langle \alpha \otimes \boldsymbol{\xi} \wedge \beta \otimes \boldsymbol{\zeta} \rangle = \langle \boldsymbol{\xi}, \boldsymbol{\zeta} \rangle \cdot \alpha \wedge \beta$. The same construction applies to $\Omega^1_{\mathrm{Ad,\ hor}}(P_K; \mathfrak{k})$. We may also endow $\Gamma(X, \mathrm{ad}(P_K))$ with a Lie algebra structure as in remark 4.6, and extend it to differential forms by linear extension of $[\alpha \otimes \boldsymbol{\xi} \wedge \beta \otimes \boldsymbol{\zeta}] := (\alpha \wedge \beta) \otimes [\boldsymbol{\xi}, \boldsymbol{\zeta}]$. Of course, everything works the same for complex Lie algebras and complex differential forms.

We want to study the *moduli space of solutions* of equation (4.1). As we will see, it is given by an infinite-dimensional symplectic reduction. To construct the moduli space rigorously, we need to see both, the space of connections and the gauge group of the bundle P_K as Banach manifolds. On their own, they are not Banach manifolds, but they can be completed using Sobolev norms. As we shall see, to consider the moduli space we need K to be a *matrix* Lie group.

4.1 Sobolev spaces and the gauge group

For this section, we follow the exposition in [61] and [48]. Let us consider a vector bundle E over a compact Kähler manifold M, endowed with a positive-definite fiber metric $\langle \cdot, \cdot \rangle$, that is, a positive-definite inner product on each fiber E_m such that for every pair of sections ψ, ϕ of E, the map $\langle \psi, \phi \rangle \colon M \to \mathbb{R}$ is smooth. Fix a connection ∇^E on E, that is compatible with the fiber metric. By this we mean that $\nabla^E \langle \phi, \psi \rangle = \langle \nabla^E \phi, \psi \rangle + \langle \phi, \nabla^E \psi \rangle$ for all sections ϕ, ψ. This connection induces a covariant derivative on all tensor bundles over M as we show below.

As we know, the connection ∇^E defines a map $\nabla^E \colon \Gamma(E) \to \Gamma(T^*M \otimes E)$. On the other hand, since (M, g) is a riemannian manifold, TM is naturally endowed with the Levi-Civita connection ∇^g. This induces a connection on the dual bundle ∇^{g*}. The bundle $T^*M \otimes E$ is therefore endowed with the connection $\nabla^{g*} \otimes \nabla^E \colon \Gamma(T^*M \otimes E) \to \Gamma(T^*M \otimes T^*M \otimes E)$. Now, $\otimes^2 T^*M \otimes E$ is endowed with a connection $\nabla^{g*} \otimes \nabla^E$, and we may use it to define a connection on $T^*M \otimes \left(\otimes^2 T^*M\right) \otimes E \cong \otimes^3 T^*M \otimes E$, namely, we define the connection $\nabla^{g*} \otimes \left(\nabla^{g*} \otimes \nabla^E\right)$ (see section A.6.1). Iterating this process, we may define a connection on each $\otimes^k T^*M \otimes E$ by

$$\nabla^{g*} \otimes \overset{k}{\cdots} \otimes \nabla^{g*} \otimes \nabla^E.$$

Under some abuse of notation, we will denote by ∇ all the so-defined connections (including ∇^E). Explicitly, if $\alpha \in \Gamma(\otimes^k T^*M \otimes E)$, then $\nabla\alpha \in \Gamma(\otimes^{k+1} T^*M \otimes E)$ is defined by [61, Eq. B2]

$$\nabla\alpha(X_0, \dots X_k) = \nabla^E_{X_0}(\alpha(X_1, \dots X_k)) - \alpha(\nabla^g_{X_0} X_1, \dots X_k) - \dots - \alpha(X_1, \dots \nabla^g_{X_0} X_k),$$

For $k \in \mathbb{N}_{\geq 1}$ and any two sections $\phi, \psi \in \Gamma(E)$, we define the L^2_k inner product

$$\langle \phi, \psi \rangle_k := \int_M \left(\langle \phi, \psi \rangle + \langle \nabla\phi, \nabla\psi \rangle + \dots + \langle \nabla^k \phi, \nabla^k \psi \rangle \right) \mathrm{d}V_g. \tag{4.3}$$

Remark 4.2. Each term in the sum inside the integral is a smooth function, so it makes sense to integrate them with respect to the volume form $\mathrm{d}V_g$, and the compactness of M ensures that the integral will be finite. As for the first claim, the symbol $\langle \nabla^\ell \phi, \nabla^\ell \psi \rangle$ *hides* the fact that we need to define an inner product on tensor products. First, the fact that g is non-degenerate (it is a riemannian metric) allows us to define an inner product of one-forms by $\langle \alpha, \beta \rangle_g := g\left(\alpha^\#, \beta^\#\right)$, where $\#$ is the *sharp* isomorphism (see [32, Ch. 2]). On the other hand, whenever $V_1, \dots V_n$ are vector spaces endowed with inner products $\langle \cdot, \cdot \rangle_i$, we may define an inner product by extending by linearity the following expression

$$\langle v_1 \otimes v_2 \otimes \cdots \otimes v_n, w_1 \otimes w_2 \otimes \cdots \otimes w_n \rangle = \langle v_1, w_1 \rangle_1 \cdot \langle v_2, w_2 \rangle_2 \cdots \langle v_n, w_n \rangle_n.$$

This defines an inner product on each $\otimes^k T^*M \otimes E$ (see [49, Lemma 1.7] for a very detailed proof that the previous expression defines an inner product).

The map

$$\begin{array}{ccc} \Gamma(E) \times \Gamma(E) & \longrightarrow & \mathbb{R} \\ (\phi, \psi) & \mapsto & \langle \phi, \psi \rangle_k \end{array}$$

defines an inner product on $\Gamma(E)$. Of course, everything works out fine if we work with complex vector bundles and a hermitian fiber metric, in which case we can define a hermitian inner product on $\Gamma(E)$. Since $\Gamma(E)$ is a vector space with an inner product, we can always consider its Hilbert completion, which we shall denote $L^2_k(M; E)$ or $L^k_2(E)$. These spaces are called *Sobolev spaces*.

Remark 4.3. Recall that the topology of a Hilbert space is the metric topology with respect to the distance induced by the norm. The inner product we defined depends on g, ∇ and $\langle \cdot, \cdot \rangle$. However, different choices of these parameters lead to equivalent norms [48, page 418]. This means that, as a topological space, $L^2_k(M; E)$ is well-defined and independent of any choice of these parameters.

Elements of a Hilbert completion are, formally, equivalence classes of Cauchy sequences. The following lemma provides a more tangible way of looking at these spaces.

Lemma 4.4 (Sobolev). Let $E \to M$ be a vector bundle over a compact smooth manifold, and let $k, p \in \mathbb{N}_{\geq 0}$ be such that $k > \dim M/2 + p$. Then, there is a continuous embedding $L^2_k(M; E) \hookrightarrow \mathscr{C}^p(M; E)$.

Proof. See [61, Th. B.2]. □

Here, $\mathscr{C}^p(M; E)$ is endowed with the metric topology induced by the norm $||\phi||_\infty := \sup_{j \leq p} \left\{ \max_{p \in M} \langle \nabla^j \phi(p), \nabla^j \phi(p) \rangle \right\}$. That is, Sobolev spaces of sections can be seen as spaces of usual sections but with the caveat that some regularity may be lost.

We present now a sketch of the construction that allows us to see $\mathscr{G}$ as a Hilbert-Lie group. For the construction, we consider G a compact, connected, *matrix Lie group*. That is, $G \subset \mathfrak{gl}(n, \mathbb{C}) = \mathrm{M}_n(\mathbb{C})$ for some $n \in \mathbb{N}$. We suppose that M is a compact Kähler manifold. Consider the vector bundle $E^n_{\mathbb{C}} := P \times_G \mathfrak{gl}(n, \mathbb{C})$ where G acts on $\mathfrak{gl}(n, \mathbb{C})$ by conjugation. We may see $\mathrm{Ad}(P)$ as a subbundle of $E^n_{\mathbb{C}}$, i.e. it is an embedded submanifold, and its projection is the restriction of the projection of $E^n_{\mathbb{C}}$. Observe that $\mathrm{Ad}(P)$ is also constructed using the action of G by conjugation. Using the riemannian metric and any hermitian fiber metric on $E^n_{\mathbb{C}}$ (those always exist, and they can be constructed through a partition of unity argument, see [33, Page 196]), we can consider $\Gamma(M; \mathrm{Ad}(P)) \subset \Gamma(M; E^n_{\mathbb{C}}) \subset L^2_k(M; E^n_{\mathbb{C}})$. We define $\mathscr{G}_k := \overline{\Gamma(M; \mathrm{Ad}(P))}$, i.e. the closure of

$\Gamma(M;\mathrm{Ad}(P))$ in $L^2_k(M;E^n_{\mathbb{C}})$ with respect to the Sobolev topology of the latter. If we choose $k\in\mathbb{N}_{\geq 0}$ satisfying $k>\dim M/2+1$, then elements of $\mathscr{G}_k$ can be seen as $\mathscr{C}^2$ sections of the bundle $\mathrm{Ad}(P)$. Hence, $\mathscr{G}_k$ has a natural group structure where the group operation is defined pointwise. Actually, $L^2_k(M;E^n_{\mathbb{C}})$ has a natural product structure. This product, call it $m\colon L^2_k(M;E^n_{\mathbb{C}})\times L^2_k(M;E^n_{\mathbb{C}})\to L^2_k(M;E^n_{\mathbb{C}})$ is bilinear and continuous (see [37, Eq. 2.2]). Hence, $m(\overline{\Gamma(M;\mathrm{Ad}(P))\times\Gamma(M;\mathrm{Ad}(P))})\subset\overline{\Gamma(M;\mathrm{Ad}(P))}$, and $\mathscr{G}_k$ is closed under products.

Theorem 4.5. Let $P(M,G)$ be a principal G-bundle over a compact manifold, where G is a compact, connected, matrix Lie group. Then, for $k>\dim M/2+1$, $\mathscr{G}_k$ is a Hilbert-Lie group whose Lie algebra is $\mathrm{Lie}(\mathscr{G}_k)\cong L^2_k(M;\mathrm{ad}(P))$, and whose exponential map is given by

$$\exp_{\mathscr{G}_k}(\boldsymbol{\xi})(p)=\big[p,\exp_G(\hat{\xi}(p))\big],\qquad \boldsymbol{\xi}\in\mathrm{Lie}(\mathscr{G}_k),\ p\in M$$

Proof. See [37]. □

Remark 4.6. $\Gamma(M;\mathrm{ad}(P))$ is a Lie algebra endowed with the pointwise Lie bracket it gets from $\mathfrak{g}$. Given $\boldsymbol{\xi},\boldsymbol{\zeta}\in\Gamma(M;\mathrm{ad}(P_G))$ we define $[\boldsymbol{\xi},\boldsymbol{\zeta}]$ as the section

$$[\boldsymbol{\xi},\boldsymbol{\zeta}](m):=\big[\ [p,\xi],[p,\zeta]\ \big]=[p,[\xi,\zeta]\ ]\qquad\text{where }\boldsymbol{\xi}(m)=[p,\xi],\boldsymbol{\zeta}(m)=[p,\zeta].$$

This is well-defined independently of the chosen representatives because Ad_g is a Lie algebra isomorphism for all $g\in G$.

Observe that the smooth gauge group $\Gamma(M;\mathrm{Ad}(P))$ can be seen as a subgroup of $\mathscr{G}_k$ for each k. Moreover, since the exponential map is smooth, it maps $\Gamma(M;\mathrm{ad}(P))$ to $\mathscr{G}$, hence the former may be seen as the Lie algebra of $\mathscr{G}$. It is quite common in the literature to omit the Sobolev completions and directly state that $\Gamma(M;\mathrm{ad}(P))$ is the Lie algebra of G.

4.2 The moduli space of solutions and the moment map picture

In this section we will lay the foundations of the moment map picture of the Hitchin-Kobayashi correspondence. We will specify how the various spaces involved have to be completed with suitable Sobolev norms to actually get infinite-dimensional manifolds. Our main references for this section are [37], [59], [11], [52] and [48]. Before diving into the constructions, let us state a lemma we will use and which motivates the definition of curvature for elements in the Sobolev completion of the space of connections.

Lemma 4.7. Let $P(M,G)$ be a smooth principal bundle, and let A be a connection. Let $\alpha\in\Omega^1_{\mathbb{R}}(M;\mathrm{ad}(P_G))$. Then

$$F_{A+\alpha}=F_A+\mathrm{d}_A\alpha+\frac{1}{2}\cdot[\alpha\wedge\alpha]\tag{4.4}$$

Proof. Let $\hat{\alpha}:=\Upsilon^{-1}(\alpha)\in\Omega^1_{\mathrm{Ad},\,\mathrm{hor}}(P;\mathfrak{g})$. By $A+\alpha$ we mean the connection one-form $A+\hat{\alpha}$. We apply Υ to equation (3.5). Then,

$$\begin{aligned}F_{A+\alpha}&=\Upsilon\left(\mathrm{D}_AA+\mathrm{D}_A\hat{\alpha}+\frac{1}{2}\cdot[\hat{\alpha}\wedge\hat{\alpha}]\right)=F_A+\Upsilon(\mathrm{D}_A\hat{\alpha})+\frac{1}{2}\cdot\Upsilon([\hat{\alpha}\wedge\hat{\alpha}])\\&=F_A+\mathrm{d}_A\Upsilon(\hat{\alpha})+\frac{1}{2}\cdot[\Upsilon(\hat{\alpha})\wedge\Upsilon(\hat{\alpha})].\end{aligned}$$

It remains to check that $\Upsilon([\hat{\alpha},\hat{\alpha}]) = [\Upsilon(\hat{\alpha}),\Upsilon(\hat{\alpha})]$. To do so, let us unveil the different definitions. Let $\alpha = \alpha^j \otimes \boldsymbol{\xi}_j$ for $\alpha^j \in \Omega^1_{\mathbb{R}}(M)$ and $\boldsymbol{\xi}_j \in \Gamma(M;\mathrm{ad}(P_G))$. A simple calculation shows that $\Upsilon^{-1}(\alpha) = \pi^*\alpha^j \otimes \hat{\xi}_j$, where $\hat{\xi}_j := \Upsilon^{-1}(\boldsymbol{\xi}_j)\colon P \to \mathfrak{g}$ is G-equivariant (see [48, Remark 1.2.13]). Then, letting $p \in P$, $v_1, v_2 \in T_pP$

$$\begin{aligned}\big(\Upsilon([\hat{\alpha}\wedge\hat{\alpha}])\big)_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2)) &= q_p\big((\pi^*(\alpha^j\wedge\alpha^\ell))_p(v_1,v_2)\cdot[\hat{\xi}_j(p),\hat{\xi}_\ell(p)]\big)\\ &= \big[p,(\pi^*(\alpha^j\wedge\alpha^\ell))_p(v_1,v_2)\cdot[\hat{\xi}_j(p),\hat{\xi}_\ell(p)]\big]\end{aligned}$$

Whereas

$$\begin{aligned}[\Upsilon(\hat{\alpha})\wedge\Upsilon(\hat{\alpha})]_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2)) &= [\alpha^j\otimes\xi_j\wedge\alpha^\ell\otimes\xi_\ell]_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2))\\ &= (\alpha^j\wedge\alpha^\ell)_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2))\cdot[p,\xi_j(\pi(p)),\xi_\ell(\pi(p))].\end{aligned}$$

And these coincide due to the way the Lie bracket on $\Gamma(X;\mathrm{ad}(P))$ is defined (cf. remark 4.6) and the way that Υ acts on sections of $\mathrm{ad}(P_G)$ (cf. section 3.4). □

Consider now a compact Lie group K and a principal K-bundle P_K over a compact Kähler manifold M. From section 3.3 and remark 3.17, we know that $\mathscr{A}(P_K)$ is an affine space modelled on $\Omega^1_{\mathbb{R}}(X;P_K\times_K\mathfrak{k})$. We consider $L^2_k(X;T^*M\otimes\mathrm{ad}(P_K))$ its Sobolev completion with respect to the connection ∇_A that any connection $A\in\mathscr{A}(P_K)$ induces on the associated bundle $\mathrm{ad}(P_K)$ and the dual of the Levi-Civita connection (see section 4.1). Since $\mathscr{A}(P_K)$ is an affine space, we can fix an element $A_0\in\mathscr{A}$. We define the space of L^k_2-connections by

$$\mathscr{A}_k(P_K) := \left\{A_0+\alpha : \alpha\in L^2_k(T^*M\otimes\mathrm{ad}(P_K))\right\}.$$

This is again an affine space, and the set is independent of the smooth connection A_0 we use to define it. In fact, since $\boldsymbol{\Omega}^1_k(M;\mathrm{ad}(P_K)) := L^2_k(M;T^*M\otimes\mathrm{ad}(P_K))$ is a Hilbert space by construction, this endows $\mathscr{A}_k(P_K)$ with the structure of a Banach manifold. Indeed, charts are given by $\alpha\mapsto\alpha+A$ for some smooth connection A, and the transition maps are given by $\alpha\mapsto\alpha+(A_0-A_1)$ for two smooth connections $A_0,A_1\in\mathscr{A}(P_K)$. This is clearly a smooth map in the sense of Banach manifolds. Finally, choosing k sufficiently large, in particular, $k>\dim_{\mathbb{R}}M/2+1$, it follows that $\mathscr{A}_k(P_K)$ is embedded in the set of $\mathscr{C}^1$ connections (connection one-forms of class $\mathscr{C}^1$ instead of smooth, see lemma 4.4). One should remark that for affine spaces, the tangent space in the sense of section C.1.1 coincides with the model vector space (more precisely, they are isomorphic). When working with elements of $\mathscr{A}_k(P_K)$ that are not necessarily smooth connections, we still want to speak about their curvature. This is done by *defining* $F_{A+\alpha}$ using (4.4).

When K is a compact *connected, matrix* Lie group, then $\mathscr{G}_k := \overline{\Gamma(X;\mathrm{ad}(P_K))}$ is a Hilbert Lie group. Choose $k+1$ to endow the gauge group with a smooth structure. Then, the action by pullback

$$\Phi : \mathscr{G}_{k+1}\times\mathscr{A}_k \longrightarrow \mathscr{A}_k,\quad (A,\phi)\mapsto(\phi^{-1})^*A \tag{4.5}$$

is smooth in the sense of Banach manifolds [37, Prop. 3.12]. See equation (3.19) for an explicit expression of the action. To set some notation, given $\phi\in\mathscr{G}_{k+1}$, we will write $\tau\colon P\to G$ the map corresponding to ϕ by the isomorphism of theorem 3.27. Observe that the construction of that theorem is still well-defined if we restrict the regularity of our maps. Observe as well that we consider the pullback with respect to the inverse to get a left-action. Had we not used the inverse, we would have obtained a right action. This is not particularly relevant, but we define a left action to be coherent with the discussions of chapters 1 and 2. To be very explicit, elements of $\mathscr{G}_k$ can be viewed as $\mathscr{C}^2$ sections of the bundle $\mathrm{Ad}(P_K)$. The group isomorphisms of theorem 3.27 hold if we consider maps

of lower regularity. Under these isomorphisms, every element of $\mathscr{G}_k$ corresponds to a $\mathscr{C}^2$ automorphism of P_K covering the identity, which can be used to define the pullback.

Back to $\mathscr{A}_k$, this Hilbert manifold is in fact symplectic. To endow it with a symplectic structure, we define

$$\Omega_A(\alpha,\beta) := \int_X \Lambda_\omega\langle\alpha\wedge\beta\rangle\,\mathrm{d}V_g = \int_X \langle\alpha\wedge\beta\rangle\wedge\frac{\omega^{n-1}}{(n-1)!} \qquad \alpha,\beta\in T_A\mathscr{A}_k = \mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K)) \tag{4.6}$$

where the second expression follows from A.18. $\langle\cdot\wedge\cdot\rangle$ is defined as in remark 4.6. Here $\langle\cdot,\cdot\rangle$ is an Ad-invariant inner product on $\mathfrak{k}$, which always exists because K is compact (this is done by averaging any positive define inner product on $\mathfrak{k}$ by means of the Haar measure, see [**21**, Lemma 10.4.14]).

Remark 4.8. It is true, and we shall assume it henceforth, that almost every property of differential forms holds when we restrict ourselves to differential forms with reduced regularity. For example, in [**30**], the theory of differential forms is constructed without requiring smoothness.

For the remainder of the section we will need to use the symbol d_A to mean the exterior differential in the sense of Banach manifolds evaluated at a point $A\in\mathscr{A}_k$. To avoid confusions with the connection induced by A on the associated bundle, we shall use boldface d's for the exterior differential, that is $\mathbf{d}_A$.

Lemma 4.9. Let K be a compact Lie group, and $P(M,K)$ a principal K-bundle over a compact Kähler manifold, and consider the Sobolev completion of its space of connections $\mathscr{A}_k := \mathscr{A}_k(P_K)$ for $k > \dim_{\mathbb{R}} M/2+1$. Then, the form Ω defined by (4.6) is symplectic. Moreover, the $\mathscr{G}_{k+1}$-action is symplectic.

Proof. For each $A\in\mathscr{A}$, Ω_A is clearly bilinear and skew-symmetric. $\mathbf{d}\Omega = 0$ follows from the fact that Ω does not depend on the point (see proposition C.3). Since this is the first time we encounter the exterior differential applied to an infinite-dimensional form, let us be more explicit. First, for a two-form, the definition given in (C.3) reads

$$\begin{aligned}\mathbf{d}\Omega(X_1,X_2,X_3) =& X_1(\Omega(X_2,X_3)) - X_2(\Omega(X_1,X_3)) + X_3(\Omega(X_1,X_2)) - \Omega([X_1,X_2],X_3)\\ &+\Omega([X_1,X_3],X_2) - \Omega([X_2,X_3],X_1).\end{aligned}$$

Since Ω does not depend on the point we evaluate it at, this expression depends only on the values of the vector fields at a point. Since all the tangent spaces can be canonically identified with $\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K))$ we can calculate the right-hand side using constant vector fields. Their commutator is then zero (the proof given [**52**, Prop. 2.6] can be adapted to this case). Hence, it suffices to show that each of the first three summands vanishes. To do so, consider $\alpha,\beta,\vartheta\in\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K))$, which we will see as constant vector fields. Observe that we may interpret $\Omega(\alpha,\beta)$ as a map $\mathscr{A}\to\mathbb{R}$. We then consider the vector field defined by assigning ϑ to every point, which we still denote ϑ. To apply the comments on remark C.5, we identify it with the map that to any $A\in\mathscr{A}$ assigns the equivalence class of the curve $t\mapsto A+t\cdot\vartheta$. Hence, at any point $A\in\mathscr{A}$, we have

$$\big(\vartheta(\Omega(\alpha,\beta))\big)_A = \mathbf{d}_A\big(\Omega(\alpha,\beta)\big)([t\mapsto A+t\vartheta]) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}\Omega_{A+t\vartheta}(\alpha,\beta) = 0$$

the latter being zero because the map $\Omega(\alpha,\beta)\colon\mathscr{A}\to\mathbb{R}$ is constant. Finally, to check non-degeneracy, let $\alpha\in\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K))$. Observe that the assignment

$$(\alpha,\beta) := \int_M\langle\alpha\wedge\star\beta\rangle \qquad \alpha,\beta\in\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K))$$

defines an inner product on $\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_G))$ (see the comment below (A.11)). Now, for a fixed α, let $\beta := \Lambda_\omega^{n-1}\star\alpha$. We have,

$$\begin{aligned}\int_M \langle \alpha\wedge\Lambda_\omega^{n-1}\star\alpha\rangle\wedge\omega^{n-1} &= \int_M \langle \alpha\wedge L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha\rangle = \int_M \langle L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha\wedge\star\star\alpha\rangle \\ &= (L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha,\star\alpha) = (\Lambda_\omega^{n-1}\star\alpha,\Lambda_\omega^{n-1}\star\alpha)^2 > 0\end{aligned}$$

and we conclude that the latter is positive using that $\Lambda_\omega^{n-1}\star\alpha = \star\, L_\omega^{n-1}\alpha$ and L_ω^{n-1} is injective [33, Lemma 9.39]. Finally, the $\mathscr{G}_{k+1}$ action is symplectic. Let $\phi(p) = p\cdot\tau(p)$, and consider $\Phi_\phi\colon \mathscr{A}_k\to\mathscr{A}_k, A\mapsto(\phi^{-1})^*A$. Let $\alpha\in\mathbf{\Omega}^1_k(M;\mathrm{ad}(P_K))$. We have (see (3.20))

$$\mathbf{d}_A\Phi_\phi(\alpha) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}(\phi^{-1})^*(A+t\alpha) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\left(\mathrm{Ad}_\tau(A+t\alpha)+(\tau^{-1})^*\theta\right) = \mathrm{Ad}_\tau(\alpha).$$

Finally,

$$(\Phi_\phi^*\Omega)_A(\alpha,\beta) = \Omega_{\phi^*A}(\mathbf{d}_A\Phi_\phi(\alpha),\mathbf{d}_A\Phi_\phi(\beta)) = \Omega_{\phi^*A}(\mathrm{Ad}_\tau(\alpha),\mathrm{Ad}_\tau(\beta)) = \Omega_A(\alpha,\beta)$$

using the fact that $\langle\cdot,\cdot\rangle$ is Ad-invariant, and the adjoint action applies to the "section of $\mathrm{ad}(P_K)$" part of the differential form. □

Once shown that the action is symplectic, let us see that it is Hamiltonian. To do so, we define $\mu\colon\mathscr{A}_k\to\mathrm{Lie}(\mathscr{G}_{k+1})^*$ by

$$(\mu(A),\boldsymbol{\xi}) := \int_M\langle\boldsymbol{\xi},(\Lambda_\omega F_A+\mathrm{i}\,c)\rangle\,\frac{\omega^n}{n!}\qquad A\in\mathscr{A}_k,\ \boldsymbol{\xi}\in\mathrm{Lie}(\mathscr{G}_{k+1}) = L^2_{k+1}(X;\mathrm{ad}(P_K)). \tag{4.7}$$

Here, the symbol $(\cdot,\cdot)$ is the pairing between $\mathrm{Lie}(\mathscr{G}_{k+1})$ and its dual as in (1.2).

Lemma 4.10. Let $P(M,K)$ be a principal bundle over a compact Kähler manifold M. The map μ defined by (4.7) is a moment map for the action defined by (4.5).

Proof. For each $A\in\mathscr{A}_k$ it is obvious that $\mu(A)$ is linear. Moreover, $F_{\phi^*A} = \mathrm{Ad}_{\tau^{-1}}\circ F_A$ (see [48, Page 462]). Hence,

$$\begin{aligned}(\mu(\phi^*A),\boldsymbol{\xi}) &= \int_M\langle\boldsymbol{\xi},(\Lambda_\omega F_{(\phi^{-1})^*A}+\mathrm{i}\,c)\rangle\,\mathrm{d}V_g = \int_M\langle\boldsymbol{\xi},\mathrm{Ad}_\tau(\Lambda_\omega F_A+\mathrm{i}\,c)\rangle\,\mathrm{d}V_g \\ &= \int_M\langle\mathrm{Ad}_{\tau^{-1}}(\boldsymbol{\xi}),\Lambda_\omega F_A+\mathrm{i}\,c\rangle\,\mathrm{d}V_g = \langle\mu(A),\mathrm{Ad}_{\tau^{-1}}(\boldsymbol{\xi})\rangle\end{aligned}$$

Finally, given $\boldsymbol{\xi}\in\mathrm{Lie}(\mathscr{G}_{k+1})$, the infinitesimal action is $X_{\boldsymbol{\xi},A} := \mathrm{d}/\mathrm{d}t|_{t=0}\ \exp_{\mathscr{G}_{k+1}}(t\boldsymbol{\xi})\cdot A = -\mathrm{d}_A\boldsymbol{\xi}$, where d_A is connection that A induces in the associated bundle $\mathrm{ad}(P_K)$. Here, we identify $\exp_{\mathscr{G}_{k+1}}(t\boldsymbol{\xi})$ with a gauge transformation by means of theorem 3.27. For a proof of that claim, see [59, Lemma 2.2]. Let us check that $\mathbf{d}\mu^{\boldsymbol{\xi}} = i_{X_{\boldsymbol{\xi}}}\Omega$. To do so, we use the expression for $F_{A+\alpha}$ given in lemma 4.7. Now,

$$\begin{aligned}\mathbf{d}_A\,\mu^{\boldsymbol{\xi}}(\alpha) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\int_M\langle\boldsymbol{\xi},(\Lambda_\omega F_{A+t\alpha}+\mathrm{i}\,c)\rangle\,\mathrm{d}V_g = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\int_M\left\langle\boldsymbol{\xi},\Lambda_\omega F_A+t\Lambda_\omega\mathrm{d}_A\alpha+\frac{t^2}{2}\Lambda_\omega[\alpha\wedge\alpha]\right\rangle\frac{\omega^n}{n!} \\ &= \int_M\langle\boldsymbol{\xi},\Lambda_\omega\mathrm{d}_A\alpha\rangle\cdot\frac{\omega^n}{n!}\overset{(1)}{=}\int_M\langle\boldsymbol{\xi}\wedge\mathrm{d}_A\alpha\rangle\wedge\frac{\omega^{n-1}}{(n-1)!}\overset{(2)}{=}\int_M\langle-\mathrm{d}_A\xi\wedge\alpha\rangle\wedge\frac{\omega^{n-1}}{(n-1)!} \\ &= i_{X_{\boldsymbol{\xi},A}}\Omega_A(\alpha).\end{aligned}$$

Hence, $\mathbf{d}\mu^\xi = i_{X_{\boldsymbol{\xi}}}\Omega$. In (1) we used lemma A.18 and in (2) we used Stokes' theorem combined with the fact that

$$\mathrm{d}\left(\langle\boldsymbol{\xi}\wedge\alpha\rangle\wedge\omega^{n-1}\right) = \langle\mathrm{d}_A\boldsymbol{\xi}\wedge\alpha\rangle\wedge\omega^{n-1}+\langle\boldsymbol{\xi}\wedge\mathrm{d}_A\alpha\rangle\wedge\omega^{n-1}.$$

Indeed, $\mathrm{d}\big(\langle \boldsymbol{\xi} \wedge \alpha\rangle \wedge \omega^{n-1}\big) = \mathrm{d}\big(\langle \boldsymbol{\xi} \wedge \alpha\rangle\big) \wedge \omega^{n-1} + \langle \boldsymbol{\xi} \wedge \alpha\rangle \wedge \mathrm{d}\omega^{n-1}$, the latter being zero because ω is closed. For the former, one can show that $\mathrm{d}\langle \boldsymbol{\xi}, \alpha\rangle = \langle \mathrm{d}_A \boldsymbol{\xi} \wedge \alpha\rangle + \langle \boldsymbol{\xi}, \mathrm{d}_A \alpha\rangle$. To do this, first observe that a straightforward computation shows that $\langle \Upsilon^{-1}(\alpha) \wedge \Upsilon^{-1}(\beta)\rangle = \pi^*\langle \alpha \wedge \beta\rangle$ for all $\mathrm{ad}(P_K)$-valued differential forms α and β. Let $\tilde{\alpha} := \Upsilon^{-1}(\alpha)$ and $\hat{\xi} := \Upsilon^{-1}(\boldsymbol{\xi})$. Then,

$$\begin{aligned}\pi^*\langle \xi \wedge \mathrm{d}_A \alpha\rangle &= \langle \hat{\xi} \wedge \mathrm{D}_A \tilde{\alpha}\rangle = \langle \hat{\xi} \wedge (\mathrm{d}\tilde{\alpha} + [A, \tilde{\alpha}])\rangle = \mathrm{d}\langle \hat{\xi} \wedge \tilde{\alpha}\rangle - \langle \mathrm{d}\hat{\xi} \wedge \tilde{\alpha}\rangle + \langle \hat{\xi} \wedge [A \wedge \tilde{\alpha}]\rangle \\ &\overset{(3)}{=} \pi^*\mathrm{d}\langle \xi \wedge \alpha\rangle - \langle (\mathrm{d}\hat{\xi} + [A, \hat{\xi}]) \wedge \tilde{\alpha}\rangle = \pi^*\mathrm{d}\langle \xi \wedge \alpha\rangle - \langle \mathrm{D}_A \hat{\xi} \wedge \tilde{\alpha}\rangle \\ &= \pi^*\big(\mathrm{d}\langle \xi \wedge \alpha\rangle - \langle \mathrm{d}_A \xi \wedge \alpha\rangle\big).\end{aligned}$$

In (3) we used the invariance of the product (see the proof of lemma 5.2). The equality holds when removing the pullbacks because π is a surjective smooth submersion (cf. lemma 2.7). □

Observe that, by construction, $\mu^{-1}(0)$ is precisely the set of connections in $\mathscr{A}_k(P_K)$ that satisfy the hermitian Yang-Mills equation.

Remark 4.11. The map

$$\begin{array}{ccc}\mathrm{Lie}(\mathscr{G}_{k+1}) & \longrightarrow & \mathrm{Lie}(\mathscr{G}_{k+1})^* \\ \boldsymbol{\xi} & \mapsto & \int_M \langle \boldsymbol{\xi}, \cdot\rangle \, \mathrm{d}V_g\end{array}$$

is a vector space isomorphism. This justifies the fact that one commonly finds in the literature the statement that the moment map is given by $\mu \colon \mathscr{A}_k \to \mathrm{Lie}(\mathscr{G}_{k+1}), A \mapsto \Lambda_\omega F_A + \mathrm{i}\, c$. When writing this, we are seeing the moment map under that isomorphism.

Once everything is set up, the *moduli space of hermitian Yang-Mills connections* on P_K is given by the symplectic reduction

$$\mathcal{M}_{\mathrm{HYM}} := \mu^{-1}(0)/\mathscr{G}_{k+1}.$$

We will assume that μ satisfies the slice condition, that 0 is a weak regular value, and that the action of $\mathscr{G}_{k+1}$ on $\mu^{-1}(0)$ is proper without further comments. Then, theorem 2.4 applies, and $\mathcal{M}_{\mathrm{HYM}}$ has the structure of a symplectic manifold. There is one last thing that we should make clear before stating that $\mathcal{M}_{\mathrm{HYM}}$ is the moduli space of solutions of equation (4.1). Observe that upon completing the spaces with respect to a Sobolev norm we are adding non-smooth connections and gauge transformations and, in principle, it could happen that we add orbits not corresponding to any smooth solution. This does not happen. One may show, using analytical techniques, that every orbit in $\mathcal{M}_{\mathrm{HYM}}$ contains a smooth connection A solving equation (4.1).

4.3 Hermitian Yang-Mills equation for non-connected groups

If we remove the connectedness hypothesis of K, equation (4.1) no longer makes sense because we will not have, in general, a constant section equal to c. We reformulate the equation as follows. Let $\hat{K}$ be a possible non-connected compact Lie group, K its identity component, and let $P_{\hat{K}} \to M$ be a principal $\hat{K}$-bundle over a compact Kähler manifold. Let $\mathfrak{z}_K$ be the Lie algebra of the center of K, $Z(K)$ (equivalently, $\mathfrak{z}_K = \mathfrak{z}(\mathfrak{k})$ is the center of the Lie algebra). This Lie algebra is invariant under the adjoint action of $\hat{K}$, but this action is usually *not trivial*. A connection A satisfies the *a-hermitian Yang-Mills equation* for some $a \in \Gamma(X; P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\,\mathfrak{z}_K)$ if $\mathrm{d}_A a = 0$ and

$$\Lambda_\omega F_A = -\mathrm{i}\, a. \tag{4.8}$$

Remark 4.12. When K is connected (that is $\hat{K} = K$), equation (4.8) reduces to the connected version of the hermitian Yang-Mills equation (4.1). Indeed, the adjoint action

of a connected Lie group on the center of its Lie algebra is trivial. Therefore, $P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\,\mathfrak{z}_K = X \times \mathrm{i}\,\mathfrak{z}_K$ is a trivial bundle. Hence, sections of this bundle can be identified with smooth maps $X \to \mathrm{i}\,\mathfrak{z}_K$. From section 3.4, to $a\colon X \to \mathrm{i}\,\mathfrak{z}_K$ we assign a K-equivariant (invariant actually, because the K-action on $\mathrm{i}\,\mathfrak{z}_K$ is trivial) map $\tilde{a}(p) := \Upsilon^{-1}(a)(p) = a(\pi(p))$. Hence, $\mathrm{d}_A a = \Upsilon(\mathrm{D}_A\tilde{a}) = 0$, and since Υ is a vector space isomorphism, we have $\mathrm{D}_A\tilde{a} = (\mathrm{d}\tilde{a})^H = 0$. However, $\mathrm{d}\tilde{a} = (\mathrm{d}\tilde{a})^H$. To see this, it suffices to check that $\mathrm{d}\tilde{a}$ vanishes when acting on vertical vectors. Indeed, given a vertical vector $X_{\xi,p}$ for $\xi \in \mathfrak{k}, p \in P$, we have

$$\mathrm{d}_p\tilde{a}(X_{\xi,p}) = \mathrm{d}_p(a \circ \pi)(\mathrm{d}_e\phi^{(p)}(\xi)) = \mathrm{d}_e(a \circ \pi \circ \phi^{(p)})(\xi) = 0$$

because $\pi \circ \phi^{(p)}$ is a constant map identically equal to p. Hence, $\mathrm{d}\tilde{a} = 0$, and this implies that $\tilde{a}$ is constant. By construction, a must also be constant. Hence, if K is connected, we recover $\Lambda_\omega F_A = -\mathrm{i}\,a$ for $a \in \mathrm{i}\,\mathfrak{z}_K = \mathrm{i}\,\mathfrak{z}(\mathfrak{k})$.

Chapter 5

The Hitchin-Kobayashi correspondence

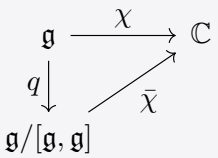
In this chapter, all the different topics we have treated come together in the Hitchin-Kobayashi correspondence. We will not treat the vector bundle version, even though it was the first version to appear, and we will jump straight to the principal bundle version. First, we give an overview of the result when the structure group is connected. We then consider the case where the structure group is non-connected, and we apply the techniques from appendix B to the study of this new situation. We also see how the moduli space of solutions of the hermitian Yang-Mills equation plays a role in an infinite-dimensional analogue of the Kempf-Ness theorem. To be fair, we will only cover here the correspondence for principal bundles over Riemann surfaces, which is Ramanathan's generalization of the Narasimhan-Seshadri theorem, but we will keep calling it the Hitchin-Kobayashi correspondence (HK correspondence from now on).

5.1 The HK correspondence for connected groups

For the remainder of the section, we fix a complex connected reductive Lie group G and let K be a maximal compact subgroup. Let $\mathcal{P}_G$ be a holomorphic principal G-bundle over some compact Riemann surface X. The statement of the Hitchin-Kobayashi correspondence is the following: a holomorphic principal G-bundle over a compact Kähler manifold is polystable if and only if there is a reduction of the structure group to K whose Chern connection satisfies the hermitian Yang-Mills equation. We discussed the equation in chapter 4, and we discuss the stability condition below. We will use extensively the notions of parabolic subgroups introduced in appendix B. It is important to remark that one may find several definitions of stability in the literature. They are all equivalent, even though it is not obvious at first glance. Our definitions will follow [46] and [16]. From now on, we fix $\langle \cdot, \cdot \rangle \colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ a non-degenerate $\mathbb{C}$-bilinear Ad-invariant pairing on $\mathfrak{g}$ that restricts to a non-degenerate pairing on $\mathfrak{k}$. We fix the notation $\mathscr{G} := \Gamma(X; P_K \times_K K)$ and $\mathscr{G}^{\mathbb{C}} = \Gamma(X; P_G \times_G G)$. We will write $\exp_{\mathscr{G}}$ and $\exp_{\mathscr{G}^{\mathbb{C}}}$ for their respective exponential maps.

Definition 5.1. Let $\mathfrak{g}$ be a complex Lie algebra.

A *character* is a complex linear map $\chi \colon \mathfrak{g} \to \mathbb{C}$ that factorizes through the quotient $q \colon \mathfrak{g} \to \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$. That is, there is a complex linear map $\bar{\chi} \colon \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \to \mathbb{C}$ such that the diagram commutes.



The following lemma gives examples of characters on parabolic algebras of the form $\mathfrak{p}_\zeta$ for $\zeta \in \mathrm{i}\,\mathfrak{k}$.

Lemma 5.2. Let $\zeta \in \mathfrak{k} \subset \mathfrak{g}$. We define $\chi_\zeta \in \mathfrak{g}^*$ by

$$\chi_\zeta(\xi) = \langle \zeta, \xi \rangle \qquad \xi \in \mathfrak{g}.$$

Then, the restriction of χ_ζ to $\mathfrak{p}_\zeta$ (which we will denote by the same symbol) is a character on $\mathfrak{p}_\zeta^*$.

Proof. χ_ζ is clearly $\mathbb{C}$-linear, and so is its restriction. Observe that $\zeta \in \mathfrak{p}_\zeta$. This follows from the definition of $\mathfrak{p}_\zeta$, and the fact that $\mathrm{Ad}_{\exp(t\zeta)}(\zeta) = \zeta$ for all $t \in \mathbb{R}$. To check that it factors through $[\mathfrak{p}_\zeta, \mathfrak{p}_\zeta]$, we must see that $\langle \zeta, [x, y] \rangle = 0$ for $x, y \in \mathfrak{p}_\zeta$. Observe that $\langle \zeta, [x, y] \rangle = \langle [x, \zeta], y \rangle$. This follows from Ad-invariance, and from the fact that $\mathrm{ad}(x)(y) = [x, y] = \mathrm{d}_e\mathrm{Ad}(x)(y)$ [**34**, Th. 20.27]:

$$\langle \zeta, [x, y] \rangle = \langle \zeta, \mathrm{ad}(x)(y) \rangle = \langle \zeta, \mathrm{d}_e\mathrm{Ad}(x)(y) \rangle = \left\langle \zeta, \left. \frac{\mathrm{d}}{\mathrm{d}t} \right|_{t=0} \mathrm{Ad}_{\exp(tx)}(y) \right\rangle$$
$$= \left\langle \left. \frac{\mathrm{d}}{\mathrm{d}t} \right|_{t=0} \mathrm{Ad}_{\exp(-tx)}(\zeta), y \right\rangle = \langle [-x, \zeta], y \rangle = \langle [\zeta, x], y \rangle.$$

Moreover, $\mathfrak{p}_\zeta = \mathfrak{l}_\zeta \oplus \mathfrak{u}_\zeta$ where $\mathfrak{u}_\zeta$ is the nilpotent radical obtained from the Levi factor $\mathfrak{l}_\zeta$ (cf. section B.2.1 and proposition B.10). Hence, by definition of $\mathfrak{l}_\zeta$, if $x \in \mathfrak{l}_\zeta$, $\chi_\zeta([x, y]) = \langle [\zeta, x], y \rangle = 0$. By the direct sum decomposition above, it suffices to see that $\chi_\zeta([x, y])$ for $x \in \mathfrak{u}_\zeta$. Now, since $\mathfrak{u}_\zeta$ is an ideal of $\mathfrak{p}_\zeta$, we have $[\mathfrak{u}_\zeta, \mathfrak{p}_\zeta] \subset \mathfrak{u}_\zeta$, hence $[x, y] \in \mathfrak{u}_\zeta$ for $x \in \mathfrak{u}_\zeta, y \in \mathfrak{p}_\zeta$. Therefore, we need only show that $\chi_\zeta(x) = 0$ for all $x \in \mathfrak{u}_\zeta$, but this is trivially true because $\mathfrak{u}_\zeta$ is the radical of the restriction of $\langle \cdot, \cdot \rangle$ to $\mathfrak{p}_\zeta$ (that is, $\langle \xi, x \rangle = 0$ for all $\xi \in \mathfrak{p}_\zeta, x \in \mathfrak{u}_\zeta$, see [**25**, Page 273]). □

Consider now any parabolic subalgebra $\mathfrak{p}$, and $\chi \in \mathfrak{p}^*$ a character. The non-degenerate pairing $\langle \cdot, \cdot \rangle$ induces a linear isomorphism $\mathfrak{g} \cong \mathfrak{g}^*, \zeta \mapsto \langle \zeta, \cdot \rangle$. Recall that we have a direct sum decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathrm{i}\mathfrak{k}$, which induces a projection $\mathfrak{g} \twoheadrightarrow \mathrm{i}\mathfrak{k}$. Finally, the pairing $\langle \cdot, \cdot \rangle$ induces a direct sum decomposition $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{p}^\perp$. This comes with a natural orthogonal projection $p_\perp \colon \mathfrak{g} \to \mathfrak{p}$, whose dual map gives an injection $\mathfrak{p}^* \hookrightarrow \mathfrak{g}^*, \chi \mapsto \chi \circ \mathsf{pr}_\perp$. We thus have the following chain of maps

$$\mathfrak{p}^* \hookrightarrow \mathfrak{g}^* \xrightarrow{\cong} \mathfrak{g} \to \mathrm{i}\mathfrak{k}.$$

Given $\chi \in \mathfrak{p}^*$ we denote by $\zeta_\chi \in \mathrm{i}\mathfrak{k}$ its image through the previous chain of maps.

Definition 5.3. Let $\mathfrak{p} \subset \mathfrak{g}$ be a parabolic subalgebra. A character $\chi \in \mathfrak{p}^*$ is said to be *antidominant* if $\mathfrak{p}_{\zeta_\chi} \subset \mathfrak{p}$ and *strictly antidominant* if $\mathfrak{p}_{\zeta_\chi} = \mathfrak{p}$.

Remark 5.4. For each $\zeta \in \mathrm{i}\mathfrak{k}$, we have $\zeta_{\chi_\zeta} = \zeta$, thus the character χ_ζ of $\mathfrak{p}_\zeta$ is always strictly antidominant. This follows from the fact that $\chi_\zeta \circ \mathsf{pr}_\perp = \chi_\zeta$ (recall that $\zeta \in \mathfrak{p}_\zeta$) the way $\langle \cdot, \cdot \rangle$ induces the pairing. The fact that each χ_ζ is antidominant is a very technical requirement, and we will not make explicit use of it in our sketch of the proof (but it is necessary nonetheless).

Consider $\mathcal{P}$ a holomorphic principal G-bundle over a Riemann surface X. Let $\zeta \in \mathrm{i}\mathfrak{k}$ and let $P_\zeta \subset G$ be the parabolic subgroup defined by ζ as in proposition B.10. Consider a reduction of the structure group of $\mathcal{P}$ to P_ζ given by a holomorphic section $\sigma \in \mathcal{O}(X; \mathcal{P}/P_\zeta)$, and let $\mathcal{P}_\sigma := \sigma^*\mathcal{P}$ be the corresponding holomorphic P_ζ-principal bundle over X. Consider L_ζ the Levi subgroup of P_ζ defined by B.10. Define $K_\zeta := K \cap L_\zeta$ which is a Lie

group whose Lie algebra is given by $\mathfrak{k}_\zeta := \mathfrak{k} \cap \mathfrak{l}_\zeta$. Then, K_ζ is a maximal compact subgroup of L_ζ and the inclusion $K_\zeta \subset L_\zeta$ is a homotopy equivalence. The inclusion $L_\zeta \subset P_\zeta$ is a homotopy equivalence as well, so K_ζ and P_ζ are homotopy equivalent. This implies that any reduction σ induces another reduction (unique up to homotopy equivalence) of the structure group to K_ζ. Call P'_σ the resulting smooth principal K_ζ-bundle. See [27, Example 5.6] and the results surrounding it for a justification of this fact.

Definition 5.5. Let $\mathcal{P}(X, G)$ be a holomorphic principal bundle and $\zeta \in \mathrm{i}\,\mathfrak{k}$. Keeping the notations of the previous paragraph, consider a connection A on $\mathcal{P}'_\sigma$, and let $F_A \in \Omega^2_{\mathbb{R}}(X; P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$ be its curvature. We define the *degree of $\mathcal{P}$ with respect to ζ and the reduction σ* by

$$\deg \mathcal{P}(\sigma, \zeta) = \frac{\mathrm{i}}{2\pi} \cdot \int_X \chi_\zeta(F_A).$$

Remark 5.6. Let us check that $\deg \mathcal{P}(\sigma, \zeta)$ is a well-defined real number. First, observe that $\chi_\zeta(\mathfrak{k}_\zeta) \subset \mathrm{i}\,\mathbb{R}$. Indeed, $\zeta = \mathrm{i}\,\zeta'$ for some $\zeta' \in \mathfrak{k}$, hence $\chi_\zeta(\xi) = \langle \mathrm{i}\,\zeta', \xi \rangle = \mathrm{i}\,\langle \zeta', \xi \rangle$ for $\zeta \in \mathfrak{k}_\zeta \subset \mathfrak{k}$ using $\mathbb{C}$-bilinearity. Moreover, since $\mathfrak{g} = \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$, $\mathfrak{g}$ is endowed with a natural conjugation map. With respect to this conjugation map, and due to $\mathbb{C}$-bilinearity, a direct computation shows that $\overline{\langle \zeta, \xi \rangle} = \langle \overline{\zeta}, \overline{\xi} \rangle$ for any pair $\zeta, \xi \in \mathfrak{g}$. Hence, if $\zeta, \xi \in \mathfrak{k}$, they coincide with their complex conjugates and their product is real.

On the other hand, F_A takes values in the associated bundle $P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta$. We define the composition of χ_ζ and any p-form in $\Omega^p_{\mathbb{R}}(X; P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$ by linear extension of $\chi_\zeta(\alpha \otimes \boldsymbol{\xi}) = \chi_\zeta(\boldsymbol{\xi}) \cdot \alpha$, where $\boldsymbol{\xi} \in \Gamma(X; P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$. Finally, if $\boldsymbol{\xi}(x) = [p, \xi] \in P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta$, with $\pi(p) = x$, we define $\chi_\zeta(\boldsymbol{\xi}(x)) = \chi_\zeta(\xi)$. This does not depend on the representative chosen. Indeed, $[p, \xi] = [p \cdot k, \mathrm{Ad}_{k^{-1}}(\xi)]$ for $k \in K_\zeta$, but

$$\chi_\zeta(\mathrm{Ad}_{k^{-1}}(\xi)) = \langle \zeta, \mathrm{Ad}_{k^{-1}}(\xi) \rangle = \langle \mathrm{Ad}_k(\zeta), \xi \rangle = \langle \zeta, \xi \rangle = \chi_\zeta(\xi)$$

using the Ad-invariance of the product and the fact that $K_\zeta \subset L_\zeta = \{g \in G : \mathrm{Ad}_g(\zeta) = \zeta\}$. Finally, let us see that the definition is independent of the chosen connection A. First, recall that since the space of connections $\mathscr{A}(P'_\sigma)$ is an affine space modelled on $\Omega^1_{\mathbb{R}}(X; \mathrm{ad}(P'_\sigma))$, if we fix a connection $A \in \mathscr{A}(P'_\sigma)$, any other connection will be given by $A + \alpha$ for some $\alpha \in \Omega^1_{\mathbb{R}}(X; P'_\sigma)$. Observe that by our definitions, and due to the fact that everything is smooth, it follows that $\chi_\zeta(\alpha) \in \Omega^1_{\mathbb{R}}(X)$. Let us assume that $\chi_\zeta(\mathrm{d}_A \alpha) = \mathrm{d}\chi_\zeta(\alpha)$ (we give a proof of this fact in lemma 5.7 below). The curvature two-forms of A and $A + \alpha$ relate as $F_{A+\alpha} = F_A + \mathrm{d}_A \alpha + 1/2 \cdot [\alpha \wedge \alpha]$, (see lemma 4.7). Since χ_ζ is a character of $\mathfrak{p}_\zeta$, it vanishes on $[\mathfrak{p}_\zeta, \mathfrak{p}_\zeta]$, hence, it vanishes on $[\mathfrak{k}_\zeta, \mathfrak{k}_\zeta]$ as well, and that is where $[\alpha \wedge \alpha]$ takes values. Hence, applying Stokes' theorem, we find

$$\int_X \chi_\zeta(F_{A+\alpha}) = \int_X \chi_\zeta(F_A) + \int_X \mathrm{d}\chi_\zeta(\alpha)$$

the latter being zero because $\partial X = \varnothing$.

Lemma 5.7. Under the conditions of definition 5.5, we have $\chi_\zeta(\mathrm{d}_A \alpha) = \mathrm{d}\chi_\zeta(\alpha)$ for all $\zeta \in \mathrm{i}\,\mathfrak{k}$ and $\alpha \in \Omega^1_{\mathbb{R}}(X; \mathrm{ad}(P'_\sigma))$.

Proof. The calculation is similar to that of lemma 4.7. Let $\pi \colon P'_\sigma \to X$ be the projection, and $\alpha \in \Omega^1_{\mathbb{R}}(X; P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$. Recall from theorem 3.16 that $\Upsilon^{-1}(\alpha) \in \Omega^1_{\text{Ad, hor}}(P'_\sigma; \mathfrak{k}_\zeta)$. Expand $\alpha = \alpha^j \otimes \boldsymbol{\xi}_j$ for some $\alpha^j \in \Omega^1_{\mathbb{R}}(X)$ and $\boldsymbol{\xi}_j \in \Gamma(X; P'_\sigma(\mathfrak{k}_\zeta))$. As in lemma 4.7 $\Upsilon^{-1}(\alpha) = \pi^* \alpha^j \otimes \hat{\xi}_j$, where $\hat{\xi}_j = \Upsilon^{-1}(\boldsymbol{\xi}_j) \colon P'_\sigma \to \mathfrak{k}_\zeta$ is K_ζ-equivariant. After setting this notation up, the result

follows from a simple computation. Let $p \in P'_\sigma$ and $v_1, v_2 \in T_p P'_\sigma$. On the one hand, it follows from our definitions that $\mathrm{d}\chi_\zeta(\alpha) = \chi_\zeta(\boldsymbol{\xi}_j) \cdot \mathrm{d}\alpha^j$. Hence,

$$\begin{aligned}\mathrm{d}_{\pi(p)}\, \chi_\zeta(\alpha)(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2)) &= \mathrm{d}_{\pi(p)}\alpha(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2)) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))) \\ &= (\pi^*\alpha^j)_p(v_1, v_2) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))).\end{aligned}$$

On the other hand,

$$\begin{aligned}\chi_\zeta((\mathrm{d}_A\alpha)_p(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2))) &= \chi_\zeta(q_p(\mathrm{D}_A\Upsilon^{-1}(\alpha))_p(v_1^H, v_2^H)) = \chi_\zeta\big(q_p(\mathrm{d}_p(\pi^*\alpha^j)(v_1^H, v_2^H) \cdot \hat{\xi}_j(p))\big) \\ &= (\pi^*\mathrm{d}\alpha^j)_p(v_1, v_2) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))).\end{aligned}$$

In the last step we used the fact that $\xi_j(\pi(p)) = [p, \hat{\xi}_j(p)]$ and the fact that $\mathrm{d}_p\pi(v) = \mathrm{d}_p\pi(v^H)$ because vertical vectors are annihilated by the projection. We thus get the identity $\mathrm{d}\chi_\zeta(\alpha) = \chi_\zeta(\mathrm{d}_A\alpha)$ but written pointwise. □

We are now ready to state the definition of stability. First, observe that parabolic subgroups P_ζ are embedded subgroups, and the action of P_ζ on G by left multiplication $P_\zeta \times G \to G, (p, g) \mapsto p \cdot g$ is holomorphic free and proper (the reasoning in the proof of [34, Th. 21.17] holds), hence, by theorem B.3, G/P_ζ has the structure of a complex manifold. Then, given a holomorphic principal bundle $\mathcal{P}$, the associated bundle $\mathcal{P}(G/P_\zeta)$ is holomorphic as well, and we can consider its holomorphic sections.

Definition 5.8 (Stability condition). Let $\mathcal{P}_G$ be a holomorphic principal bundle over a Riemann surface, and let $c \in \mathrm{i}\,\mathfrak{z}_K$. We say that $\mathcal{P}_G$ is

1. *c-stable* (resp. *c-semistable*) if for any $\zeta \in \mathrm{i}\,\mathfrak{k}$, and any $\sigma \in \mathcal{O}(X; \mathcal{P}(G/P_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle > 0 \qquad \text{(resp. } \deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle \geq 0).$$
2. *c-polystable* if it is c-semistable and or any $\zeta \in \mathrm{i}\,\mathfrak{k}$ and any $\sigma \in \mathcal{O}(X; \mathcal{P}(G/P_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle = 0$, there is a holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(P/L_\zeta))$.

The arguments of remark 5.6 apply to show that $\langle c, \zeta\rangle \in \mathbb{R}$, hence the inequalities above make sense. With the correct notion of stability, we can now state the correspondence.

Remark 5.9. If X is not a Riemann surface but a compact Kähler manifold, the definition is slightly modified because the sections one has to consider are not defined on all of X, but on open sets X_0 such that $X \setminus X_0$ is a complex submanifold satisfying $\mathrm{codim}_{\mathbb{C}}(X \setminus X_0) \geq 2$.

Theorem 5.10 (Hitchin-Kobayashi for principal bundles). Let G be a connected complex reductive Lie group, $K \subset G$ a maximal compact subgroup, and $\mathcal{P}_G$ a holomorphic principal G-bundle over a compact Riemann surface X. Then, $\mathcal{P}_G$ is c-polystable if and only if there is a reduction of the structure group to K, P_K whose Chern connection A satisfies the equation

$$\Lambda_\omega F_A = -\mathrm{i}\, c \tag{5.1}$$

for some $c \in \mathrm{i}\,\mathfrak{z}_K$.

Equation (5.1) is called the *hermitian Yang-Mills equation for principal bundles*. We insist on the fact that $F_A \in \Omega^2_{\mathbb{R}}(X; P_K \times_K \mathfrak{k})$ is the curvature of the Chern connection on the *reduced* bundle.

5.1.1 Infinite-dimensional analogue of the Kempf-Ness theorem

We have claimed several times that the Hitchin-Kobayashi correspondence can be seen as an infinite-dimensional version of the Kempf-Ness theorem. Section 4.2 was devoted to establish, with some level of detail, the moment map picure for the moduli space of solutions of the hermitian Yang-Mills equation. In this section, we shall explore how this situation can be seen as a formal analogue of the Kempf-Ness theorem when the base manifold is a Riemann surface X. We will not try to give proofs for most of the claims stated in this section because their proofs require very advanced techniques coming from the fields of analysis and elliptic differential equations, and we have not developed the tools required to deal with them. There are two points we should clarify. First, at the date of writing of this document *there is no infinite-dimensional version of the Kempf-Ness theorem*. Hence, by saying that the Hitchin-Kobayashi correspondence is an infinite-dimensional analogue of said theorem, we mean that it is a proof of the statement of the Kempf-Ness theorem for the very specific case of the action of the gauge group on the space of connections. The second point is that this theorem is a *formal analogue*: it should not be interpreted as a word-for-word version of the finite-dimensional version, but rather as a philosophy, as we shall discuss below. Along the lines of this second point, there are a few things we should remark. The first one is that when we say that $\mathscr{G}^{\mathbb{C}} = \Gamma(X; \mathrm{ad}(P_G))$ is a complexification of $\mathscr{G} = \Gamma(X; \mathrm{Ad}(P_K))$ we do not mean it literally in the sense of satisfying a universal property as in definition B.1. We can *formally* say that it is a complexification because fiberwise, it is. Secondly, if one looks at the definition of stability for principal bundles we have given here, and the definition we gave for the finite-dimensional version of the Kempf-Ness theorem (cf. definitions 5.8 and 1.6) it does not look at all like one is a generalization of the other. There are ways to write these conditions that make them look more similar to each other, but that does not matter much to see how *the philosophy* of the theorem is preserved. For the rest of the section, we fix a principal K-bundle, and set $P_G := P_K \times_K G$, which is a principal G-bundle.

For the algebraic side, recall that the Chern-Singer correspondence induces a bijection between the set of holomorphic structures on P_G, $\mathscr{J}(P_G)$, and the space of connections on P_K, $\mathscr{A}(P_K)$. Polystability is a condition depending on the holomorphic structure, but by means of the Chern-Singer correspondence, we may define a connection $A \in \mathscr{A}(P_K)$ as polystable if $\mathsf{CS}^{-1}(A) = J$ makes P_G into a polystable holomorphic principal bundle. Hence, we can define $\mathscr{A}^{ps}(P_K) \subset \mathscr{A}(P_K)$. Since the action of $\mathscr{G}^{\mathbb{C}}$ on $\mathscr{A}(P_K)$ is induced by the action of $\mathscr{G}^{\mathbb{C}}$ on $\mathscr{J}(P_G)$, and polystability is preserved under biholomorphism, we deduce that $\mathscr{G}^{\mathbb{C}}$ preserves $\mathscr{A}^{ps}(P_K)$. Hence, we may consider $\mathscr{A}^{ps}(P_K)/\mathscr{G}^{\mathbb{C}}$. This quotient then represents the set of holomorphic polystable structures on P_G up to biholomorphism.

Now, the Hitchin-Kobayashi correspondence can be interpreted as defining a bijection

$$\mu^{-1}(0)/\mathscr{G}_{k+1} \longleftrightarrow \mathscr{A}^{ps}(P_K)/\mathscr{G}^{\mathbb{C}}, \tag{5.2}$$

and this very clearly mimics the Kempf-Ness theorem: we have a Hamiltonian action of a group on a manifold, and there is a bijection between the symplectic quotient and the quotient of a set of polystable points by the complexification of the gauge group. Observe that the Kempf-Ness theorem can be interpreted as saying that the symplectic quotient is homeomorphic to the usual topological quotient we get from the action of the complexification of the group when the set of "bad points" (i.e. non-polystable points) is removed. Here we have the same thing: there is a bijection between the symplectic quotient and the set of connections quotiented by the complexified gauge group where the set of "bad points", that is, non-polystable points, has been removed.

It remains to see how theorem 5.10 defines the bijection in (5.2). Given an equivalence class of connections in $\mu^{-1}(0)/\mathscr{G}$, choose a smooth connection A. This connection defines an integrable almost holomorphic structure on P_G through the Chern-Singer correspondence, and A is clearly the Chern connection with respect to that almost holomorphic structure. Since A satisfies equation (5.1), the Hitchin-Kobayashi correspondence implies that the almost holomorphic structure it endows is polystable. Any other smooth connection A' we may find in the same orbit is related to A by $A' = \phi \cdot A$ for some $\phi \in \mathscr{G}$. Since $\mathscr{G}$ is a subgroup of $\mathscr{G}^{\mathbb{C}}$, and the action of the latter on the space of connections extends the action of $\mathscr{G}$, we get two integrable almost holomorphic structures in the same $\mathscr{G}^{\mathbb{C}}$, hence, the map is well-defined. Observe that we are basically sending $\mathscr{G}$-orbits to $\mathscr{G}^{\mathbb{C}}$-orbits.

To see that the construction above actually defines a bijection, we need to shift our viewpoint in the Hitchin-Kobayashi correspondence. The statement we have given in theorem 5.10 fixes a holomorphic structure on P_G and looks for reductions of the structure group to K whose Chern connection satisfies equation (5.1). However, the bijection given by (5.2) is phrased differently. There, we *are fixing a reduction of the structure group*, P_K, and we *are looking for polystable holomorphic structures* on P_G, or equivalently, connections $A \in \mathscr{A}(P_K)$ satisfying equation (5.1). Theorem (5.10) can be stated as follows (this is similar to theorem 5.11 below): Fix P_K a principal K-bundle, and $P_G = P_K \times_K G$ a principal G-bundle. Let $A \in \mathscr{A}(P_K)$ be a connection. Then, there is a gauge transformation $\phi \in \mathscr{G}^{\mathbb{C}}$ such that $\phi \cdot A$ satisfies (5.1) if and only if P_G is polystable with respect to the holomorphic structure given by $\mathsf{CS}^{-1}(A)$.

We can now show that (5.2) is a bijection. Given $A \in \mathscr{A}^{ps}(P_K)$, by the formulation above, there is $\phi \in \mathscr{G}^{\mathbb{C}}$ such that $A' := \phi \cdot A$ satisfies equation (5.1), hence $A' \in \mu^{-1}(0)$. Then the equivalence class of A' is mapped to that of A. For injectivity, let $A', A'' \in \mu^{-1}(0)$ be in the same $\mathscr{G}^{\mathbb{C}}$-orbit, thus, there is $\psi \in \mathscr{G}^{\mathbb{C}}$ such that $A' = \psi \cdot A''$, but one may actually show that $\psi \in \mathscr{G}$, hence they are in the same $\mathscr{G}$-orbit. Again, this is along the lines of theorem 5.11 below.

When we consider a compact Kähler manifold instead of a Riemann surface, we look for connections satisfying $\Lambda_\omega F_A = -\mathrm{i}\, c$ *and* $F_A^{(0,2)} = 0$ (so that they induce a holomorphic structure on P_G through the Chern correspondence). When constructing the moduli space of solutions of the hermitian Yang-Mills equations there is a very important difference from what we did in chapter 4: the space $\mathscr{A}^{(1,1)}$ is not an affine space. In fact, it is not even a manifold because it may have singular points. It is a Banach manifold outside its singularity set, and there $\mathscr{G}$ acts symplectically for the restriction of the moment map. The moduli space of solutions can still be constructed using Banach-symplectic reduction.

5.1.2 A sketch of the proof

We will sketch a proof of theorem 5.10 following Mundet i Riera, García Prada and Gothen [46] and [16]. The results stated in these articles are more general: they are given in the context of Kähler fibrations and Higgs bundles respectively, but we may reduce their arguments to ours by ignoring the extra structures. For an alternative proof of this result which relies heavily on the vector bundle theorem, we refer the reader to [3]. Our strategy for the proof is to reduce the structure group further to another subgroup in such a way that another stronger condition is satisfied: simplicity. Then, the results will follow from the arguments in [46]. The results in these articles are quite involved; giving detailed proofs would take a lot of work, and it is beyond the scope of this work. We include a sketch for the sake of completeness.

First of all, we say that an element $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G}^{\mathbb{C}}) = \Gamma(X; P_G \times_G \mathfrak{g})$ is *semisimple* if, for any $x \in X$, $\boldsymbol{\xi}(x) \in \mathfrak{g}$ is semisimple (i.e. its adjoint diagonalizes). We see $\boldsymbol{\xi}(x) \in \mathfrak{g}$ through any of the isomorphisms $(P_G \times_G \mathfrak{g})_x \cong \mathfrak{g}$. The condition of being semisimple does not depend on the chosen representative (because any other representative will be its image through the adjoint action, and Ad_g is an isomorphism for every $g \in G$). Finally, recall that the action of $\mathscr{G}$ on $\mathscr{A}$ can be extended to an action of $\mathscr{G}^{\mathbb{C}}$ using the Chern-Singer correspondence (see section 3.8). Then, a connection $A \in \mathscr{A}(P_G)$ is said to be *simple* if no element $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G})$ fixes A under the infinitesimal action of $\mathscr{G}^{\mathbb{C}}$. More explicitly, given $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G}^{\mathbb{C}})$ we can construct the associated vector field $X_{\boldsymbol{\xi}}$. Then, the *infinitesimal action* is the vector $X_{\boldsymbol{\xi},A} \in \Omega^1(M; \mathrm{ad}(P_G))$. As shown in [**59**, Prop. 2.3], this is given, explicitly, by

$$X_{\boldsymbol{\zeta}_1 + \mathrm{i}\,\boldsymbol{\zeta}_2, A} = -\mathrm{d}_A \boldsymbol{\zeta}_1 - \star\, \mathrm{d}_A \boldsymbol{\zeta}_2$$

for $\boldsymbol{\xi} = \boldsymbol{\zeta}_1 + \mathrm{i}\,\boldsymbol{\zeta}_2$ and $\boldsymbol{\zeta}_1, \boldsymbol{\zeta}_2 \in \mathrm{Lie}(\mathscr{G})$. Then, [**46**] shows the following version of the Hitchin-Kobayashi correspondence. By "not fixing A under the infinitesimal action" we mean $X_{\boldsymbol{\xi},A} \neq 0$.

Theorem 5.11 (Hitchin-Kobayashi for simple connections). Let P_K be a principal K-bundle over a Riemann surface, and $P_G := P_K \times_K G$. Let $A \in \mathscr{A}(P_K)$ be a simple connection and $c \in \mathrm{i}\,\mathfrak{z}_K$. Then, there is a gauge transformation $\psi \in \mathscr{G}^{\mathbb{C}}$ such that

$$\Lambda_\omega F_{\psi \cdot A} = -\mathrm{i}\, c \tag{5.3}$$

if and only if P_G is c-stable with respect to the integrable almost holomorphic structure $\mathsf{CS}^{-1}(A)$. If $\psi \cdot A$ and $\psi' \cdot A$ solve equation (5.3), then there is some $\phi \in \mathscr{G}$ such that $\psi' = \phi \circ \psi$.

*Sketch of proof [**46**].* The main idea is to construct a functional $\Psi \colon \mathscr{A}_k(P_K) \times \mathscr{G}^{\mathbb{C}}_{k+1} \to \mathbb{R}$ whose critical points are solutions of (5.3) after completing these spaces with suitable Sobolev norms as in section 4.2. We fix a *hermitian metric* $\langle \cdot, \cdot \rangle$ in $\mathfrak{g}$. This is done by means of a representation as follows. First, since K is compact, it admits a faithful representation $\rho \colon K \to \mathrm{U}(V)$ where V is some hermitian vector space. By the universal property of complexifications, this extends to a faithful representation $\rho_{\mathbb{C}} \colon G \to \mathrm{GL}(V)$. We may then define a hermitian inner product on $\mathfrak{g}$ using this representation as follows

$$\langle \xi, \zeta \rangle := \mathrm{Tr}(\mathrm{d}_e \rho_{\mathbb{C}}(\xi) \cdot \mathrm{d}_e \rho_C(\zeta)^*).$$

This extends to a hermitian fiber metric on $P_G \times_G \mathfrak{g}$.

First, one defines the *weight* ($(\cdot, \cdot)$ is the pairing between $\mathrm{Lie}(\mathscr{G}_{k+1})$ and its dual)

$$\lambda_t(A; \boldsymbol{\zeta}) := \mu^{\boldsymbol{\zeta}}(\exp_{\mathscr{G}_{k+1}}(t\ \boldsymbol{\zeta}) \cdot A) = \big(\mu(\exp_{\mathscr{G}_{k+1}}(t\ \boldsymbol{\zeta}) \cdot A),\ \boldsymbol{\zeta}\big) \in \mathbb{R}, \qquad \boldsymbol{\zeta} \in \mathrm{Lie}(\mathscr{G}_{k+1}).$$

One may show that λ_t increases with $t \in \mathbb{R}$, hence, we define (and it always exists) the *maximal weight*

$$\lambda(A; \boldsymbol{\zeta}) = \lim_{t \to 0} \lambda_t(A; \boldsymbol{\zeta}) \in \mathbb{R} \cup \{\infty\}\,.$$

Then, one shows that there is a unique functional $\Psi \colon \mathscr{A}_k(P_K) \times \mathscr{G}^{\mathbb{C}}_{k+1} \to \mathbb{R}$, called *the integral of the moment map* satisfying:

1. For any $A \in \mathscr{A}_k(P_K)$, the critical points of the restriction $\Psi_A \colon \mathscr{G}_{k+1} \to \mathbb{R}$ of Ψ to $\{A\} \times \mathscr{G}_{k+1}$ coincide with the points of the orbit $\mathscr{G}_{k+1} \cdot A$ on which μ vanishes.
2. The restriction of Ψ_A to lines of the form $\left\{\exp_{\mathscr{G}_{k+1}}(t\boldsymbol{\zeta}) : t \in \mathbb{R}\right\}$ for $\boldsymbol{\zeta} \in \mathrm{Lie}(\mathscr{G}_{k+1})$ is convex.

One then shows that an element $\psi \in \mathscr{G}_{k+1}$ is a critical point of Ψ_A for some $A \in \mathscr{A}_k(P_K)$ if and only if $\mu(\psi \cdot A) = 0$, i.e. if $\psi \cdot A$ is a solution of (5.3).

After everything is set up, we can show that stability implies the existence of solutions. Let A be a simple connection. One may show that $\Psi(A, \phi \circ \psi) = \Psi(A, \psi)$ for $\phi \in \mathscr{G}_{k+1}$ and $\psi \in \mathscr{G}_{k+1}^{\mathbb{C}}$ [46, Prop. 3.4]. Then, Ψ_A factors through $\mathscr{G}_{k+1}$:

$$\Psi_A \colon \mathscr{G}_{k+1}^{\mathbb{C}}/\mathscr{G}_{k+1} \to \mathbb{R}.$$

We want to minimize that functional. To do so, since $\mathfrak{g}/\mathfrak{k} \cong \mathrm{i}\,\mathfrak{k}$, we may use the exponential map to identify $\mathscr{G}^{\mathbb{C}}/\mathscr{G}$ with $\Gamma(M; P_K \times_K \mathrm{i}\,\mathfrak{k})$. Choosing $p > 2$, we define $\mathscr{M}_2^p = L_2^p(P_K \times_K \mathrm{i}\,\mathfrak{k})$ (the L_2^p norm is defined as usual with respect to the hermitian fiber product defined as above), which we shall identify with the quotient of the Sobolev completions. When A is a simple connection, we can restrict ourselves to looking for minima on the subset

$$\mathscr{M}_{2,B}^p := \left\{ \boldsymbol{\zeta} \in \mathscr{M}_2^p : ||\mu(\exp(\boldsymbol{\zeta}) \cdot A)||_{L_2^p}^p \leq B \right\}$$

for some positive constant B. We see the moment map as taking values in $\mathrm{Lie}(\mathscr{G}_{k+1})$ to make sense of the norm (cf. remark 4.11), that is

$$||\mu(\exp(\boldsymbol{\zeta}) \cdot A)||_{L_2^p}^p = ||\Lambda_\omega F_{\exp_{\mathscr{G}_{k+1}}(\boldsymbol{\zeta}) \cdot A} + \mathrm{i}\, c||_{L_2^p}^p.$$

Finally, one shows that there is an element of $\mathscr{M}_{2,B}^p$ that minimizes Ψ_A and this minimizing element is smooth. Then, it corresponds to an equivalence class of the quotient $\mathscr{G}_{k+1}^{\mathbb{C}}/\mathscr{G}_{k+1}$, so there is a solution, and any two different solutions are related by a transformation of $\mathscr{G}_{k+1}$. This is rather technical, and it is the content of section [46, Sec. 6.2].

Conversely, choose a simple connection A and a gauge transformation $\psi \in \mathscr{G}_{k+1}^{\mathbb{C}}$ such that $\psi \cdot A$ satisfies (5.3). To show that A is stable, let $\zeta \in \mathrm{i}\,\mathfrak{k}$, and choose any holomorphic reduction of the structure group to P_ζ, $\sigma \in \mathcal{O}(X; \mathcal{P}(G/P_\zeta))$. By definition, one should check that $\deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta \rangle > 0$, however, [46, Lemma 4.3] shows that one may equivalently check that $\lambda(A; -\mathrm{i}\, g_{\sigma,\zeta}) > 0$, where $g_{\sigma,\zeta} \in \mathrm{i}\,\mathrm{Lie}(\mathscr{G}) = \Gamma(X; P_K \times_K \mathrm{i}\,\mathfrak{k})$ is fiberwise dual to the character χ_ζ with respect to the hermitian fiber product induced by $\langle \cdot, \cdot \rangle$ (see [46, Sec. 2.7]). We proceed by contradiction. First, one can show the following bound [46, Sec. 6.3]

$$t \sup |g_{\sigma,\zeta}| \leq C_1 \cdot \Psi_A(\exp_{\mathscr{G}_{k+1}}(t g_{\sigma,\zeta})) + C_2, \qquad C_1, C_2 > 0, \tag{5.4}$$

and for sufficiently large $t > 0$. Moreover, the following relation can be proven

$$\Psi_A(\exp_{\mathscr{G}_{k+1}}(t g_{\sigma,\zeta})) = \int_0^1 \lambda_{t \cdot \ell}(A; -\mathrm{i}\, g_{\sigma,\zeta})\, \mathrm{d}\ell.$$

If $\lambda(A; -\mathrm{i}\, g_{\sigma,\zeta}) < 0$, then the previous integral would be negative for every t (because λ_t increases with t). Finally, (5.4) would imply $t \cdot \sup |g_{\sigma,\zeta}| < C_2$, but the left-hand side tends to $+\infty$, so we get a contradiction.

We refer to [46] for the uniqueness claim. □

Since our goal is to use theorem 5.11 to prove theorem 5.10, we will need a way to translate the stability condition into a condition dealing with simplicity. We do this by means of the following theorem, which associates to each c-polystable principal bundle a c-stable principal bundle over a different group.

Theorem 5.12 (Jordan-Hölder reduction). Let $\mathcal{P}_G$ be a holomorphic principal G-bundle over a compact Riemann surface, and let $c \in \mathrm{i}\mathfrak{z}_K$. Then, there is a compact subgroup $K' \subset K$ and a holomorphic reduction of the structure group of $\mathcal{P}_G$ to $G' := (K')^{\mathbb{C}}$, that is c-stable. Any two such reductions $\mathcal{P}^{(1)}_{G'}$ and $\mathcal{P}^{(2)}_{G'} \subset \mathcal{P}_G$ are related by $\mathcal{P}^{(1)}_{G'} = \mathcal{P}^{(2)}_{G''} \cdot g$ for some $g \in G$.

Proof. The proof is not trivial at all. We refer the reader to section 2.10 of [16]. □

Sketch of proof of theorem 5.10. Suppose first that $\mathcal{P}_G$ is c-stable, and fix a reduction of the structure group to K, P_K. Then, one can show that it admits a simple connection $A \in \mathscr{A}(P_K)$. The holomorphic structure induced by A on P_G via the Chern-Singer correspondence coincides with that of $\mathcal{P}_G$. By theorem 5.11, there is $\varphi \in \mathscr{G}^{\mathbb{C}}$ such that $A' := \varphi \cdot A$ satisfies equation 5.1. Now, since A and A' are on the same $\mathscr{G}^{\mathbb{C}}$-orbit, they induce the same structure of holomorphic principal bundle on P_G (see (3.21)). Finally, A' is the Chern connection with respect to $\varphi \cdot J$, $J = \mathsf{CS}^{-1}(A)$, and $\varphi \cdot J$ induces the same structure of holomorphic principal bundle on P_G as J.

Suppose now that $\mathcal{P}_G$ is c-polystable, but not stable. We use theorem 5.12 and obtain a c-stable reduction of the structure group to some subgroup G'. Now, $\mathcal{P}_{G'}$ is stable, and the previous reasoning applies. Hence, there is a reduction of the structure group to $P_{K'}$ whose Chern connection satisfies (5.3). Now we define $P_K := P_{K'} \times_{K'} K \subset P_{K'} \times_{K'} G = P_G$ (cf. 3.21) for the action of K' on K by inverse left multiplication. This is a reduction of the structure group to K, and one can show that its Chern connection satisfies (5.1).

For the converse, one shows first that if $\mathcal{P}_G$ admits a reduction of the structure group to K satisfying (5.1), then it is c-semistable (see [9]). Let P_K be the corresponding K-bundle. Now, to prove polystability, let $\sigma \in \Gamma(X; P(G/K))$ be a reduction of the structure group satisfying (5.1). We know that the bundle is semistable, thus, for some parabolic subgroup $P_\zeta \subset G$, we consider a holomorphic reduction of the structure group $\sigma \in \mathcal{O}(X, \mathcal{P}(G/P_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) = \langle c, \zeta \rangle$. We seek a further restriction of the structure group to the Levi subgroup L_ζ. Let P_K be the reduction of the structure group whose Chern connection satisfies (5.1). This reduction produces a section $\boldsymbol{\zeta}$ of the bundle $P_K \times_K \mathrm{i}\mathfrak{k}$. Since $\mathrm{Lie}(\mathscr{G}) = \Gamma(X; P_K \times_K \mathfrak{k})$, we can identify $-\mathrm{i}\,\boldsymbol{\zeta}$ with an element of $\mathrm{Lie}(\mathscr{G})$. Now, let J be the almost complex structure of $\mathcal{P}_G$, and let $A = \mathsf{CS}(J) \in \mathscr{A}(P_K)$. One can see that the condition $\deg \mathcal{P}(\sigma, \zeta) = \langle c, \zeta \rangle$ is equivalent to

$$\lambda(A, -\mathrm{i}\,\boldsymbol{\zeta}) := \lim_{t \to \infty} \lambda_t(A, -\mathrm{i}\,\boldsymbol{\zeta}) = \lim_{t \to \infty} (\mu(\exp_{\mathscr{G}}(t\boldsymbol{\zeta}) \cdot A), -\mathrm{i}\,\boldsymbol{\zeta}) = 0.$$

From [46, Lemma 2.2], we know that the function $\lambda_t(A, -\mathrm{i}\,\boldsymbol{\zeta})$ is non-decreasing. Moreover, by the way the moment map μ is defined, equation (5.1) is equivalent to the vanishing of the moment map. Hence, at $t = 0$, we have $\mu(\exp_{\mathscr{G}}(t\boldsymbol{\zeta}) \cdot A)|_{t=0} = \mu(A)$, and the latter vanishes because the Chern connection A satisfies (5.1). Therefore, we have $\lambda_0(A, -\mathrm{i}\,\boldsymbol{\zeta}) = 0$. Now, the fact that the weight is non-decreasing implies that it is constant and equal to 0 for any $t \geq 0$. One may deduce from this fact that $\exp_{\mathscr{G}}(t\boldsymbol{\zeta}) \cdot J = J$ for all $t \geq 0$. This implies that $\boldsymbol{\zeta}$ induces a holomorphic section $\sigma_\zeta \in \mathcal{O}(M; \mathcal{P}_\sigma(P_\zeta / L_\zeta))$, hence a reduction of the structure group to L_ζ, and this is precisely the definition of polystability. □

5.2 The HK correspondence for non-connected groups

This section contains the most original part of the document. It formulates a Hitchin-Kobayashi correspondence for principal bundles with non-connected complex reductive

structure groups. Most of the section is based on the unpublished article [8] that my supervisor kindly shared with me. His work is more general, but with his ideas, we try to outline a generalization of the theorems by Narasimhan and Seshadri [43] and Ramanathan [44]. This section relies heavily on the concepts introduced in appendix B.3. We will not give a detailed proof of the non-connected version of the correspondence, but we will see how it can be reduced to a statement concerning a twisted principal bundle over a covering of the original manifold. We will not give a proof of the latter. To sum up, what this section intends to do is to formulate a version of the Hitchin-Kobayashi for bundles with non-connected structure groups in terms of an equivalent statement concerning a twisted principal bundle on a covering. We sketch a proof of the fact that these two statements are equivalent.

For the whole section we fix a complex reductive Lie group $\hat{G}$, and $\hat{K} \subset \hat{G}$ a maximal compact subgroup. K and G will be their respective identity components. We also set $\Gamma := \hat{G}/G \cong \hat{K}/K$ the group of connected components. Then, K is a maximal compact subgroup of G. We fix $\langle \cdot, \cdot \rangle$ a non-degenerate $\mathbb{C}$-bilinear pairing on $\mathfrak{g}$ invariant under the adjoint $\hat{G}$ action. If $\hat{G}$ is semisimple, then the Killing form $\kappa_{\mathfrak{g}}$ does the job. If $\hat{G}$ is only reductive, an invariant pairing still exists [6]. Indeed, reductive Lie algebras can be decomposed as $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. On the semisimple part, the Killing form gives a $\mathbb{C}$-bilinear, $\hat{G}$-invariant, non-degenerate pairing. On the center, we may choose *any* non-degenerate bilinear form, and it will be invariant under the adjoint action because the center is abelian. The direct sum of these forms yields the desired pairing. If a $\hat{G}$-invariant pairing exists, we can make it invariant under the Γ-action on $\mathfrak{g}$ (cf. equation (B.10)) through a standard averaging procedure, that is, we define a new $\mathbb{C}$-bilinear pairing:

$$\langle \cdot, \cdot \rangle' := \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \langle \cdot \gamma, \cdot \gamma \rangle.$$

Observe that Γ is finite, hence so is the sum. Recall from the appendix that the action of Γ on G and K is given by a lift of the characteristic homomorphism $\bar{\theta} \colon \Gamma \to \mathrm{Out}(K)$, and its extension to G. In particular, K is invariant under the Γ action. Since the differential of any Lie group homomorphism is a Lie algebra homomorphism, the induced action on $\mathfrak{g}$ leaves the Lie algebra of $Z(K)$, $\mathfrak{z}_K$, invariant. Indeed, given $\zeta \in \mathfrak{z}_K, \xi \in \mathfrak{k}$, we have

$$\big[\mathrm{d}_e\theta_\gamma(\zeta), \xi\big] = \mathrm{d}_e\theta_{\gamma^{-1}}\left(\big[\zeta, \mathrm{d}_e\theta_\gamma(\xi)\big]\right) = \mathrm{d}_e\theta_\gamma(0) = 0. \tag{5.5}$$

5.2.1 The hermitian Yang-Mills equation revisited

We saw in section 4.3 how the hermitian Yang-Mills equation should be generalized to principal bundles with non-connected structure groups. Let $\mathcal{P}_{\hat{G}} \to X$ be a holomorphic principal $\hat{G}$-bundle, and consider a reduction $P_{\hat{K}}$ of the structure group to $\hat{K}$. We say that $P_{\hat{K}}$ is an *a-hermitian Yang-Mills reduction of the structure group* for $a \in \Gamma(X; P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\,\mathfrak{z}_K)$ satisfying $\mathrm{d}_A a = 0$ if the Chern connection of the reduction satisfies the a-hermitian Yang-Mills equation, that is

$$\Lambda_\omega F_A = -\mathrm{i}\, a. \tag{5.6}$$

Equation (5.6) should be understood as an equality between sections of the vector bundle $P_{\hat{K}} \times_{\hat{K}} \mathfrak{k}$ (observe that it contains $P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K$ as a vector subbundle).

Sometimes, equation (5.6) can be written in a similar manner to equation (5.1), by which we mean that it may be written with a *constant section* on the right-hand side of the equation even if $\hat{G}$ is *not connected*. This is related to the concept of *monodromy*, which we present briefly. Consider $p \colon Y \to X$ a Galois Γ-covering between topological spaces,

and suppose that X is path-connected. Fix a basepoint $x \in X$. The fundamental group $\pi_1(X,x)$ acts on $p^{-1}(x)$ (which is finite) from the right as follows: for every equivalence class of loops $[\beta] \in \pi_1(X,x)$, and $y \in p^{-1}(x)$, we define $y \cdot [\beta] := \tilde{\beta}_y(1)$, where $\tilde{\beta}$ is the unique lift of any representative β of $[\beta]$ whose value at zero is y. Since $\tilde{\beta}$ is a lift, $\beta(1) \in p^{-1}(x)$, and everything is well-defined (see [**35**, Th. 11.22]). This is what we call the *monodromy action*. The *monodromy* of the covering is the morphism of groups $\pi_1(X,x) \to \mathrm{Sym}(Y_0)$ where $Y_0 := p^{-1}(x)$ and $\mathrm{Sym}(Y_0)$ is the group of permutations of its elements. The image M of this morphism is what we call the *monodromy group*. Different choices of the basepoint give rise to isomorphic monodromy groups. Finally, whenever we have a principal $\hat{G}$-bundle $P_{\hat{G}} \to X$, we define its monodromy as the monodromy of the covering $Y := P_{\hat{G}}/G \to X$. The covering space Y is connected if and only if the monodromy is surjective.

Remark 5.13. If $\hat{G}$ is not connected but $\mathcal{P}_{\hat{G}}$ has surjective monodromy, then the section $a \in \Gamma(X; P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\mathfrak{z}_K)$ can be identified with a single element $\alpha \in \mathrm{i}\mathfrak{z}_K^\Gamma$. Then, solving (5.6) is equivalent to solving $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$. To see why, first observe that, since $\hat{G}$ acts on $P_{\hat{G}}$, proposition B.21 implies that there is a natural Γ-action on $Y := P_{\hat{G}}/G$. In fact, $Y \to X$ has the structure of a principal Γ-bundle. We thus have a natural vector bundle isomorphism given by

$$P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K \longrightarrow Y \times_\Gamma \mathfrak{z}_K,\ [p,\zeta] \mapsto [\hat{\pi}(p),\zeta]$$

where $\hat{\pi}\colon P_{\hat{G}} \to Y$ is the natural projection (which we restrict to $P_{\hat{K}}$). Moreover, the fact that $\mathrm{d}_A\alpha = 0$ means that the section a is parallel along every curve in X. What this means, in particular, is that the image of a at any point is determined uniquely by its image at one point by parallel transport (see [**33**, Prop. 7.6]). Now, fix any $x_0 \in X$, and let Y_0 be as in the comments above. a descends to a Γ-equivariant map $\delta_a\colon Y_0 \to \mathfrak{z}_K$ given by $\delta_a(y_0) = \xi_0$ if and only if $a(x_0) = [y_0,\xi_0]$ (in fact, δ_a can be defined in all Y using the same argument). Since Y_0 is a Γ-orbit, and $[y_0\cdot\gamma, \xi_0\cdot\gamma] = [y_0,\xi_0]$ we see that the map is indeed Γ-equivariant. If we show that δ_a is constant, its image must be Γ-invariant, that is, an element of $\mathfrak{z}_K^\Gamma$. Hence, we will be able to identify δ_a with some $\alpha \in \mathfrak{z}_K^\Gamma$ and conversely, given α we will be able to recover a using parallel transport. The map δ_a descends to a Γ-equivariant map $M \setminus Y_0 \to \mathfrak{z}_K$. This follows from the fact that $\mathrm{d}_A a = 0$ forces δ_a to be constant on each connected component of Y, and the quotient $M \setminus Y_0$ parametrizes connected components of Y_0. Since the monodromy action is defined by liftings of paths of the fundamental group, y and $y\cdot[\beta]$ are on the same connected component. Hence, δ_a takes the same value on both of them. If the monodromy is surjective, then $M = \mathrm{Sym}(Y_0)$, so $M \setminus Y_0$ is a point. Then the action of Γ is trivial, and since the map is Γ-equivariant, its image must be an element $\alpha \in \mathrm{i}\mathfrak{z}_K$ that is Γ-invariant. That is $\alpha \in \mathrm{i}\mathfrak{z}_K^\Gamma$.

A reduction of the structure group to $\hat{K}$ satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$ is called an *α-hermitian Yang-Mills reduction of the structure group*. Observe that α defines a constant section. Indeed, for every $\hat{k} \in \hat{K}$, there is $\gamma \in \Gamma$ such that $\mathrm{Ad}_{\hat{k}}(\alpha) = \mathrm{d}_e\theta_\gamma(\alpha) = \alpha$.

5.2.2 Stability condition

The first step towards the generalization of the correspondence is to give a notion of stability. In fact, we will need two notions: one for principal bundles with non-connected structure group, and another for twisted equivariant bundles.

Observe that the statement of lemma 5.2 holds in the non-connected case. Moreover, if the pairing $\langle\cdot,\cdot\rangle$ is Γ-invariant, and $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$, then χ_ζ is Γ-invariant, i.e. $\chi_\zeta(\xi\cdot\gamma) = \chi_\zeta(\xi)$ for all $\xi \in \mathfrak{g}$. Note as well that the definition of the degree of a holomorphic principal bundle with respect to a reduction and an element of $\mathrm{i}\mathfrak{k}$ applies verbatim.

Before giving the definition of stability, observe that if we choose $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$, then the parabolic subgroup P_ζ defined by proposition B.10 is a Γ-invariant subgroup of G (see [8, Prop. 2.12]). From proposition B.24 and the comments below, we deduce that G/P_ζ is a (θ, c)-twisted (G, Γ)-equivariant manifold. Finally, by proposition B.22, we deduce that, given a (θ, c)-twisted Γ-equivariant bundle, $P_G \to Y$, the associated bundle $P_G(G/P_\zeta) \to Y$ is a Γ-equivariant principal bundle. Hence, we may consider Γ-equivariant sections or, equivalently, Γ-equivariant reductions of the structure group to P_ζ. Similarly, consider $L_\zeta \subset P_\zeta$. It is Γ-invariant, and given a Γ-invariant reduction of the structure group of P to P_ζ, call it P_σ, we know from proposition B.25 that P_σ is endowed with the structure of a (θ, c)-twisted Γ-equivariant principal P_ζ-bundle. Since $L_\zeta \subset P_\zeta$ is also Γ-invariant, and $\hat{L}_\zeta \subset \hat{P}_\zeta$ meets all the connected components of the latter, we have an isomorphism $P_\zeta/L_\zeta \cong \hat{P}_\zeta/\hat{L}_\zeta$. Thus, the same arguments apply, $P_\sigma(P_\zeta/L_\zeta)$ is a Γ-equivariant principal bundle, and it makes sense to consider Γ-equivariant reductions of the structure group to L_ζ. With this in mind, we are now ready to define stability for twisted principal bundles.

Definition 5.14. Let $\mathcal{P}$ be a (θ, c)-twisted Γ-equivariant principal bundle over a Riemann surface Y, and let $\alpha \in \mathrm{i}\mathfrak{z}_K^\Gamma$. We say that $\mathcal{P}$ is

1. Γ-*invariantly* α-*stable (resp. semistable)* if for any $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$ and any Γ-invariant holomorphic section $\sigma \in \mathcal{O}(Y; \mathcal{P}(G/P_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle > 0 \qquad (\text{resp.} \quad \deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle \geq 0)\,.$$
2. α-*invariantly* Γ-*polystable* if it is Γ-invariantly α-semistable and, for any $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$ and any Γ-invariant holomorphic section $\sigma \in \mathcal{O}(Y; \mathcal{P}(G/P_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle = 0$, there is a Γ-invariant holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(P_\zeta/L_\zeta))$.

Now we need a notion of stability for principal bundles with non-connected structure group. The definition we are about to give is quite similar to the definition for twisted equivariant principal bundles, and as we shall see, these definitions are closely related.

Definition 5.15. Let X be a compact Riemann surface, and let $\mathcal{P}(X, \hat{G})$ be a holomorphic principal $\hat{G}$-bundle with surjective monodromy, and let $\alpha \in \mathrm{i}\mathfrak{z}_K^\Gamma$. We say that the principal bundle is

1. α-*stable (resp. semistable)* if for any $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$ and any holomorphic section $\sigma \in \mathcal{O}(X; \mathcal{P}(\hat{G}/\hat{P}_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle > 0 \qquad (\text{resp.} \quad \deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle \geq 0)$$
2. α-*polystable* if it is α-semistable and for any $\zeta \in \mathrm{i}\mathfrak{k}^\Gamma$ and $\sigma \in \mathcal{O}(X; \mathcal{P}(\hat{G}/\hat{P}_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle = 0$, there is a holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(\hat{P}_\zeta/\hat{L}_\zeta))$.

When $G = \hat{G}$ is connected, we recover the usual stability criteria (see definition 5.8).

5.2.3 The theorem

The aim of this section is to state the version of the correspondence for principal bundles with non-connected structure group and sketch how it can be reduced to a Γ-equivariant correspondence. We fix a complex reductive non-connected Lie group $\hat{G}$, its identity

component G, $\hat{K}$ a maximal compact subgroup, and $K := \hat{K} \cap G$ its identity component. We thus have the following diagram of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & G & \longrightarrow & \hat{G} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \uparrow & & \uparrow & & \| & & \\ 1 & \longrightarrow & K & \longrightarrow & \hat{K} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array}$$

We also fix a Riemann surface X, and a lift of the characteristic homomorphism $\theta\colon \Gamma \to \mathrm{Aut}(K)$ (which we extend to a lift to $\mathrm{Aut}(G)$), and a two-cocycle $c\colon \Gamma \times \Gamma \to Z(K)$ as in appendix B. Under these hypotheses, the version of the Hitchin-Kobayashi correspondence for principal bundles with non-connected structure group reads as follows.

Theorem 5.16 (Hitchin-Kobayashi, non-connected version). Let $\alpha \in \mathrm{i}\,\mathfrak{z}_K^\Gamma$. A holomorphic principal $\hat{G}$-bundle $\mathcal{P}_{\hat{G}}$ admits a reduction of the structure group to $\hat{K}$, say $P_{\hat{K}}$, satisfying equation (5.6) for a section $a \in \Gamma(X; P_{\hat{K}} \times_K \mathfrak{z}_K)$ satisfying $\mathrm{d}_A a = 0$, where A is the Chern connection in $\hat{K}$ if and only if it is α-polystable for $\alpha \in \mathrm{i}\,\mathfrak{z}_K$ defined as in remark 5.13.

Our strategy for the proof is to translate the problem into one concerning principal bundles with connected structure groups. This is done by means of the correspondence of categories given by theorem B.26. More explicitly, given a holomorphic principal $\hat{G}$-bundle over X, $\mathcal{P}_{\hat{G}}$, we define $Y := \mathcal{P}_{\hat{G}}/G$, and consider the quotient map $\mathcal{P}_{\hat{G}} \to Y$, which endows $\mathcal{P}_{\hat{G}}$ with the structure of a (θ, c)-twisted Γ-equivariant principal G-bundle over Y. We denote it $\mathcal{P}_G$, even though, as a set, and as a manifold, it coincides with $\mathcal{P}_{\hat{G}}$. We will suppose that the monodromy of the covering is surjective, so that Y is connected and remark 5.13 applies. If the monodromy is not surjective, one may consider intermediate reductions of the structure group to make it connected, but we take the surjectivity as an assumption and not delve deeper into the issue.

To prove theorem 5.16, the idea is to show that it is equivalent to the following theorem.

Theorem 5.17 (Hitchin-Kobayashi for twisted bundles). Let $P_G \to X$ be a (θ, c)-twisted Γ-equivariant principal G-bundle, and let $\alpha \in \mathrm{i}\,\mathfrak{z}_K^\Gamma$. Then, P is Γ-invariantly α-polystable if and only if there exists a Γ-equivariant reduction of the structure group $\sigma \in \Gamma(X; P_G(G/K))$ satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$, where A is the Chern connection on P_K.

A Γ-equivariant reduction of the structure group satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$ is called a *Γ-equivariant α-hermitian Yang-Mills reduction of the structure group* to K. Moreover, observe that a Γ-invariant reduction of the structure group is, in particular, a reduction of the structure group, hence it makes sense to consider its Chern connection.

We are not claiming that theorem 5.17 is straightforward, or a direct consequence of the Hitchin-Kobayashi correspondence for principal bundles with connected structure group because we have to take into account the twisted structure somehow. However, since we are now working with a principal bundle with connected structure group over a connected Riemann surface, it seems reasonable to think that a proof in terms of a functional Ψ, as we did in section 5.1.2 will work here as well.

It remains to sketch how theorem 5.16 can be reduced to theorem 5.17. The first

thing one should check is that the condition of α-polystability for the bundle $\mathcal{P}_{\hat{G}} \to X$ is equivalent to the condition of α-polystability of the twisted equivariant bundle $P_G \to Y$. More precisely:

Proposition 5.18. Let $\mathcal{P}_{\hat{G}}$ be a holomorphic principal $\hat{G}$-bundle, and let $\mathcal{P}_G$ be the corresponding (θ, c)-twisted Γ-equivariant principal bundle over $Y := \mathcal{P}_{\hat{G}}/G$ given by theorem B.26. Let $\alpha \in \mathrm{i}\mathfrak{z}_K^{\Gamma}$, then $\mathcal{P}_{\hat{G}}$ is α-polystable if and only if $\mathcal{P}_G$ is Γ-invariantly α-polystable.

Sketch of the proof. Fix $\zeta \in \mathrm{i}\mathfrak{k}$. One may show that $G/P_\zeta \cong \hat{G}/\hat{P}_\zeta$ and that there is a bijection between holomorphic sections of the bundle $\mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta) \to X$ (that is, holomorphic reductions of the structure group to $\hat{P}_\zeta$), and Γ-equivariant sections of the bundle $\mathcal{P}_G(G/P_\zeta) \to Y$ (that is, Γ-equivariant reductions of the structure group to P_ζ). Call σ the section, under some abuse of notation, we will use the same notation for a section seen from both sides of the previous bijection. A similar discussion applies to the Levi subgroup, which yields a bijection between holomorphic sections of $(\mathcal{P}_{\hat{G}})_\sigma(\hat{P}_\zeta/\hat{L}_\zeta) \to X$ and Γ-equivariant sections of $(\mathcal{P}_G)_\sigma(P_\zeta/L_\zeta) \to Y$. This reasoning is used to show that $\deg \mathcal{P}_G(\sigma, \zeta) = \deg \mathcal{P}_{\hat{G}}(\sigma, \zeta)$, and from this identity the claim follows directly. □

Finally, one shows that $\mathcal{P}_{\hat{G}} \to X$ admits an α-hermitian Yang-Mills reduction of the structure group to $\hat{K}$ if and only if $P_G \to Y$ admits a Γ-equivariant α-hermitian Yang-Mills reduction of the structure group to K.

5.2.4 Moment map picture for the non-connected version

This version of the correspondence also has a moment map interpretation. Let $P_{\hat{G}}$ be a smooth principal $\hat{G}$-bundle, and consider a reduction of the structure group $P_{\hat{K}}$ to a principal $\hat{K}$-bundle. By the Chern-Singer correspondence, each $A \in \mathscr{A}\left(P_{\hat{K}}\right)$ defines a holomorphic structure on $P_{\hat{G}}$. Let $\mathscr{G}$ be the gauge group of $P_{\hat{K}}$. Upon completing these spaces with Sobolev norms, $\mathscr{A}\left(P_{\hat{K}}\right)$ and $\mathscr{G}$ are a Banach manifold and a Banach-Lie group respectively. Then, given a section $a \in \Gamma(X; P_{\hat{K}} \times_K \mathfrak{z}_K)$ satisfying $\mathrm{d}_A a = 0$, the action of $\mathscr{G}$ on $\mathscr{A}\left(P_{\hat{K}}\right)$ given by (4.5) is Hamiltonian, and the associated moment map is

$$\begin{array}{rccc} \mu\colon & \mathscr{A}\left(P_{\hat{K}}\right) & \longrightarrow & \mathrm{Lie}(\mathscr{G}) \\ & A & \mapsto & \Lambda_\omega F_A + \mathrm{i}\, a \end{array}$$

is a moment map (we are identifying $\mathrm{Lie}(\mathscr{G})$ and its dual as in remark 4.11). Hence, the moduli space of solutions to the hermitian Yang-Mills equations for $P_{\hat{G}}$ is given by

$$\mu^{-1}(0)/\mathscr{G}.$$

We could make these constructions rigorous as we did for the connected version.

5.3 Conclusions and future work

The aim of this work has been to present an account of the Hitchin-Kobayashi correspondence for principal bundles with connected structure groups and to set the bases of a version of the correspondence for principal bundles with non-connected structure groups. To that end, we have worked mostly from a differential-geometric viewpoint. Aside from the preliminaries and prerequisites we have covered in the appendices and chapters 1 and 3, we have presented a somewhat detailed and self-contained exposition of the principal bundle version of the correspondence. These are relatively standard

results today, and they are part of the shared mathematical knowledge of the research community, and we do not claim any originality of the results presented. However, there are not many sources from which one could try to get acquaintance with the topic aside from the original papers and some scattered lecture notes. We hope this work may be useful for future readers trying to learn the material. We shall remark, however, that we have not discussed the algebraic point of view. Recall that the notion of stability arises from Mumford's GIT, but we have presented all our definitions from a differential-geometric point of view. We have also not covered with many details how these results may be phrased rigorously in terms of moduli spaces (aside from our comments in section 5.1.1).

Our second objective was to lay the foundations for a proof of a correspondence in the non-connected case. We have not given a complete proof of the result because we do not have a proof of 5.17 and, to our knowledge, this result is not proven in the literature. However, we think that reducing the problem to the twisted version is quite an advancement as the twisted version looks much more manageable. Indeed, it seems quite reasonable to think that the techniques outlined in section 5.1.2 may be applied to obtain a proof of the twisted version.

To sum up, since the publishing of the theorem by Narasimhan and Seshadri in 1965, the discipline has evolved significantly, and several generalizations of the correspondence have appeared in recent years. We hope that the results we present here will be of interest to improve our understanding of the Hitchin-Kobayashi correspondence.

Appendix A

Vector bundles and complex manifolds

This appendix gives a summary of the theory of complex and Kähler manifolds, as well as the theory of smooth and holomorphic vector bundles. We refer the reader to the books [33] and [24] and [34, Ch. 10] for proofs. We also introduce complex differential forms, and bundle-valued differential forms. Let us set the notations $\mathrm{i} := \sqrt{-1}$ and $\Omega^p_{\mathbb{R}}(M)$ for the $\mathscr{C}^\infty(M;\mathbb{R})$-module of differential p-forms on a smooth manifold M. Finally, whenever we have a product of spaces $A \times B$ we will write $\pi_A \colon A \times B \to A$ for the projection.

A.1 Complex manifolds

Let $W \subset \mathbb{C}^n$ be open and $f \colon W \to \mathbb{C}^m$ a continuous map defined on W. If, for each $p \in W$, the following limit exists

$$\frac{\partial f}{\partial z^j} := \lim_{h\to 0} \frac{f(p^1,\dots,p^j+h,\dots,p^n) - f(p^1,\dots,p^n)}{h} \tag{A.1}$$

then we say that f is holomorphic in W. The Cauchy-Riemann equations still give a characterization of holomorphicity (but now they have to be satisfied for all coordinates). We define $\partial_{z^j} := \partial/\partial z^j := 1/2 \cdot (\partial_{x^j} - \mathrm{i}\,\partial_{y^j})$ and $\partial_{\bar{z}^j} := \partial/\partial \bar{z}^j := 1/2 \cdot (\partial_{x^j} + \mathrm{i}\,\partial_{y^j})$. Observe that $\partial_{z^j} f = \partial_{x^j} f = -\mathrm{i}\,\partial_{y^j} f$ which coincides with the limit in equation (A.1) when calculated at a point where f is holomorphic. We will use, when needed, the identification $\mathbb{R}^{2n} \equiv \mathbb{C}^n$ via $(x^1, y^1, \dots, x^n, y^n) \leftrightarrow (x^1 + \mathrm{i}\,y^1, \dots, x^n + \mathrm{i}\,y^n)$, and we set $z^j := x^j + \mathrm{i}\,y^j$. The definition of a complex manifold is the obvious modification of the definition of a smooth manifold:

Definition A.1. A *complex manifold* M is a second countable Hausdorff topological space, endowed with a complex atlas, that is, a family $\mathscr{A}$ of pairs (complex charts) (U,φ) where each $U \subset M$ is open, the open sets cover M, $\varphi \colon U \to \mathbb{C}^n$ is a homeomorphism, and whenever $(U,\varphi),(V,\psi) \in \mathscr{A}$ satisfy, $U \cap V \neq \varnothing$ the maps,

$$\psi \circ \varphi^{-1} \colon \varphi(U \cap V) \to \psi(U \cap V) \qquad \varphi \circ \psi^{-1} \colon \psi(U \cap V) \to \varphi(U \cap V)$$

are holomorphic.

Then, a map between complex manifolds $f \colon M \to N$ is holomorphic, if its *localization* is holomorphic, that is, for all charts (U,φ) and (V,ψ) of M and N respectively, the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic. A holomorphic bijective map whose inverse is also holomorphic is called a *biholomorphism*. Whenever M is a complex manifold, $\mathcal{O}(U)$ denotes the ring of all holomorphic functions on the open set U. Observe that all complex manifolds are even-dimensional when considered as real manifolds. Moreover, one may show that complex manifolds are always orientable (see [33, Prop. 1.49] for a proof). The concepts of *submersion, immersion* and *embedding* admit direct generalizations to complex man-

ifolds: a holomorphic map is said to have either of these properties if it has them when considered as a smooth map between the smooth underlying manifolds.

A.2 Vector bundles

We will use both smooth and holomorphic vector bundles, but given how similar these concepts are, we will introduce them simultaneously. To do so, we shall mimic the notation used in [15]. We will use the symbol $\mathcal{S}$ to denote either smooth or holomorphic functions, sections, maps, etc. For example, $\mathcal{S}(M)$ denotes either the set of smooth functions on a manifold, $\mathscr{C}^\infty(M;\mathbb{K})$ ($\mathbb{K}=\mathbb{R}$ or $\mathbb{C}$, depending on the context), or the set of holomorphic functions on a complex manifold, $\mathcal{O}(M)$. Similarly, an $\mathcal{S}$-manifold is either a smooth manifold, or a holomorphic one, an $\mathcal{S}$-morphism between $\mathcal{S}$-manifolds is either a smooth map, or a holomorphic one, and an $\mathcal{S}$-isomorphism is either a diffeomorphism or a biholomorphism.

Definition A.2 (Vector bundle). Let M be an $\mathcal{S}$-manifold. A real (or complex) $\mathcal{S}$-vector bundle of rank k over M is another $\mathcal{S}$-manifold, E, together with a surjective $\mathcal{S}$-morphism $\pi\colon E\to M$ satisfying the following conditions:

1. For each $p\in M$, $E_p:=\pi^{-1}(p)$ is a k-dimensional real (or complex) vector space.
2. For each $p\in M$ there is an open neighborhood U of p, and an $\mathcal{S}$-isomorphism $\Phi\colon \pi^{-1}(U)\to U\times\mathbb{R}^k$ (or $\Phi\colon \pi^{-1}(U)\to U\times\mathbb{C}^k$), called a *local trivialization*, satisfying the following conditions:
   (a) $\pi_U\circ\Phi=\pi$, where $\pi_U\colon U\times\mathbb{R}^k\to U$ is the usual projection;
   (b) $\Phi|_{E_q}\colon E_q\to\{q\}\times\mathbb{R}^k$ (or $\Phi|_{E_q}\colon E_q\to\{q\}\times\mathbb{C}^k$) is a linear isomorphism.

The space M is often called the *base space*, and E is called the *total space*. The map $\pi\colon E\to M$ is called the *projection*. Let us make clear that the field may still be $\mathbb{C}$ even when we consider smooth not holomorphic bundles.

It is often the case that we want to elevate some linear constructions to vector bundles. To name a few examples, we want to have tensor products, direct sums and quotients of vector bundles, by which we mean vector bundles whose fiber at each point is a tensor product, direct sum or quotient of fibers. The set on which we define those structures is the disjoint union of the fibers. For example, given two vector bundles E,F over the same base manifold M, we define the tensor product bundle

$$E\otimes F:=\bigsqcup_{m\in M}E_m\otimes F_m.$$

Thus far, we only have a set; we need a topology, a manifold structure and a vector bundle structure. These can be obtained for all the usual algebraic constructions, and we shall assume this without further comment throughout the text. We refer the reader to [34, Lemma 10.6] for a way to rigorously endow these sets with bundle structures.

Example A.3. Some algebraic constructions that can be elevated to vector bundles are:

1. The tensor product of bundles as constructed above. It can be extended to more than two bundles.
2. Given a vector bundle E, we can construct its *dual bundle* as the bundle whose fiber at each point $m\in M$ is the dual E_m^* of E_m.

3. Given two vector bundles E and F one may construct their *Whitney sum*, i.e. the vector bundle $E \oplus F$ whose fiber at each point $p \in M$ is $E_p \oplus F_p$. See [34, Example 10.7] for an explicit construction.

There is a natural notion of subbundle.

Definition A.4 (Subbundle). Let $\pi\colon E \to M$ be an $\mathcal{S}$-bundle over an $\mathcal{S}$-manifold M. An $\mathcal{S}$-subbundle is an embedded submanifold $D \subset E$ such that $D_p := D \cap E_p$ is a vector subspace of E_p for each $p \in M$, all of them of the same dimension, together with the restriction $\pi|_D\colon D \to M$, such that $\pi|_D\colon D \to M$ is a vector bundle.

In the usual tangent bundle, vector fields are thought of as maps sending each point to a vector and varying smoothly. More formally, they are sections of TM. The analogous concept for vector bundles is that of a section.

Definition A.5 (Section). Let $\pi\colon E \to M$be an $\mathcal{S}$-bundle. A *global section* of E is a continuous map $\sigma\colon M \to E$ such that $\pi \circ \sigma = \mathrm{id}_M$. A local section is a continuous map defined on an open set $U \subset M$, $\sigma\colon U \to E$ such that $\pi \circ \sigma = \mathrm{id}_U$. $\mathcal{S}$-sections are sections that are $\mathcal{S}$-morphisms.

We will write σ_p or $\sigma(p)$ interchangeably for the value of the section at a point p. We will denote by $\Gamma(M;E)$ and $\Gamma(U;E)$ the sets of global and local smooth sections on a smooth manifold, and $\mathcal{O}(U;E)$ and $\mathcal{O}(M;E)$ the corresponding sets for holomorphic sections. Observe that $\Gamma(M;E)$ (the same applies to holomorphic sections) is a $\mathbb{K}$-vector space under pointwise addition and scalar multiplication. In fact, it is an $\mathcal{S}(M)$-module where the product of a function $f \in \mathcal{S}(M)$ and an $\mathcal{S}$-section is defined pointwise. A *local frame* on an open set U is a set of sections $\{s_1, \dots, s_r\}$ such that $\{s_1(p), \dots, s_r(p)\}$ is a basis of E_p for each $p \in U$.

A.3 The complexified Tangent and Cotangent Bundles

Observe that under the identification $\mathbb{R}^{2n} \equiv \mathbb{C}^n$, every complex manifold is a smooth manifold. Hence, it makes sense to speak about vector fields and covector fields (one-forms) on any complex manifold. However, we would like to extend those definitions to complex vector fields and one-forms (we will introduce higher-order complex differential forms later on).

A.3.1 Complexification of real vector spaces

Firstly, given a real vector space V, we may always consider its *complexification*, which is the complex vector space $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$. Its elements can always be written as $u \otimes 1 + v \otimes \mathrm{i}$, and we shall omit the tensor product and simply write $u + \mathrm{i}\, v \in V_{\mathbb{C}}$. Every $\mathbb{R}$-linear map $f\colon V \to W$ defines a $\mathbb{C}$-linear map $f_{\mathbb{C}}\colon V_{\mathbb{C}} \to W_{\mathbb{C}}$ (which we will usually denote f as well) by linear extension of $f_{\mathbb{C}}(u \otimes \alpha) := f(u) \otimes \alpha$ for $\alpha \in \mathbb{C}$. Complexifications are endowed with natural conjugation maps $\overline{u + \mathrm{i}\, v} := u - \mathrm{i}\, v$. This allows us to speak about real vectors, which are those $w \in V_{\mathbb{C}}$ for which $\overline{w} = w$. Observe that if we set $(V_{\mathbb{C}})^{\mathbb{R}}$ the underlying real subspace, then $(V_{\mathbb{C}})^{\mathbb{R}} \neq V$ (for example, $\mathbb{R}_{\mathbb{C}} \cong \mathbb{C}$, but $\mathbb{C}^{\mathbb{R}} \cong \mathbb{R}^2$). Moreover, one has $(V_{\mathbb{C}})^{\mathbb{R}} = V \oplus \mathrm{i}\, V$. We usually write $V_{\mathbb{C}} = V \oplus \mathrm{i}\, V$ making some abuse of notation, but the right-hand side is not a direct sum decomposition of the left-hand side as complex vector spaces.

A.3.2 Complexification of real vector bundles

Given a real vector bundle E over a smooth manifold M, the *complexification* $E_{\mathbb{C}} := E \otimes \mathbb{C}$ of E is the vector bundle whose fiber at each point is the complexification of E_p. This produces a complex (in general smooth) vector bundle over M (see [**33**, Ch. 1] for a more explicit construction). In particular, the *complexified tangent and cotangent bundles* are the complexifications of TM and T^*M which we denote $T_{\mathbb{C}}M$ and $T^*_{\mathbb{C}}M$ respectively. Complex vector and covector fields are just sections of these bundles. Observe that $T_{\mathbb{C}}M$ can be interpreted as a very natural object. Recall from basic differential geometry that for a smooth manifold, T_pM is the vector space of derivations acting on (germs of) smooth real-valued maps. Then, there is a complex vector space isomorphism between $T_pM \otimes_{\mathbb{R}} \mathbb{C}$ and the space of $\mathbb{C}$-linear derivations acting on germs of complex-valued smooth maps. What that means for practical purposes is that, given $Z = X + \mathrm{i}\, Y \in \Gamma(M; T_{\mathbb{C}}M)$ and $f = u + \mathrm{i}\, v \in \mathscr{C}^\infty(M; \mathbb{C})$, we have

$$Z(f) = (X(u) - Y(v)) + \mathrm{i}\, (Y(u) + X(v))\,.$$

We can define a Lie bracket on $\Gamma(M; T_{\mathbb{C}}M)$ by complex-linear extension of the original Lie bracket on TM which turns $(\Gamma(M; T_{\mathbb{C}}M), [\cdot, \cdot])$ into a Lie algebra. Similarly, elements of $\Gamma(M; T^*_{\mathbb{C}}M)$ are called complex one-forms. Since $V^*_{\mathbb{C}}$ is naturally isomorphic to $(V^*)_{\mathbb{C}}$ (see [**24**, Lemma 1.2.6]), for each $p \in M$, we have that the fiber of $T^*_{\mathbb{C}}M$ at p, $T^*_pM \otimes_{\mathbb{R}} \mathbb{C}$ is naturally isomorphic to $(T_pM \otimes_{\mathbb{R}} \mathbb{C})^*$. In the language of vector bundles, this means that $T^*_{\mathbb{C}}M \cong \mathrm{Hom}_{\mathbb{C}}(T_{\mathbb{C}}M, \mathbb{C})$. Then, the action of a complex one-form $\alpha + \mathrm{i}\, \beta$ on a complex vector field $X + \mathrm{i}\, Y$ is just given by $(\alpha + \mathrm{i}\, \beta)(X + \mathrm{i}\, Y) = (\alpha(X) - \beta(Y)) + \mathrm{i}\, (\alpha(Y) + \beta(X))$. Any $f \in \mathscr{C}^\infty(M; \mathbb{C})$ now defines a complex one-form $\mathrm{d}f := \mathrm{d}u + \mathrm{i}\, \mathrm{d}v$ (we will generalize the exterior differential when we introduce higher order complex differential forms). Finally, given a smooth map between complex manifolds $F\colon M \to N$, we can extend its differential $\mathrm{d}_m F$ to a map $(T_mM)_{\mathbb{C}} \to (T_{F(m)}N)_{\mathbb{C}}$ by complexifying it.

A.4 Almost complex structures

Let V be a real vector space. An endomorphism $J\colon V \to V$ is called an *almost complex structure* if $J^2 := J \circ J = -\mathrm{id}_V$. Almost complex structures receive their name because, when they exist, they endow V with the structure of a complex vector space. Indeed, for $a + \mathrm{i}\, b \in \mathbb{C}$ and $v \in V$ their product can be defined as $(a + \mathrm{i}\, b) * v := a \cdot v + b \cdot J(v)$. As seen in proposition A.6 below, only even-dimensional real vector spaces can have almost complex structures.

Given a real vector space endowed with an almost complex structure, the fact that $J^2 = -\mathrm{id}_V$ means that J has no real eigenvalues. If we extend J to the complexified space, then the only possible eigenvalues of J are i and $-\mathrm{i}$. If we use the symbol J acting on vectors of a complexified vector space, we will always understand that we are working with the complexified map.

Proposition A.6. Let J be a complex structure on a real finite-dimensional vector space V. Then $V_{\mathbb{C}}$ has a complete eigenspace decomposition $V_{\mathbb{C}} = V^{(1,0)} \oplus V^{(0,1)}$, where $V^{(1,0)}$ is the i-eigenspace and $V^{(0,1)}$ is the $(-\mathrm{i})$-eigenspace. Moreover, the decomposition is given, for $w \in V_{\mathbb{C}}$ by

$$w^{(1,0)} = \frac{1}{2} \cdot (w - \mathrm{i}\, J(w)) \qquad w^{(0,1)} = \frac{1}{2} \cdot (w + \mathrm{i}\, J(w)).$$

Furthermore, $\dim_{\mathbb{C}} V^{(1,0)} = \dim_{\mathbb{C}} V^{(0,1)}$, and therefore, $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$ is even.

Proof. See [33, Prop. 1.52]. □

Remark A.7. Let V be an even-dimensional real vector space endowed with an almost complex structure J_V. We have seen that this induces on V the structure of a complex vector space. On the other hand, the complexification $V_{\mathbb{C}}$ induces another linear complex structure on V. *These structures are not the same!* Indeed, $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V = 2 \cdot \dim_{\mathbb{C}}(V, J_V)$. Another way of seeing this is that for $v \in V^{(0,1)}$, $J_V(v)$ and $\mathrm{i}\, v$ are opposites, even though both of them can be thought of as "multiplying by the imaginary unit".

These constructions can be applied to vector bundles. Given a real vector bundle E over a smooth manifold, a vector bundle endomorphism $J\colon E \to E$ is an *almost complex structure* if $J^2 = -\mathrm{id}_E$. Every complex manifold admits a complex structure on TM, which is given, in terms of any holomorphic coordinate system $(z^1 = x^1 + \mathrm{i}\, y^1, \dots z^n = x^n + \mathrm{i}\, y^n)$ by equation $J(\partial_{x^j}) = \partial_{y^j}$ and $J(\partial_{y^j}) = -\partial_{x^j}$ (see [33, Prop. 1.55]). In that case, J induces a decomposition in the fibers of $T_{\mathbb{C}}M$ as in proposition A.6. This can be elevated to a vector bundle decomposition: indeed, for every complex manifold there is a Whitney sum decomposition

$$T_{\mathbb{C}}M = T^{(1,0)}M \oplus T^{(0,1)}M, \tag{A.2}$$

where the fibers at each point of $T^{(1,0)}M$ and $T^{(0,1)}M$ are the i and $(-\mathrm{i}\,)$-eigenspaces of J at each point. Moreover, local frames for those complex smooth subbundles are $\{\partial_{z^1}, \dots \partial_{z^n}\}$ and $\{\partial_{\bar{z}^1}, \dots \partial_{\bar{z}^n}\}$ respectively (we define them as in section A.1). The canonical almost complex structures on complex manifolds characterize holomorphicity of smooth maps as follows.

Theorem A.8. Let M, N be two complex manifolds, and let J_M and J_N be their canonical almost complex structures. Let $F\colon M \to N$ be a smooth map. Then, F is holomorphic if and only if $\mathrm{d}F \circ J_M = J_N \circ \mathrm{d}F$.

Proof. See [33, Prop. 1.55]. □

We have said that every complex manifold admits an almost complex structure, but the converse is false in general. There are even-dimensional manifolds that admit an almost complex structure, but not the structure of a complex manifold. Those almost complex structures that are actually induced by a complex manifold structure are called *integrable*. A very deep result, called the *Newlander-Nirenberg theorem* states that an almost complex structure is integrable if and only if the distribution $T^{(0,1)}M$ is integrable, i.e. $[T^{(0,1)}M, T^{(0,1)}M] \subset T^{(0,1)}M$.

A.5 The Dolbeault complex

An almost complex structure J on a vector space V induces an almost complex structure on its dual by $\big[J(f)\big](v) := f(J(v))$, which induces a natural decomposition on $V^*_{\mathbb{C}} = (V^*)^{(1,0)} \oplus (V^*)^{(0,1)}$ and there are isomorphisms $(V^*)^{(1,0)} \cong \big(V^{(1,0)}\big)^*$ and $(V^*)^{(0,1)} \cong \big(V^{(0,1)}\big)^*$. This decomposition carries over to the complexified cotangent bundle, and yields $T^*_{\mathbb{C}}M = (T^*M)^{(1,0)} \oplus (T^*M)^{(0,1)} \cong (T^{(1,0)}M)^* \oplus (T^{(0,1)}M)^*$. Moreover, local frames for $(T^*M)^{(1,0)}$ and $(T^*M)^{(0,1)}$ are given by $\{\mathrm{d}z^1, \dots \mathrm{d}z^n\}$ and $\{\mathrm{d}\bar{z}^1, \dots \mathrm{d}\bar{z}^n\}$ respectively. We have defined $\mathrm{d}z^j := \mathrm{d}x^j + \mathrm{i}\,\mathrm{d}y^j$ and $\mathrm{d}\bar{z}^j := \mathrm{d}x^j - \mathrm{i}\,\mathrm{d}y^j$.

Let M be a complex manifold, we define

$$\bigwedge\nolimits^{(p,q)}_{\mathbb{C}} M := \Big(\bigwedge\nolimits^p (T^{(1,0)}M)^*\Big) \otimes_{\mathbb{C}} \Big(\bigwedge\nolimits^q (T^{(0,1)}M)^*\Big) \qquad \text{and} \qquad \bigwedge\nolimits^k_{\mathbb{C}} M := \bigwedge\nolimits^k T^*_{\mathbb{C}}M.$$

A *complex (p,q)-differential form* is a section of this complex vector bundle. We will denote $\Omega^{(p,q)}_{\mathbb{C}}(M) := \Gamma(M; \bigwedge^{(p,q)}_{\mathbb{C}} M)$. Moreover, each $\bigwedge^{(p,q)}_{\mathbb{C}} M$ can be seen in a canonical way as a vector subbundle of $\bigwedge^{p+q} T^*_{\mathbb{C}} M$ by extending by linearity the map $\alpha \otimes \beta \mapsto \alpha \wedge \beta$. What is more, we actually have a vector bundle Whitney sum decomposition $\bigwedge^k_{\mathbb{C}} M = \oplus_{p+q=k} \left(\bigwedge^{(p,q)}_{\mathbb{C}} M \right)$. It carries over to a Whitney sum decomposition of the spaces of complex differential forms

$$\Omega^k_{\mathbb{C}}(M) = \bigoplus_{p+q=k} \Omega^{(p,q)}_{\mathbb{C}}(M).$$

There is another canonical isomorphism $\bigwedge^k_{\mathbb{C}} M \cong (\bigwedge^k T^*M) \otimes_{\mathbb{R}} \mathbb{C}$, which means that every complex k-form can be seen as a sum $\alpha + \mathrm{i}\,\beta$, where $\alpha, \beta \in \Omega^k_{\mathbb{R}}(M)$. In particular, this means that there is a conjugation map in the space of complex differential forms, which is given by $\overline{\alpha + \mathrm{i}\,\beta} := \alpha - \mathrm{i}\,\beta$. Finally, we also have $\Omega^k_{\mathbb{C}}(M) = \Omega^k_{\mathbb{R}}(M) \otimes_{\mathbb{R}} \mathbb{C}$.

Observe that every vector space can be embedded into its complexification as a real vector subspace $V \hookrightarrow V_{\mathbb{C}}, v \mapsto v \otimes 1$. In particular, we may see $\bigwedge^k T^*M$ as a real subbundle of $\bigwedge^k T^*_{\mathbb{C}} M = \bigwedge^k_{\mathbb{C}} M$. This means that we can see $\Omega^k_{\mathbb{R}}(M)$ as a real vector subspace of $\Omega^k_{\mathbb{C}}(M)$ by simply extending real forms to complex vectors by complex bilinearity.

Remark A.9. When M is a complex manifold, we have holomorphic coordinates $(z^1, \dots z^n)$ the local coframes $\{\mathrm{d}z^1, \dots, \mathrm{d}z^n\}$ and $\{\mathrm{d}\bar{z}^1, \dots, \mathrm{d}\bar{z}^n\}$ are available. More importantly, (p,q)-forms are characterized by the fact that they can be written as sums of the following terms

$$\omega = f \cdot \mathrm{d}z^{i_1} \wedge \dots \wedge \mathrm{d}z^{i_p} \wedge \mathrm{d}\bar{z}^{j_1} \wedge \dots \wedge \mathrm{d}\bar{z}^{j_q}, \tag{A.3}$$

for some $f \in \mathscr{C}^\infty(M; \mathbb{C})$.

A.5.1 Bundle-Valued forms

Let M be a smooth manifold, and let $E \to M$ be a smooth real vector bundle. A *real E-valued p-form* is a section of the real vector bundle $\bigwedge^p T^*M \otimes E$. We define

$$\Omega^p_{\mathbb{R}}(M; E) := \Gamma\left(M; \bigwedge^p T^*M \otimes E\right)$$

the space of all E-valued p-forms defined on M. Similarly, if E is a smooth complex vector bundle, we define a *complex E-valued k-form* as a section of the bundle $\bigwedge^k_{\mathbb{C}} M \otimes E$, and a *complex E-valued (p,q)-form* as a section of the bundle $\bigwedge^{(p,q)}_{\mathbb{C}} M \otimes E$. We define

$$\Omega^k_{\mathbb{C}}(M; E) := \Gamma\left(M; \bigwedge^k_{\mathbb{C}} M \otimes E\right) \qquad \text{and} \qquad \Omega^{(p,q)}_{\mathbb{C}}(M; E) := \Gamma\left(M; \bigwedge^{(p,q)}_{\mathbb{C}} M \otimes E\right)$$

Observe that an element $\sigma \in \Omega^k_{\mathbb{C}}(M; E)$ is a sum of elements $\omega \otimes \psi$, where $\omega \in \Omega^k_{\mathbb{C}}(M)$ and $\psi \in \Gamma(U; E)$. The Dolbeault direct sum decomposition of the space of complex k-forms implies that ω can be uniquely decomposed as a sum of (p,q)-forms satisfying $k = p + q$. This then implies that any E-valued k-form can be uniquely decomposed as a sum of E-valued (p,q)-forms. In conclusion, we have the following Whitney sum decomposition

$$\Omega^k_{\mathbb{C}}(M; E) = \bigoplus_{p+q=k} \Omega^{(p,q)}_{\mathbb{C}}(M; E). \tag{A.4}$$

By convention, we set $\Omega^0_{\mathbb{C}}(M; E) := \Gamma(M; E)$.

A.6 Connections on vector bundles

Let us briefly sketch the notion of a connection on a vector bundle and some of its properties. The notion of a connection itself is familiar from riemannian geometry, and the definition can be adapted to abstract vector bundles with minor modifications. We will consider the case of complex vector bundles, but everything applies to real vector bundles (we do not consider the complexified tangent and cotangent bundles in that case).

Definition A.10. Let $\pi\colon E \to M$ be a complex vector bundle, a *connection* on E is a $\mathbb{C}$-linear map $\nabla\colon \Gamma(E) \to \Gamma(T^*_{\mathbb{C}}M \otimes E)$ such that for every $f \in \mathscr{C}^\infty(M;\mathbb{C})$, $\sigma \in \Gamma(E)$, we have

$$\nabla(f\cdot\sigma) = \mathrm{d}f \otimes \sigma + f\cdot\nabla\sigma \tag{A.5}$$

Moreover, observe that $\Omega^1_{\mathbb{C}}(M;E) \cong \Gamma(\bigwedge^1 T^*_{\mathbb{C}}M \otimes E) = \Gamma(T^*_{\mathbb{C}}M \otimes E)$, hence a connection is a map $\nabla\colon \Omega^0_{\mathbb{C}}(M;E) \to \Omega^1_{\mathbb{C}}(M;E)$ satisfying the "Leibniz rule" of definition A.10.

The operator ∇ sometimes needs to be extended to act on higher-dimensional differential forms. Some references still call this operator ∇, but to make the distinction clear (for example, [26]), we will use the symbol d_∇.

Theorem A.11 (The Exterior Covariant Derivative). Let M be a complex manifold, and let $\pi\colon E \to M$ be a smooth complex vector bundle over M. Let ∇ be a connection on E. Then, there are unique operators called *exterior covariant derivatives* $\mathrm{d}_\nabla\colon \Omega^q_{\mathbb{C}}(M;E) \to \Omega^{q+1}_{\mathbb{C}}(M;E)$ for each $q \geq 0$, satisfying:

1. For every $\sigma \in \Gamma(E) = \Omega^0_{\mathbb{C}}(M;E)$, $\mathrm{d}_\nabla\sigma = \nabla\sigma$.
2. For every $\alpha \in \Omega^q_{\mathbb{C}}(M)$ and $\beta \in \Omega^p_{\mathbb{C}}(M;E)$,

$$\mathrm{d}_\nabla(\alpha\wedge\beta) = \mathrm{d}\alpha\wedge\beta + (-1)^q\alpha\wedge\mathrm{d}_\nabla\beta.$$

Proof. We define $\mathrm{d}_\nabla\sigma = \nabla\sigma$ for every $\sigma \in \Gamma(E)$ and for higher-degree forms, we consider $(s_1,\dots,s_r)$ a local frame, and let $\alpha \in \Omega^q_{\mathbb{C}}(M;E)$ whose local expression is given by $\alpha = \alpha^j \otimes s_j$. We define $\mathrm{d}_\nabla\alpha := \mathrm{d}\alpha^j \otimes s_j + (-1)^q\alpha^j \wedge \mathrm{d}_\nabla s_j$. One then checks that this is well-defined and independent of the local frame. See [33, Prop. 7.11] for details. □

A.6.1 Induced connections

Connections on vector bundles give rise to connections in other vector bundles, provided these are constructed using the ones we originally had. We will be concerned with two cases:

1. Let E and F be two vector bundles over a smooth manifold, endowed with connections ∇^E and ∇^F. Then, we may define a connection on $E\otimes F$ by extending by linearity the following expression

$$\left(\nabla^E \otimes \nabla^F\right)(\xi\otimes\eta) = \nabla^E\xi\otimes\eta + \xi\otimes\nabla^F\eta, \tag{A.6}$$

   as a direct computation shows.

2. Let ∇ be a connection on E, then, it induces a connection on E^* as follows. There is a unique connection ∇^* on E^* such that for every $t \in \Gamma(E^*)$, and $\xi \in \Gamma(E)$,

$$\left[\nabla^*(t)\right](\xi) = \mathrm{d}(t(\xi)) - t(\nabla\xi), \tag{A.7}$$

where $t(\alpha \otimes s) := t(s) \cdot \alpha$.

Sometimes, complex vector bundles are endowed with some extra structure.

Definition A.12 (Hermitian fiber metric). Let $\pi\colon E \to M$ be a smooth complex vector bundle over a smooth manifold M. A *hermitian fiber metric* h is an assignment of a hermitian inner product h_x to each fiber E_x and such that for every pair of local smooth sections $\sigma, \tau \in \Gamma(U; E)$, the map $U \to \mathbb{C}, x \mapsto h_x(\sigma_x, \tau_x)$ is smooth. A vector bundle that is endowed with a hermitian fiber metric is called a *hermitian vector bundle*

A.7 Kähler manifolds

Let M be a complex manifold, and recall that its canonical almost complex structure J induces a complex vector bundle structure on TM. Let us denote $T_J M$ the resulting complex vector bundle. Observe that every hermitian fiber metric on $T_J M$ defines a riemannian metric $g := \mathrm{Re}(h)$. Something slightly less obvious is the fact that $\omega = -\mathrm{Im}(h)$ is a real form of type $(1,1)$, that is, $\omega \in \Omega^2_{\mathbb{R}}(M) \cap \Omega^{(1,1)}_{\mathbb{C}}(M)$ [33, Lemma 8.2].

A *hermitian metric* on a complex manifold M is a riemannian metric g for which J is orthogonal, i.e. $g(JX, JY) = g(X, Y)$. Then, we can *define* $\omega(\cdot, \cdot) := g(J\cdot, \cdot)$, and $h := g - \mathrm{i}\,\omega$, and it turns out that h is a hermitian fiber metric as defined above. Under these assumptions, we call ω the *fundamental two-form* of h. The forms ω and g are uniquely determined by each other. Indeed, $\omega(\cdot, \cdot) = g(J\cdot, \cdot)$ and $g(\cdot, \cdot) = \omega(\cdot, J\cdot)$ (see [33, lemma 8.3 and lemma 8.7]).

A $(1,1)$-form ω on a complex manifold M is said to be *positive* if $\omega_p(v, J_p v) > 0$ for every $v \in T_p M \setminus \{0\}$ and $p \in M$. Given a complex manifold M endowed with a two-form ω, if we define $g(\cdot, \cdot) := \omega(\cdot, J\cdot)$, then g is a hermitian form whose fundamental form is ω if and only if ω is a positive $(1,1)$-form (see [33, Prop. 8.9])

Definition A.13 (Kähler manifold). Let M be a complex manifold, and g be a hermitian metric on M. We say that g is a *Kähler metric* if its fundamental two-form ω is closed. In that case, we say that M is a Kähler manifold.

Observe that every Kähler manifold is a symplectic manifold, the fundamental form being its symplectic form. Indeed, ω_p defines an injective map $\omega_p^\flat\colon T_p M \to T_p^* M, v \mapsto \omega(v, \cdot)$ because if $\omega_p^\flat(v) = 0$, then $\omega_p(v, Jv) = 0$, so $v = 0$.

Example A.14. Some examples are $\mathbb{CP}^n$ and all Riemann surfaces (complex manifolds of complex dimension equal to 1). Indeed, one simply fixes a hermitian metric g (this can be done by [33, Lemma 8.5]), and then defines $\omega(\cdot, \cdot) = g(\cdot, J\cdot)$. This is a positive form of type $(1,1)$ [33, Prop. 8.9], and it is closed because every two-form is closed on a Riemann surface (its dimension as a smooth manifold is 2).

A.8 Hodge Theory

A.8.1 The Hodge inner product

The aim of this section is to define a hermitian inner product on the set of differential forms on a complex manifold. This can be used to define the Lefschetz operator and its

dual, which is the operator appearing on the hermitian Yang-Mills equation. Let (M, g) be an n-dimensional compact, oriented riemannian manifold. The riemannian metric can be complexified by extending it by complex bilinearity so that it acts on the complexified tangent bundle $T_{\mathbb{C}}M$. We can use the musical isomorphism $\sharp\colon T^*_{\mathbb{C}}M \to T_{\mathbb{C}}M$ to induce a hermitian fiber metric on $\Omega^1_{\mathbb{C}}(M)$, which is given as follows:

$$\langle \alpha, \beta \rangle := g\left(\alpha^{\sharp}, \overline{\beta^{\sharp}}\right). \tag{A.8}$$

Firstly, we will extend this inner product to higher order forms. For smooth complex-valued functions, we simply define $\langle f, h\rangle := f \cdot \overline{h}$. For higher degree forms, consider any local orthonormal frame $\{\varepsilon^1, \dots \varepsilon^n\}$ for the hermitian fiber metric (A.8). The hermitian inner product on $\bigwedge^k_{\mathbb{C}} M$ is defined such that $\{\varepsilon^{i_1} \wedge \ldots \wedge \varepsilon^{i_k} : i_1 < \ldots < i_k\}$ is orthonormal. More precisely, there is a unique hermitian fiber product on $\bigwedge^k_{\mathbb{C}} M$ which satisfies

$$\langle \alpha_1 \wedge \ldots \wedge \alpha_k,\ \beta_1 \wedge \ldots \wedge \beta_k \rangle = \det\left(\left(\langle \alpha_i, \beta_j \rangle\right)_{1 \le i,j \le k}\right),$$

for any locally defined one-forms [33, Lemma 9.1]. When M is compact, this yields an inner product on $\Omega^k_{\mathbb{C}}(M)$ given as follows:

$$(\alpha, \beta) := \int_M \langle \alpha, \beta \rangle \, \mathrm{d}V_g, \qquad \alpha, \beta \in \Omega^k_{\mathbb{C}}(M). \tag{A.9}$$

Compactness of M ensures the convergence of the integral above. Equation (A.9) defines a hermitian product on the complex vector space of complex k-forms. This is called the *(global) Hodge inner product* or L_2 product. The symbol $\mathrm{d}V_g$ in equation (A.9) denotes the riemannian volume on M. For a Kähler manifold, $\mathrm{d}V_g = \omega^n/n!$, and the orientation with respect to which we integrate is determined by ω [33, Lemma 8.17].

A.8.2 The Hodge star operator

Let M be a compact complex manifold. The Hodge star operator is a vector bundle isomorphism taking k-forms to $(n-k)$-forms. Just as the Hodge inner product, the construction does not use anything particular to vector bundles: it is purely linear, and everything can be done in euclidean vector spaces with a fixed orientation.

Proposition A.15. Let M be an oriented Kähler n-dimensional manifold. Then, there is a unique smooth vector bundle isomorphism

$$\star\colon \bigwedge^{(p,q)}_{\mathbb{C}} M \longrightarrow \bigwedge^{(n-p,n-q)}_{\mathbb{C}} M,$$

called the *Hodge star operator* satisfying, for every $\alpha, \beta \in \bigwedge^k_{\mathbb{C}} M$, the following relation

$$\alpha \wedge \star\overline{\beta} = \langle \alpha, \beta \rangle \, \mathrm{d}V_g. \tag{A.10}$$

Proof. See [33, Prop. 9.8] and [33, Lemma 9.23-(b)]. . □

Moreover, the star operator satisfies [33, Prop. 9.8]:

$$\overline{\star\, \alpha} = \star\, \overline{\alpha} \qquad \star\star\ \alpha = (-1)^{k\cdot(N-k)}\alpha \qquad \langle \star\, \alpha, \star\, \beta \rangle = \langle \alpha, \beta \rangle.$$

The global inner product can be equivalently written as

$$(\alpha, \beta) = \int_M \alpha \wedge \star\overline{\beta}. \tag{A.11}$$

Remark A.16. Everything we have done can be done on riemannian manifolds without requiring complexification of the cotangent bundle. In that case, the pointwise inner product is defined directly on T^*M by $\langle \alpha, \beta \rangle := g(\alpha^\sharp, \beta^\sharp)$ (no conjugation), the Hodge star operator satisfies $\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \cdot \mathrm{d}V_g$, and the global inner product can be defined as $(\alpha, \beta) = \int_M \alpha \wedge \star\beta$.

With these definitions in mind, given a Kähler manifold (M, ω) we define the *Lefschetz operator* as $L_\omega \colon \Omega_{\mathbb{C}}^{(p,q)}(M) \to \Omega_{\mathbb{C}}^{(p+1,q+1)}(M), \alpha \mapsto \omega \wedge \alpha$. There is a unique operator, called the *dual Lefschetz operator* or the *contraction with the Kähler form*, denoted Λ_ω or $L_\omega^\star$, which is defined by the property

$$\langle L_\omega \alpha, \beta \rangle = \langle \alpha, \Lambda_\omega \beta \rangle \qquad \alpha \in \Omega_{\mathbb{C}}^q(M), \beta \in \Omega_{\mathbb{C}}^{q+2}(M).$$

Explicitly, we have $\Lambda_\omega = \star^{-1} \circ L_\omega \circ \star$ (see [24, Lemma 1.2.23]). Finally, these operators can be extended to bundle valued forms [24]. Consider a complex vector bundle $E \to M$, and define $L_\omega \otimes \mathrm{id}_E$ and $\Lambda_\omega \otimes \mathrm{id}_E$ on the spaces $\bigwedge^{(p,q)} M$. That is, these maps are the complex-bilinear extension of

$$(L_\omega \otimes \mathrm{id}_E)(\alpha \otimes \xi) = (L_\omega \alpha) \otimes \xi \qquad (\Lambda_\omega \otimes \mathrm{id}_E)(\beta \otimes \eta) = (\Lambda_\omega \beta) \otimes \eta.$$

Remark A.17. For two-forms on a Riemann surface X, one has $\star\alpha = \Lambda_\omega \alpha$. This follows from the fact that $\star\omega = 1$ (see [41, p. 6.2.1] and recall that $\mathrm{d}V_g = \omega$ for Riemann surfaces), so for $f \in \mathscr{C}^\infty(X; \mathbb{C}), \alpha \in \Omega_{\mathbb{C}}^2(X)$

$$\langle L_\omega f, \alpha \rangle = \langle f \cdot \omega, \alpha \rangle = f \cdot \langle \star 1, \star\star\alpha \rangle = f \cdot \langle 1, \star\alpha \rangle = f \cdot \overline{\star\alpha} = \langle f, \star\alpha \rangle.$$

Finally, a lemma we will need.

Lemma A.18. Let M be a Kähler manifold of complex dimension n and $\alpha \in \Omega_{\mathbb{C}}^2(M)$. Then,

$$\Lambda_\omega \alpha \cdot \omega^n = n \cdot L_\omega^{n-1} \alpha.$$

Proof. From [33, Lemma 9.39] we know that $[L_\omega, \Lambda_\omega] = (k-n) \cdot \mathrm{id}$ acting on k-forms, where $[\cdot, \cdot]$ is the commutator of operators. Inductively, one shows, $[L_\omega^p, \Lambda_\omega]\alpha = p \cdot (k - n + p - 1) \cdot L_\omega^{p-1} \alpha$ for each k-form α. Indeed, $p = 1$ is the lemma. For the inductive step (we do not write the k-form)

$$\begin{aligned}
[L_\omega^p, \Lambda_\omega] &= L_\omega^p \Lambda_\omega - \Lambda_\omega L_\omega^p = L_\omega^p \Lambda_\omega - \Lambda_\omega L_\omega L_\omega^{p-1} \overset{(1)}{=} L_\omega^p \Lambda_\omega - (L_\omega \Lambda_\omega - (k + 2(p-1) - n) \cdot \mathrm{id}) L_\omega^{p-1} \\
&= L_\omega (L_\omega^{p-1} \Lambda_\omega - \Lambda_\omega L_\omega^{p-1}) + (k + 2(p-1) - n) \cdot L_\omega^{p-1} \\
&= (p-1) \cdot (k - n + p - 2) L_\omega^{p-1} + (k + 2(p-1) - n) L_\omega^{p-1} \\
&= p \cdot (k - n + p - 1) L_\omega^{p-1}
\end{aligned}$$

In (1) we used that $\Lambda_\omega L_\omega$ now acts on a $k + 2(p-1)$-form because it acts on $L_\omega \alpha$. Once this formula has been proven, we apply it to a two-form and $p = n$. Thus, $[L_\omega^n, \Lambda_\omega]\alpha = n \cdot L_\omega^{n-1} \alpha$. Hence (observe that $\omega^n \wedge \alpha = 0$ because it is a form whose degree is higher than the (real) dimension of the manifold)

$$L_\omega^n \Lambda_\omega \alpha - \Lambda_\omega L_\omega^n \alpha = \omega^n \wedge \Lambda_\omega \alpha = \Lambda_\omega \alpha \cdot \omega^n = n \cdot \omega^{n-1} \wedge \alpha.$$

□

Appendix B

Some Lie theory

We devote this appendix to giving a brief exposition of some aspects of Lie theory that we will need for the development of the main text. First, we give the definition of a complex Lie group, and we recall the notion of a complexification of a Lie group. We then give a brief introduction to parabolic subgroups and subalgebras (they will be essential to define the notion of stability of a principal bundle). The appendix ends by introducing some tools from the theory of non-connected Lie groups and the way these can be applied to bundle theory. We assume basic familiarity with the theory of Lie groups. The main references we used are [21] for basic Lie theory, and [5] for everything related to twisted equivariant bundles. Even though in other chapters we have used Greek letters ξ, ζ to name elements of the Lie algebra of a Lie group, we will use in this appendix the more standard notation of roman capital letters X, Y. We will write e_G for the unit element of the group G. We will, however, use the notation 1 for the trivial group.

B.1 Complex Lie groups

Unsurprisingly, complex Lie groups are defined as follows:

> **Definition B.1 (Complex Lie group).** A *complex Lie group* is a complex manifold G which is endowed with a group structure such that the product $G \times G \to G, (g, h) \mapsto g \cdot h$ and inversion $G \to G, g \mapsto g^{-1}$ maps are holomorphic.

Clearly, every complex Lie group is a (real) Lie group in the usual sense. Observe that G being a complex manifold, it is endowed with an almost complex structure J_G. This induces a linear almost complex structure on its tangent spaces, in particular on T_eG (considering T_eG as the tangent space of the underlying smooth manifold). Since $T_eG \cong \mathfrak{g}$, the Lie algebra is endowed with a complex structure $J_\mathfrak{g}$, i.e. $\mathfrak{g}$ has the structure of a complex vector space with the product by complex scalars defined by $(a + \mathrm{i}\, b) \cdot X := a \cdot X + b \cdot J_\mathfrak{g}(X)$. In fact, $\mathfrak{g}$ is a complex Lie algebra (equipped with the Lie bracket it gets from the smooth Lie group structure of G) [25, Page 55]. This amounts to checking that

$$J_\mathfrak{g}[X, Y] = [J_\mathfrak{g}X, Y] = [X, J_\mathfrak{g}Y] \qquad X, Y \in \mathfrak{g}. \tag{B.1}$$

Of course, there is an underlying real Lie algebra, which we shall denote $\mathfrak{g}^\mathbb{R}$.

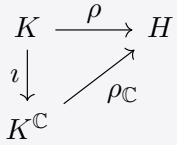
An important remark is the fact that the adjoint representation $\mathrm{Ad}\colon G \to \mathrm{Aut}(\mathfrak{g})$ is holomorphic. A thorough treatment of complex Lie groups is given in [31]. Moreover, complex Lie groups come equipped with an exponential map, which is simply the exponential map they have as real Lie groups. It turns out that the exponential of a complex Lie group is a holomorphic map. Some examples of complex Lie groups are $\mathrm{GL}(n, \mathbb{C})$ and $\mathrm{SL}(n, \mathbb{C})$.

Every Lie algebra $\mathfrak{g}$ can be complexified. This is done by setting $\mathfrak{g}^\mathbb{C} := \mathfrak{g} \otimes_\mathbb{R} \mathbb{C} \cong \mathfrak{g} \oplus \mathrm{i}\, \mathfrak{g}$

and extending the Lie bracket by $[X \otimes \alpha, Y \otimes \beta] := \alpha\beta \cdot [X, Y]$ for $\alpha, \beta \in \mathbb{C}$ and $X, Y \in \mathfrak{g}$. Actually, complexifications can be extended to Lie groups, although the definition is not as straightforward.

Definition B.2 (Complexification of a Lie group). Let K be a real Lie group. A *complexification* of K is a complex Lie group, denoted $K^{\mathbb{C}}$, together with an injective Lie group homomorphism (a smooth group homomorphism) $\imath\colon K \to K^{\mathbb{C}}$ satisfying the following universal property:

For every complex Lie group H, and every (real) Lie group homomorphism $\rho\colon K \to H$, there exists a unique holomorphic Lie group homomorphism $\rho_{\mathbb{C}}\colon K^{\mathbb{C}} \to H$ such that $\rho = \rho_{\mathbb{C}} \circ \imath$. That is, the diagram is commutative



The previous definition is given in terms of a universal property, which automatically means that, if such a complexification exists, it is unique up to isomorphism [21, Remark 15.1.3]. With some work, one can show that every real Lie group admits a complexification [21, Th. 15.1.4].

When K is compact, the Lie algebra of the complexification is the complexification of the Lie algebra of K. That is, let $G := K^{\mathbb{C}}$ with Lie algebra $\mathfrak{g}$, and let $\mathfrak{k}$ be the Lie algebra of K. Then, $\mathfrak{g} = \mathfrak{k}^{\mathbb{C}} \cong \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$ (we should actually write $\mathfrak{g}^{\mathbb{R}} = \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$, see section A.3.1). In fact, the case in which K is compact is the most interesting: under those assumptions, the inclusion is injective, and $\imath(K) \subset K^{\mathbb{C}}$ is a maximal compact subgroup of $K^{\mathbb{C}}$ [21, Prop. 15.2.1]. In particular, $\imath(K) \subset G$ is closed, so it is a real Lie subgroup of G. In general, one has $\dim_{\mathbb{R}} G = 2 \cdot \dim K$ (see [21, Prop. 15.2.1]). As an example, $\mathrm{U}(n)$ is a compact real Lie group whose complexification is $\mathrm{GL}(n, \mathbb{C})$.

Finally, there is a quotient manifold theorem for holomorphic actions on complex manifolds, which we state here for convenience.

Theorem B.3 (Complex quotient manifold theorem). Let G be a complex Lie group and let M be a complex manifold. Suppose that there is a free and proper holomorphic action on M. Then, the quotient space M/G has a unique complex manifold structure such that the natural projection $\pi\colon M \to M/G$ is a holomorphic submersion.

Proof. See [24, Prop. 2.1.13] or [20, Lemma 3.2.6]. □

Remark B.4. For real Lie groups, the closed subgroup theorem [21, Th. 9.3.7] states that whenever H is a closed subgroup of a Lie group G, then, it is an embedded Lie subgroup of G. By an embedded Lie subgroup, we mean a subgroup that is also an embedded submanifold of G and whose Lie group structure is given by restricting the group operations in G. For complex Lie groups, this is no longer true, we may have closed subgroups of a complex Lie group that are embedded *real* Lie groups, but not complex Lie groups. However, if $H \leq G$ is a closed subgroup, and $\mathfrak{h}$ is a complex Lie subalgebra (i.e. it is $J_{\mathfrak{g}}$-invariant), then it is a complex Lie subgroup. That is, a complex Lie group that is an embedded complex submanifold and whose group operations are obtained by restricting those of G. See [28, Ch. IX, Example 2.1].

A very important class of complex Lie groups is the class of linearly reductive groups.

Definition B.5 (Linearly reductive complex Lie group). A complex Lie group G is *(linearly) reductive* if it is the complexification of a real compact Lie group.

We will often call linearly reductive Lie groups just reductive Lie groups. This is not standard, as some authors reserve reductive for a more general type of group, but since we will only refer to linearly reductive groups, no confusion will arise. This definition works regardless of connectedness assumptions, for instance $\mathrm{O}(n)^{\mathbb{C}} = \mathrm{O}(n,\mathbb{C})$ and $\mathrm{SO}(n)^{\mathbb{C}} = \mathrm{SO}(n,\mathbb{C})$ are reductive, but only the latter is connected. Moreover, complex reductive Lie groups (with finitely many connected components) are characterized by the fact that they admit faithful holomorphic finite-dimensional representations, and they are all completely reducible [21, Th. 15.3.11]. This fact can be used to endow every reductive complex Lie group with the structure of an algebraic group [10, p. III. 8].

B.2 Parabolic subgroups

This section discusses parabolic subgroups of Lie groups. This type of group will be essential to define polystability on principal bundles. We assume that the reader is familiar with root systems. A good reference is [21, Ch. 6].

Let $\mathfrak{g}$ be a complex semisimple Lie algebra, $\mathfrak{h}$ a Cartan subalgebra, and denote by $\lambda \in \mathfrak{h}^*$ the roots, $\Delta := \Delta(\mathfrak{g},\mathfrak{h})$ the set of all roots, and $\mathfrak{g}_\lambda$ the root spaces. Under these hypotheses, we have a *root space decomposition* $\mathfrak{g} = \mathfrak{h} \oplus (\oplus_{\lambda\neq 0}\, \mathfrak{g}_\lambda)$. If $\mathfrak{g}$ is not semisimple, but reductive (i.e. for each ideal $\mathfrak{a} \subset \mathfrak{g}$, there is another ideal $\mathfrak{b}$ such that $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$), then $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g},\mathfrak{g}]$, where $\mathfrak{z}(\mathfrak{g})$ is the center of the Lie algebra, and $[\mathfrak{g},\mathfrak{g}]$ is semisimple [21, Lemma 5.7.2]. Then, the root space decomposition reads

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g},\mathfrak{g}] = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus \left(\bigoplus_{\lambda\neq 0} \mathfrak{g}_\lambda \right).$$

B.2.1 Parabolic subalgebras

Let $\mathfrak{g}$ be a complex semisimple Lie algebra and $\mathfrak{h}$ a Cartan subalgebra. Fixing a basis or set of simple roots, it makes sense to speak about the set of positive roots, which we shall denote Δ^+. In that situation, we say that a *Borel subalgebra* is a subalgebra:

$$\mathfrak{b} := \mathfrak{h} \oplus \left(\bigoplus_{\alpha\in\Delta^+} \mathfrak{g}_\alpha \right). \tag{B.2}$$

Observe that $\mathfrak{b}$ is indeed a subalgebra. This follows from two facts. First, for a complex semisimple Lie algebra, if α,β are two positive roots (or zero), then $[\mathfrak{g}_\alpha,\mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$, and $\alpha+\beta \in \Delta^+$ [21, Lemma 6.3.5] and $\mathfrak{h} = \mathfrak{g}_0$. Note that Borel subalgebras are not unique; they depend on the chosen basis.

Definition B.6 (Parabolic subalgebra). Let $\mathfrak{g}$ be a complex semisimple Lie algebra. A *parabolic subalgebra* is a Lie subalgebra $\mathfrak{p}$ of $\mathfrak{g}$ containing a Borel subalgebra.

All the parabolic subalgebras containing $\mathfrak{b}$ (defined as in (B.2)) are given by

$$\mathfrak{p} = \mathfrak{h} \oplus \left(\bigoplus_{\lambda\in\Gamma} \mathfrak{g}_\lambda \right), \tag{B.3}$$

for some $\Delta^+ \subset \Gamma \subset \Delta$. More precisely, let Π be a basis of the root system. For every subset $\Pi' \subset \Pi$, define

$$\Gamma := \Delta^+ \cup \left\{ \alpha \in \Delta \cap \operatorname{span}(\Pi') \right\}. \tag{B.4}$$

Then, there is a one-to-one correspondence between all parabolic subalgebras containing $\mathfrak{b}$ and sets Γ defined as above, which associates to each set Γ the algebra (B.3) [**25**, Prop. 5.90].

Let $-\Gamma$ be the set of opposites of elements in Γ. Observe that the second set in (B.4) is closed under taking opposites. Each parabolic subalgebra admits a decomposition $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}$ where

$$\mathfrak{l} := \mathfrak{h} \oplus \left(\bigoplus_{\lambda \in \Gamma \cap -\Gamma} \mathfrak{g}_\lambda \right) \qquad \text{and} \qquad \mathfrak{u} := \bigoplus_{\lambda \in \Gamma \cap (\Gamma \setminus -\Gamma)} \mathfrak{g}_\lambda. \tag{B.5}$$

$\mathfrak{u}$ is called the *nilpotent radical*, and $\mathfrak{l}$ is called the *Levi factor*. $\mathfrak{u}$ is an ideal, and a nilpotent Lie algebra, and $\mathfrak{l}$ is a reductive Lie algebra [**25**, Cor. 5.94].

Remark B.7. When the Lie algebra $\mathfrak{g}$ is not semisimple, the definition is generalized as follows: consider the decomposition $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. $[\mathfrak{g}, \mathfrak{g}]$ is semisimple, so the previous discussion applies. Then, parabolic subalgebras are

$$\mathfrak{p} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus \left(\bigoplus_{\lambda \in \Gamma} \mathfrak{g}_\lambda \right),$$

for subsets Γ defined as above. The center is included in the Levi factor $\mathfrak{l} := \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus (\oplus_{\lambda \in \Gamma \cap -\Gamma} \mathfrak{g}_\lambda)$.

Observe that both parabolic and Borel subalgebras are complex subalgebras. We are now ready to define parabolic subgroups.

B.2.2 Definition and first properties of parabolic subgroups

Definition B.8 (Parabolic subgroups). Let G be a connected complex reductive Lie group. A *Borel subgroup* of G is any connected complex embedded Lie subgroup B whose Lie algebra is a Borel subalgebra. A *parabolic subgroup* P is any connected subgroup of G whose Lie algebra is parabolic.

If we fix a parabolic subalgebra $\mathfrak{p}$ and consider the Lie subgroup P such that $\mathfrak{p}$ is its Lie algebra, then, any other parabolic subgroup is obtained from this one by conjugation. Observe that parabolic subgroups are complex closed subgroups of G, hence embedded Lie subgroups because their Lie algebras are complex subalgebras of $\mathfrak{g}$. Equivalently, parabolic subgroups are precisely those containing a Borel subgroup. A very useful characterization of parabolic subgroups is given by the following result:

Theorem B.9. Let G be a connected complex reductive Lie group. A subgroup $P \subset G$ is parabolic if and only if the quotient space G/P is compact.

Proof. See [**23**, Sec. 29.3] □

Every parabolic subgroup P of a connected complex reductive Lie group G can be written as $P = LU$ where L and U are the connected Lie groups whose Lie algebras are

given by $\mathfrak{l}$ and $\mathfrak{u}$ in (B.5) respectively. L is called the *Levi subgroup* and U the *unipotent radical*. U is a normal subgroup of P, see [12, Page 316] and [16, Sec. 2.1].

Moreover, parabolic subgroups of a complex reductive Lie group G are parametrized precisely by elements of the Lie algebra $\mathrm{i}\,\mathfrak{k}$. The precise statement is as follows:

Proposition B.10. Let $G = K^{\mathbb{C}}$ be a complex reductive Lie group, and $\mathfrak{g} = \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$ its Lie algebra. For any $\zeta \in \mathrm{i}\,\mathfrak{k}$, define

$$P_\zeta := \{g \in G : \exp(t\zeta)\ g\ \exp(-t\zeta) \text{ is bounded as } t \to \infty\} \qquad L_\zeta := \{g \in G : \mathrm{Ad}_g(\zeta) = \zeta\}$$
$$\mathfrak{p}_\zeta := \left\{\xi \in \mathfrak{g} : \mathrm{Ad}_{\exp(t\zeta)}(\xi) \text{ is bounded as } t \to \infty\right\} \qquad \mathfrak{l}_\zeta := \{\xi \in \mathfrak{g} : [\xi, \zeta] = 0\}.$$

Then P_ζ and L_ζ are connected, closed complex Lie subgroups of G whose Lie algebras are $\mathfrak{p}_\zeta$ and $\mathfrak{l}_\zeta$ respectively. Moreover, $\mathfrak{p}_\zeta$ is parabolic (in the sense of (B.3)) so P_ζ is a parabolic subgroup, and L_ζ is reductive.

Proof. See [16, Lemma 2.5] and [59, Lemma 2.12]. □

L_ζ is what we call a *Levi subgroup of* P_ζ. Its Lie algebra $\mathfrak{l}_\zeta$, is a Levi factor in the sense of (B.5) (see [59, Lemma 2.14]).

Remark B.11. Since $\mathfrak{g}$ is a finite-dimensional vector space, and all norms on finite-dimensional vector spaces are equivalent, the notion of being bounded is well-defined. Now, for the definition of P_ζ, observe that there is no natural notion of boundedness on G. We say that the map $t \mapsto g(t) := \exp(t\zeta) \cdot g \cdot \exp(-t\zeta)$ is bounded if the limit when $t \to \infty$ exists, that is, if there is some $h \in \hat{G}$ such that for every neighborhood $\mathcal{U}$ of h there exists some $M > 0$ such that for $t > M$, $g(t) \in \mathcal{U}$.

We conclude the section by drawing attention to the fact that, in general, parabolic subgroups *are not* normal subgroups.

B.2.3 Examples

Let us conclude by giving some examples of parabolic subgroups. Let $K = \mathrm{U}(n)$ (which is compact [21, Lemma 2.1.4]), and $G = \mathrm{GL}(n, \mathbb{C})$ its complexification. Even though proposition B.10 characterizes all parabolic subgroups of G, it is always instructive to look at a concrete example. To describe these parabolic subgroups, we briefly introduce the concept of flag. Given a complex vector space, say $\mathbb{C}^n$ for concreteness, a *(partial) flag* is a chain of vector subspaces $0 \subsetneq V_1 \subsetneq V_2 \subsetneq \ldots \subsetneq V_r = \mathbb{C}^n$. Then, one may show that every parabolic subgroup $P \subset G$ is the stabilizer of a partial flag [23, Sec. 21.3 Cor. B]. That is, $p \cdot V_i = V_i$ for each $1 \leq i \leq r$ and $p \in P$. Hence, up to an adequate choice of a basis (that is, up to conjugation) all parabolic subgroups of G are given by

$$P = \left\{ \begin{pmatrix} A_{11} & A_{12} & \ldots & A_{1k} \\ 0 & A_{22} & \ldots & A_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & A_{kk} \end{pmatrix} : A_{ii} \in \mathrm{GL}(n_i, \mathbb{C}), A_{jk} \in \mathrm{M}_{n_i \times n_j}(\mathbb{C}) \right\}$$

where $n_1 = \dim V_1$, and $n_i = \dim V_i - \dim V_{i-1}$ for $i \geq 2$.

B.3 Non-connected Lie groups

Many important results in the theory of Lie groups rely heavily on connectedness to work. There are, however, some ways to deal with non-connected groups, which constitutes the

content of this section. References for this section are [21, Ch. 18] and especially [5]. In fact, later sections are a summary of the key results of the latter. Our main setting will be the following: consider a non-connected Lie group $\hat{G}$ with finitely many connected components, and let G be the connected component containing the identity (identity component, from now on). Let $\Gamma = \hat{G}/G$. Since the connected component of the identity is always a normal closed subgroup [34, Prop. 7.15], the quotient has a natural Lie group structure [34, Th. 21.26]. This is the group of connected components of $\hat{G}$, and it fits in an exact sequence of Lie groups

$$1 \to G \hookrightarrow \hat{G} \twoheadrightarrow \Gamma = \hat{G}/G \to 1, \tag{B.6}$$

The fact that Γ is the group of connected components of $\hat{G}$ follows from the fact that each L_g is a diffeomorphism. Hence, $L_g(G) = gG$ is a connected component of $\hat{G}$ for each $g \in G$. Since $gG = \tilde{g}G$ if and only if $g\tilde{g}^{-1} \in G$, the latter being the relation defining Γ. Thus, it follows that $gG = \tilde{g}G$ if and only if g and $\tilde{g}$ are in the same connected component of $\hat{G}$. Since $G \triangleleft \hat{G}$, for each $\hat{g} \in \hat{G}$, $C_{\hat{g}}$ (recall that $C_{\hat{g}}(h) = \hat{g}h\hat{g}^{-1}$) restricts to a Lie group isomorphism of G. This defines the *characteristic homomorphism* $\bar{\theta}\colon \Gamma \to \mathrm{Out}(G)$. More precisely, first, for any Lie group, define $\mathrm{Aut}(G)$ to be the group of Lie group isomorphisms, and $\mathrm{Int}(G) := \{C_g : g \in G\}$. Then $\mathrm{Int}(G) \triangleleft \mathrm{Aut}(G)$, and we may consider $\mathrm{Out}(G) := \mathrm{Aut}(G)/\mathrm{Int}(G)$. The characteristic homomorphism is defined as follows:

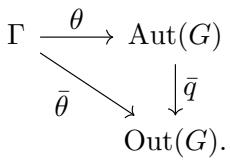
$$\begin{array}{rccc} \bar{\theta}\colon & \Gamma & \longrightarrow & \mathrm{Out}(G) \\ & \gamma = \hat{g}G & \mapsto & [C_{\hat{g}}], \end{array} \tag{B.7}$$

where $[\cdot]$ denotes the equivalence class in $\mathrm{Out}(G)$. Observe that for $\hat{g} \in \hat{G} \setminus G$, $C_{\hat{g}} \notin \mathrm{Int}(G)$.

Lemma B.12. Let $\hat{G}$ be a non-connected Lie group and consider an exact sequence as in (B.6). Then, the map $\bar{\theta}$ defined by (B.7) is well-defined.

Proof. Let $\hat{g}_1 G = \hat{g}_2 G$. Then, $\hat{g}_1^{-1}\hat{g}_2 \in G$. The result follows from $(C_{\hat{g}_1})^{-1} \circ C_{\hat{g}_2} = C_{(\hat{g}_1)^{-1}\hat{g}_2} \in \mathrm{Int}(G)$. □

We will assume for now that $\bar{\theta}$ admits a lift $\theta\colon \Gamma \to \mathrm{Aut}(G)$ such that the diagram is commutative ($\bar{q}$ denotes the quotient map). We will write $\theta_\gamma = \theta(\gamma)$ and $\theta_\gamma(g) = g^\gamma$.



The lift θ, together with an element $c \in Z^2_\theta(\Gamma, Z(G))$ will give us another way to look at $\hat{G}$. c is what we call a two-cocycle, that is, a map $c\colon \Gamma \times \Gamma \to Z(G)$ satisfying

$$c(\gamma_1, \gamma_2)^{\gamma_0} c(\gamma_0, \gamma_1\gamma_2) = c(\gamma_0, \gamma_1) c(\gamma_0\gamma_1, \gamma_2), \qquad \gamma_0, \gamma_1, \gamma_2 \in \Gamma \tag{B.8}$$

and $c(e_\Gamma, \gamma) = c(\gamma, e_\Gamma) = e_G$ for all $\gamma \in \Gamma$. This condition comes from closedness in group cohomology (see [21, Appendix 18.3] for details).

Definition B.13. Let G, θ and $c \in Z^2_\theta(\Gamma, Z(G))$ be as above, we define the group $G \times_{(\theta,c)} \Gamma$, whose underlying set is $G \times \Gamma$, and whose group operation is given by

$$(g_1, \gamma_1)\ (g_2, \gamma_2) := (g_1 g_2^{\gamma_1}\ c(\gamma_1, \gamma_2), \gamma_1\gamma_2).$$

A direct computation shows that it is indeed a group (associativity follows from (B.8) and the fact that c is $Z(G)$-valued) whose unit element is (e_G, e_Γ), and such that the

inverse of (g,γ) is given by $((g^{-1}\ c(\gamma,\gamma^{-1})^{-1})^{\gamma^{-1}},\gamma^{-1})$ [21, Lemma 18.1.9]. Observe that $c\equiv 1$ implies $G\times_{(\theta,c)}\Gamma = G\rtimes_\theta\Gamma$. This group fits in an exact sequence

$$1\to G\xrightarrow{f} G\times_{(\theta,c)}\Gamma\xrightarrow{k}\Gamma\to 1,$$

where $f(g)=(g,e_\Gamma)$ and $k(g,\gamma)=\gamma$. Equipped with the product structure we just defined, one may show that $G\times_{(\theta,c)}\Gamma$ is a Lie group. Indeed, $G\times\Gamma$ is a smooth manifold endowed with the product manifold structure (Γ is discrete hence a zero dimensional manifold, and θ and c are smooth). For adequate choices of c and θ, we actually have $G\times_{(\theta,c)}\Gamma\cong\hat{G}$. To construct such an isomorphism, we start by choosing a section $s\colon\Gamma\to\hat{G}$ (i.e. $q\circ s=\mathrm{id}_\Gamma$) that is *normalized*, meaning that the composition of the conjugation map $\hat{G}\to\mathrm{Aut}(G)$ with s is equal to θ. In particular, $s(e_\Gamma)=e_{\hat{G}}$. In other words $\theta_\gamma(g)=s(\gamma)\ g\ s(\gamma)^{-1}$ for all $\gamma\in\Gamma, g\in G$. The section can always be chosen like that because for each $\gamma\in\Gamma$, conjugation, seen as a map $q^{-1}(\gamma)\to\mathrm{Aut}(G)$ has as its image the whole class of $\bar{\theta}(\gamma)$. In that case, we can *define* the cocycle c by the relation

$$c(\gamma_1,\gamma_2)=s(\gamma_1)\ s(\gamma_2)\ s(\gamma_1\gamma_2)^{-1}.$$

Hence, c measures the failure of s to be a group homomorphism. This obviously satisfies $c(e_\Gamma,\gamma)=c(\gamma,e_\Gamma)=e_G$. A direct calculation shows that c satisfies (B.8). Moreover, it is $Z(G)$-valued:

$$\begin{aligned} g\ c(\gamma_1,\gamma_2)\ g^{-1} &= s(\gamma_1)\ s(\gamma_1)^{-1}g\ s(\gamma_1)\ s(\gamma_2)\ s(\gamma_1\gamma_2)^{-1}g^{-1}s(\gamma_1\gamma_2)\ s(\gamma_1\gamma_2)^{-1}\\ &= s(\gamma_1)\ \theta_{\gamma_1^{-1}}(g)\ s(\gamma_2)\ \theta_{(\gamma_1\gamma_2)^{-1}}(g^{-1})\ s(\gamma_1\gamma_2)^{-1}\\ &= s(\gamma_1)\ s(\gamma_2)\ \theta_{\gamma_2^{-1}}(\theta_{\gamma_1^{-1}}(g))\theta_{(\gamma_1\gamma_2)^{-1}}(g^{-1})\ s(\gamma_1\gamma_2)^{-1}\\ &= c(\gamma_1,\gamma_2). \end{aligned}$$

Then, the map

$$\Phi\colon G\times_{(\theta,c)}\Gamma\longrightarrow\hat{G},(g,\gamma)\mapsto g\ s(\gamma) \tag{B.9}$$

is a Lie group isomorphism [21, Remark 18.1.8].

For what is to come, we fix $\hat{G}$ a complex reductive Lie group, and $\hat{K}\subset\hat{G}$ a maximal compact Lie subgroup. By a theorem of Mostow $\Gamma\cong\hat{K}/K$, where $K:=\hat{K}\cap G$ is the identity component of $\hat{K}$, and it is a maximal compact subgroup of G (see [38, Th. 3.1]). Since every compact manifold has finitely many connected components, Γ is finite. Moreover, observe that the previous reasoning holds for any extension of Lie groups, for instance $1\to K\to\hat{K}\to\Gamma\to 1$. Given $\bar{\theta}\colon\Gamma\to\mathrm{Out}(K)$, by the results in [53, See Ch I, Sec 2], there is a lift $\theta\colon\Gamma\to\mathrm{Aut}(K)$. More explicitly, that theorem states that letting $K\subset G$ be a maximal compact subgroup, then the short exact subsequence

$$1\to\mathrm{Int}(K)\to\mathrm{Aut}(K)\to\mathrm{Out}(K)\to 1$$

splits. Hence, the composition of $\bar{\theta}\colon\Gamma\to\mathrm{Out}(K)$ with a left splitting $\mathrm{Out}(K)\to\mathrm{Aut}(K)$ yields a lift of θ. Using the universal property of complexifications, we may obtain a lift of $\bar{\theta}\colon\Gamma\to\mathrm{Out}(G)$. By the universal property of complexification, we may see it as $\theta\colon\Gamma\to\mathrm{Aut}(G)$. Moreover, again by the universal property of complexifications, $Z(K)\subset Z(G)$ so any cocycle in K can be seen as a cocycle in G. Then, upon fixing a normalized section, the Lie group isomorphism $\hat{G}\cong G\times_{(\theta,c)}\Gamma$ restricts to an isomorphism $\hat{K}\cong K\times_{(\theta,c)}\Gamma$ if we see $K\times_{(\theta,c)}\Gamma$ as a subgroup of $G\times_{(\theta,c)}\Gamma$ [5, Prop. 2.2]. In particular, $K\subset G$ is Γ-invariant because the lift θ comes from a lift of the homomorphism $\Gamma\to\mathrm{Out}(K)$.

Remark B.14. Let $\hat{G}$ be a complex reductive Lie group as before, which is a complexification of a maximal compact subgroup $\hat{K}$, and let G be the identity component of $\hat{G}$ and $K := \hat{K} \cap G$. First, since the connected component of the identity is always closed [33, Lemma 7.11 and Prop. 7.15], and $\hat{K}$ is compact, it follows that K is compact. Moreover, from the universal property of complexifications, $K^{\mathbb{C}} \cong G$, and G is diffeomorphic to $K \times \mathfrak{k}$ [21, Prop. 15.2.1-(c)]. Since G is connected, K must be connected as well.

B.3.1 Twisted equivariant actions

This section is devoted to introducing what we will call *twisted equivariant actions*. As we shall see, introducing this concept will allow us to obtain principal bundles with connected structure group out of principal bundles with non-connected structure group. Our main reference is [5]. The lift θ induces left actions of Γ on G and $\mathfrak{g}$ given by

$$\Gamma \times G \longrightarrow G, (\gamma, g) \mapsto \theta_\gamma(g), \qquad \Gamma \times \mathfrak{g} \to \mathfrak{g}, (Y, \gamma) \mapsto \mathrm{d}_e\theta_\gamma(Y). \tag{B.10}$$

Definition B.15. Let M be an $\mathcal{S}$-manifold. A (θ, c)-*twisted left (resp. right)* (G, Γ)-*action* is a pair given by:

1. a left (resp. right) G-action by $\mathcal{S}$-isomorphisms,
2. an $\mathcal{S}$-isomorphism $\Gamma \times M \to M, (\gamma, m) \mapsto \gamma \cdot m$ satisfying (for a left action),

$$e_\Gamma \cdot m = m \qquad \forall\ m \in M, \tag{B.11}$$
$$\gamma \cdot (g \cdot m) = g^\gamma \cdot (\gamma \cdot m) \qquad \forall\ m \in M, \gamma \in \Gamma, g \in G, \tag{B.12}$$
$$\gamma_1 \cdot (\gamma_2 \cdot m) = c(\gamma_1, \gamma_2) \cdot ((\gamma_1\gamma_2) \cdot m) \qquad \forall\ m \in M, \gamma_1, \gamma_2 \in \Gamma. \tag{B.13}$$

We also say that M is a (θ, c)*twisted* (G, Γ)-*manifold*.

For a right action, (B.11) is swapped for $m \cdot e_\Gamma = m$, (B.12) and (B.13) for the following

$$(m \cdot g^\gamma) \cdot \gamma = (m \cdot \gamma) \cdot g \qquad \forall\ m \in M, \gamma \in \Gamma, g \in G, \tag{B.14}$$
$$(m \cdot \gamma_1) \cdot \gamma_2 = (m \cdot c(\gamma_1, \gamma_2)) \cdot (\gamma_1\gamma_2) \qquad m \in M, \gamma_1, \gamma_2 \in \Gamma. \tag{B.15}$$

If $G = Z(G)$ is abelian, we simply speak of a (θ, c)-*twisted* Γ-*action*. Given two $\mathcal{S}$-manifolds endowed with (θ, c)-twisted (G, Γ)-equivariant left actions, M and N, an $\mathcal{S}$-map $f\colon M \to N$ is called (θ, c)-*twisted* (G, Γ)-*equivariant* if it satisfies

$$f(g \cdot m) = g \cdot f(m) \quad \text{and} \quad f(\gamma \cdot m) = \gamma \cdot f(m) \qquad \forall\ m \in M, \gamma \in \Gamma. \tag{B.16}$$

When the data (θ, c) and (G, Γ) are clear from the context, we will speak directly of twisted equivariant actions and manifolds.

Since $\hat{G} \cong G \times_{(\theta,c)} \Gamma$, one may think that $\hat{G}$ actions and twisted actions are related. Indeed:

Proposition B.16. Let $\hat{G} = G \times_{(\theta,c)} \Gamma$. There is a bijective correspondence between $\hat{G}$-actions on an $\mathcal{S}$-manifold and (θ, c)-twisted (G, Γ)-actions given as follows:

1. Given a $\hat{G}$-action on M, we define the G-action by restriction and $\gamma \cdot m := (e_G, \gamma) \cdot m$ (cf. (B.9)).
2. Given a (θ, c)-twisted (G, Γ)-action, we define $(g, \gamma) \cdot m = g \cdot (\gamma \cdot m)$ for a right action, and $m \cdot (g, \gamma) = (m \cdot g) \cdot \gamma$.

Moreover, a map $f \colon M \to N$ between sets with (θ, c)-twisted (G, Γ)-actions is (θ, c)-twisted (G, Γ)-equivariant if and only if it is $\hat{G}$-equivariant under the previous correspondence.

Proof. See [5, Prop. 2.7 and Prop. 2.12], the proof is just a sequence of lengthy computations. □

Remark B.17. In our discussion, the lift θ has been fixed, but it may not be unique. The choice of lift, however, does not matter in the end. Without further detail, different choices lead to equivalent categories. We refer the reader to [5, Sec. 4] for details.

B.3.2 Twisted equivariant bundles

Twisted equivariant actions allow us to define analogues to principal bundles, where instead of a regular action we now have a (θ, c)-twisted (G, Γ)-equivariant action. Chapter 3 contains the basic features of principal bundles needed to understand this section. If M is an $\mathcal{S}$-manifold endowed with a left Γ-action by $\mathcal{S}$-isomorphisms, we will say that it is a $(\mathcal{S}, \Gamma)$-manifold.

Both general fiber bundles and principal bundles admit generalizations to twisted equivariant actions.

Definition B.18 (Twisted equivariant fiber bundles). Let M be an $(\mathcal{S}, \Gamma)$-manifold. A *(θ, c)-twisted Γ-equivariant bundle* is an $\mathcal{S}$-fiber bundle $F \to M$ endowed with a (θ, c)-twisted action of Γ on F such that $Z(G)$ acts fiberwise (i.e. for $z \in Z(G), f \in \pi^{-1}(m)$, we have $z \cdot f \in \pi^{-1}(m)$) and $\pi \colon F \to M$ is Γ-equivariant, that is $\pi(\gamma \cdot f) = \gamma \cdot \pi(f)$ for a left action, and $\pi(f \cdot \gamma) = \pi(f) \cdot \gamma$ for a right action.

We can then define *twisted Γ-equivariant sections* $s \colon M \to F$ as usual sections satisfying the Γ-equivariance condition: $s(m \cdot \gamma) = s(m) \cdot \gamma$ for a right action and $s(\gamma \cdot m) = \gamma \cdot s(m)$ for a left action. Observe that this is well-defined thanks to equivariance of the projection π.

Definition B.19 (Twisted equivariant principal G-bundles). Let M be an $(\mathcal{S}, \Gamma)$-manifold. A *(θ, c)-twisted Γ-equivariant principal G-bundle* is a right (θ, c)-twisted equivariant bundle $P \to M$ with a right G-action such that

1. The G and Γ actions make P into a (θ, c)-twisted (G, Γ)-manifold.
2. The right G-action makes $P \to M$ into an $\mathcal{S}$-principal G-bundle.

As usual, if we work with holomorphic bundles, both the Lie group and the base manifold must be complex. Let us remark that, in the previous definition, the G-action

restricts to an action of $Z(G)$ that is the same as the one induced by the twisted equivariant Γ-action. Observe that by proposition B.16, P may be seen as a manifold endowed with a $\hat{G}$-action.

Remark B.20. Just to be very clear, a twisted principal G-bundle is a G-bundle endowed with a right Γ-action, such that the projection map is Γ-equivariant, and the following relations hold

$$(p \cdot g^\gamma) \cdot \gamma = (p \cdot \gamma) \cdot g \qquad (p \cdot \gamma_1) \cdot \gamma_2 = (p \cdot c(\gamma_1, \gamma_2)) \cdot (\gamma_1 \gamma_2) \tag{B.17}$$

for all $p \in P, g \in G, \gamma \in \Gamma$. These are just (B.14) and (B.15) applied to our particular situation.

The notion of morphism has to be modified as well. Given two twisted Γ-equivariant principal bundles with the same structure group G, say P and P', over two Γ-manifolds M and N, a *morphism* $\vartheta\colon P \to P'$ is a twisted equivariant $\mathcal{S}$-morphism such that the induced $\mathcal{S}$-morphism $\vartheta'\colon M \to N$ is Γ-equivariant. That is $\vartheta(p \cdot \gamma) = \vartheta(p) \cdot \gamma$, $\vartheta(p \cdot g) = \vartheta(p) \cdot g$ and $\vartheta'(m \cdot \gamma) = \vartheta'(m) \cdot \gamma$. By proposition B.16, we may equivalently ask ϑ to be $\hat{G}$-equivariant.

$$\begin{array}{ccc} P & \xrightarrow{\vartheta} & P' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ M & \xrightarrow{\vartheta'} & N \end{array}$$

Pullbacks under Γ-equivariant maps behave well: if $P \to Y$ is a twisted equivariant principal G-bundle, and $\varphi\colon X \to Y$ is a Γ-equivariant $\mathcal{S}$-morphism between Γ-manifolds, then $\varphi^* P \to X$ defined as in (3.1) has a natural structure of twisted equivariant principal G-bundle. The Γ action on $\varphi^* P$ is defined component-wise, i.e. $(n, p) \cdot \gamma = (n \cdot \gamma, p \cdot \gamma)$.

Finally, something unexpected happens when we try to apply this construction to associated bundles: the twisting disappears and we get a "genuine" Γ-equivariant bundle, i.e. a fiber bundle with a Γ-action over a Γ-manifold such that the projection is Γ-invariant. Observe that (θ, c)-twisted Γ-equivariant fiber bundles are *not* Γ-equivariant bundles precisely because the twisted Γ-action is *not* an action in the usual sense. This result is based on the following proposition (we state it in general terms).

Proposition B.21. Let $1 \to G \xrightarrow{i} \hat{G} \xrightarrow{\pi} \Gamma \to 1$ be a $\mathcal{S}$-Lie group extension, and suppose that there is a free and proper $\mathcal{S}$-action of $\hat{G}$ on an $\mathcal{S}$-manifold M. Then, there is an $\mathcal{S}$-action of Γ on M/G and the quotient map $\hat{\pi}\colon M \to M/G$ is equivariant for the homomorphism $q\colon \hat{G} \to \Gamma$.

Proof. Consider right actions and define $[m] \cdot \gamma := [m \cdot \hat{g}]$ for any $\hat{g} \in \hat{\pi}^{-1}(\gamma)$. Choosing $\hat{g}_1, \hat{g}_2 \in \pi^{-1}(\gamma)$, we have $\hat{g}_2^{-1} \hat{g}_1 \in \ker \pi = i(G)$ so $[m \cdot \hat{g}_1] = [m \cdot \hat{g}_2 \hat{g}_2^{-1} \hat{g}_1] = [m \cdot \hat{g}_2]$, so the action does not depend on the representative of γ we choose. Similarly, $[m] \cdot \gamma = [m \cdot g] \cdot \gamma$ for $g \in G$ because $i(G) \triangleleft \hat{G}$. Indeed, $[m \cdot \hat{g}] = [m \cdot g \cdot \hat{g}]$ because $g\hat{g} = \hat{g}g'$ for some $g' \in i(G)$. Finally, $\pi\colon M \to M/G$ is equivariant by definition of the action. □

If $\hat{G} = G \times_{(\theta,c)} \Gamma$, and we have a (θ, c)-twisted (G, Γ)-action on M, then we also have a $\hat{G}$ action on M (see proposition B.16). The previous proposition applies and yields a Γ-action on M/G.

Proposition B.22. Let F be a (θ, c)-twisted right (G, Γ)-manifold and $\pi\colon P(M, G) \to M$ be a (θ, c)-twisted Γ-equivariant principal G-bundle over a Γ-manifold M. Then, the associated bundle $P \times_G F$ is a Γ-equivariant fiber bundle with the action $[p, f] \cdot \gamma = [p \cdot \gamma, f \cdot \gamma]$. That is, a fiber bundle with a Γ-action such that the projection is Γ-equivariant.

Proof. Endow $P \times F$ with the component-wise (θ, c)-twisted (G, Γ)-action obtained from that of P and G and apply proposition B.21 to the induced $\hat{G} = G \times_{(\theta,c)} \Gamma$-action. It follows that $[p, f] \cdot \gamma = [p \cdot \gamma, f \cdot \gamma]$ is a right Γ-action, and the projection is equivariant by definition (cf. definition B.18 and section 3.1.1). □

Remark B.23. If F had been a (θ, c)-twisted *left* (G, Γ)-manifold, we would have needed to modify the left twisted equivariant action to obtain a right twisted equivariant action. This is done by setting $f \cdot g := g^{-1} \cdot f$ and $f \cdot \gamma := c(\gamma^{-1}, \gamma)^{-1} \cdot (\gamma^{-1} \cdot f)$.

B.3.3 Reductions of the structure group on twisted equivariant bundles

The aim of this section is to adapt the notion of reduction of the structure group to twisted equivariant principal bundles. More precisely, we want to see how reductions of the structure group of a twisted equivariant principal G-bundle may be endowed with a twisted equivariant structure. Under our usual situation (B.6), let $H \subset G$ be a closed Lie subgroup, and suppose that it is invariant under the left Γ-action induced by θ, i.e. $\gamma \cdot H = H$ for all $\gamma \in \Gamma$.

Proposition B.24. Let $H \subset G$ be a Γ-invariant closed Lie subgroup. The action $G \times G/H \to G/H, (g, g'H) \mapsto gg'H$, and the map $\Gamma \times G/H \to G/H, (\gamma, gH) \mapsto \theta_\gamma(g) \cdot H$ define a θ-twisted left (G, Γ)-action on G/H.

Proof. Let $g, g' \in G$ and $\gamma, \gamma' \in \Gamma$. We have $\gamma \cdot (g'gH) = \theta_\gamma(g') \cdot (\theta_\gamma(g)H) = (g')^\gamma \cdot (\gamma \cdot gH)$, and trivially $\gamma\gamma' \cdot (gH) = (\theta_\gamma \circ \theta_{\gamma'}(g))H = \gamma \cdot (\gamma' \cdot gH)$. □

The previous proposition can be phrased by saying that there is an action of $G \rtimes_\theta \Gamma$ on G/H. If $\hat{G} = G \times_{(\theta,c)} \Gamma$ and the cocycle c takes values in $Z(H)$ (this happens if we choose $H = K$ when the lift θ is obtained from a lift on K), and we define $\hat{H} \cong H \times_{(\theta,c)} \Gamma$ then $G/H \cong \hat{G}/\hat{H}$ (the isomorphism is given by $gH \to g\hat{H}$). On $\hat{G}/\hat{H}$ we have a natural $\hat{G}$- action. Under the isomorphism, this induces a $\hat{G}$-action on G/H, and by proposition B.16, this is equivalent to a (θ, c)-twisted (G, Γ)-action on G/H. Now, proposition B.22 together with remark B.23 implies that $P(G/H)$ is a Γ-equivariant fiber bundle. Then, there is a true action of Γ on $\Gamma(M; P(G/H))$ given by $(\gamma \cdot \sigma)(m) := \sigma(m \cdot \gamma^{-1}) \cdot \gamma$. This is Γ-equivariant with respect to the Γ actions on M and $P(G/H)$ if and only if σ is Γ-invariant for this action. Moreover, the $\hat{G}$-action coincides with that of $G \rtimes_\theta \Gamma$ when we identify the two groups as sets by means of the bijection given by $\hat{G} \cong G \times_{(\theta,c)} \Gamma$.

Proposition B.25. Let $H \subset G$ be a Γ-invariant embedded $\mathcal{S}$-subgroup, and $P(M, G)$ a (θ, c)-twisted Γ-equivariant principal G-bundle over a Γ-manifold M. Let $\sigma \in \Gamma(M; P(G/H))$ be a reduction of the structure group. The bundle $P_\sigma \subset P$ is Γ-invariant if and only if σ is Γ-equivariant. If any of these two equivalent conditions is met, then P_σ has a unique structure of (θ, c)-twisted Γ-equivariant principal H-bundle.

Proof. See [5, Prop 3.21] and the surrounding comments. □

B.3.4 Twisted equivariant bundles and coverings

We present here the most important result proven in [5]. Plainly stated, it says that whenever we have a non-connected complex reductive Lie group $\hat{G}$, and a principal $\hat{G}$-bundle $P_{\hat{G}}$ over an $\mathcal{S}$-manifold X, we may find a covering $Y \to X$, whose deck group is

$\Gamma = \hat{G}/G$, and a twisted equivariant principal bundle $P_G \to Y$. That is, at the expense of working on a covering Y of our original manifold, and acquiring a "twisting", we may get a principal bundle whose structure group is connected out of one whose structure group is non-connected.

Fix a connected $\mathcal{S}$-manifold Y endowed with an $\mathcal{S},\Gamma$-action $\Gamma \to \mathrm{Aut}(Y)$, and let $P(Y,G) \to Y$ be a (θ,c)-twisted equivariant principal G-bundle. By remark 3.2, $P/G \cong Y$, and the G-action on P induces a Γ-action on Y (see proposition B.21). Assuming this Γ-action to be free and proper, $P \to X := Y/\Gamma$ is a principal $\hat{G}$-bundle [**5**, Sec. 4.1]. In that case, $X := Y/\Gamma$ has unique $\mathcal{S}$-structure such that $q\colon Y \to X$ is an $\mathcal{S}$-covering map [**34**, Th. 21.13].

Thus, fix an $(\mathcal{S},\Gamma)$-covering $Y \to X$ (that is, a covering whose deck group is $\mathrm{Deck}_q(Y) = \Gamma$), and suppose that Γ acts on Y on the right. We define two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ as follows.

1. $\mathcal{C}_1$ is the category whose
   - **objects** are principal $\hat{G}$-bundles P over X such that there is an isomorphism $P/G \cong Y$ covering the identity on X (see diagram B.1a) and whose
   - **morphisms** are morphisms of principal $\hat{G}$-bundles.
2. $\mathcal{C}_2$ is the category whose
   - **objects** are (θ,c)-twisted Γ-equivariant principal G-bundles $P \to Y$ such that the twisted Γ-action descends to the action of Γ as covering transformations of the fixed Γ-covering $P/G \cong Y \to X$ (see diagram B.1b, γ means the map induced by the Γ (twisted) action) and whose
   - **morphisms** are (θ,c)-twisted (G,Γ)-equivariant maps $\phi\colon P \to P'$ such that the diagram B.1c commutes and the induced map $\bar{\phi}\colon Y \to Y$ is a covering map that belongs to $Z(\Gamma) \subset \Gamma = \mathrm{Deck}_q(Y)$.

$$\begin{array}{ccc} P/G & \xrightarrow{\phi} & Y \\ {\scriptstyle\hat{\pi}}\downarrow & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{\mathrm{id}_X} & X = Y/\Gamma \end{array}$$

(a) Objects on $\mathcal{C}_1$.

$$\begin{array}{ccc} P & \xrightarrow{\gamma} & P \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ Y & \xrightarrow{\gamma} & Y \end{array}$$

(b) Objects on $\mathcal{C}_2$.

$$\begin{array}{ccc} P & \xrightarrow{\phi} & P' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ Y & \xrightarrow{\bar{\phi}} & Y \end{array}$$

(c) Morphisms in $\mathcal{C}_2$.

Diagram B.1: Diagrams for the equivalence of categories.

In diagram B.1a, $\hat{\pi}$ is the $\mathcal{S}$-morphism defined by $\hat{\pi}([p]) = \pi(p)$ where $[\cdot]$ is the equivalence class for the relation obtained from the quotient by G, and q is the quotient map for $Y \to Y/\Gamma$. Then, the key theorem is the following.

Theorem B.26. There is an isomorphism of categories between $\mathcal{C}_1$ and $\mathcal{C}_2$.

Proof. See [**5**, Sec. 4.1]. □

Remark B.27. In practice, one usually wants to use this result as follows: let (X,ω_X) be a compact Kähler manifold, and let $P_{\hat{G}} \to X$ be a principal $\hat{G}$-bundle. We define $Y := P_{\hat{G}}/G$. Then, we have a Γ-action on Y induced by the $\hat{G}$-action, and $p\colon Y \to X$

is a holomorphic covering (recall that Γ is finite). We endow Y with a Kähler structure via pullback $\omega_Y := p^*\omega_X$. Observe that Y is compact because it is a general fact that finite-sheeted coverings of compact spaces are compact [**40**, Page 341]. Observe that Y is endowed with a natural Γ-action (see proposition B.21). Then, the (θ, c)-twisted Γ principal G-bundle over Y is $P_{\hat{G}} \to Y$. It is common to write this new bundle as $P_G \to Y$ even though as sets, and as manifolds, $P_G = P_{\hat{G}}$. What we get is a *Galois covering* $Y \to X$ with Galois group Γ, i.e. Γ acts on Y by deck transformations, and Y/Γ is isomorphic to X.

B.3.5 Parabolic subgroups of non-connected groups

For non-connected groups $\hat{G}$, we want to extend the definition of parabolic subgroup to allow for non-connected parabolic subgroups. By theorem B.9, a reasonable definition is as follows: we say that a closed subgroup $\hat{P}$ of a complex reductive group $\hat{G}$ is *parabolic* if $\hat{G}/\hat{P}$ is compact. $\hat{P}$ is a parabolic subgroup of $\hat{G}$ if and only if $P := \hat{P} \cap G$ is a parabolic subgroup of G. We will work in the context of an extension of Lie groups given as in (B.6). For a non-connected Lie group $\hat{G}$, and any element $\zeta \in \mathfrak{i}\mathfrak{k}$, we define $\hat{P}_\zeta, \hat{L}_\zeta, \mathfrak{p}_\zeta$ and $\mathfrak{l}_\zeta$ by the same expressions given in proposition B.10, but allowing $g \in \hat{G}$ for the groups. Then, each $\hat{P}_\zeta$ is a parabolic subgroup whose Levi subgroup is $\hat{L}_\zeta$, and their respective Lie algebras are $\mathfrak{p}_\zeta$ and $\mathfrak{l}_\zeta$. Moreover, if we define $P_\zeta \subset G$ as in B.10, then P_ζ is the identity component of $\hat{P}_\zeta$.

Appendix C

The basics of Banach manifolds

The aim of this appendix is to introduce the basics of infinite-dimensional manifolds. This will be used in chapter 2 to give an infinite-dimensional version of the Marsden-Weinstein theorem. This is in turn useful because it provides a rigorous framework to deal with the moment map picture of the Hitchin-Kobayashi correspondence, which is introduced in chapter 4. Some results in this appendix have proofs that are analogous to those for finite-dimensional manifolds, and we shall omit them. We refer the reader to [30], which is a book that develops the theory of smooth manifolds in a very general setting, giving most proofs and definitions for Banach manifolds, and particularizing results to finite-dimensional ones when there are substantial differences.

C.1 Basic definitions

Banach manifolds are manifolds that are locally modelled on Banach spaces. To make sense of smoothness, we first need derivatives on Banach spaces. These are Fréchet derivatives. Let $\mathbf{E}, \mathbf{F}$ be two Banach spaces, we write $\mathcal{L}(\mathbf{E}, \mathbf{F})$ for the Banach space of linear maps defined from $\mathbf{E}$ to $\mathbf{F}$, which is a Banach space when endowed with the operator norm. We define $\mathrm{Aut}(\mathbf{E})$ the space of *toplinear isomorphisms*, that is, vector space isomorphisms of $\mathbf{E}$ that are continuous and have continuous inverses.

Definition C.1 (Fréchet derivative). Let $(\mathbf{E}, ||\cdot||_{\mathbf{E}}), (\mathbf{F}, ||\cdot||_{\mathbf{F}})$ be two Banach spaces, $U \subset \mathbf{E}$ an open set, and $f\colon U \to \mathbf{F}$ a continuous map. We say that f is *Fréchet differentiable* at $x \in U$ if there is a linear map $Df(x) \in \mathcal{L}(\mathbf{E}, \mathbf{F})$ such that

$$\lim_{h \to 0} \frac{||f(x+h) - f(x) - Df(x)(h)||_{\mathbf{F}}}{||h||_{\mathbf{E}}} = 0.$$

Observe that this definition is just the direct generalization to a Banach space of the usual differential of a map; that is, a continuous map is Fréchet differentiable if it has a "best linear approximation". Observe that if f is Fréchet differentiable at every point $x \in U$, then we may define a map $Df\colon U \to \mathcal{L}(\mathbf{E}, \mathbf{F})$. If this map is continuous, we say that f is a Fréchet-$\mathscr{C}^1$ map. We now have a continuous map $Df\colon U \to F'$, where $\mathbf{F}' = \mathcal{L}(\mathbf{E}, \mathbf{F})$, so we may ask whether this new map is Fréchet differentiable. If it is, we define $D^2 f := D(Df)$, and if this new map is continuous, we say that f is a Fréchet-$\mathscr{C}^2$ map. Inductively, we define $D^k f := D^{k-1}(Df)$, $k \in \mathbb{N}$, and we say that f is Fréchet-smooth or a Fréchet-$\mathscr{C}^\infty$ map if it is a Fréchet-$\mathscr{C}^k$ map for every $k \in \mathbb{N}$. Whenever it is clear from the context that we refer to Fréchet regularity, we will omit the word Fréchet. Observe that this reduces to the usual definition of differential if the Banach spaces are eucliean space. If a Fréchet derivative exists, then it is unique.

Definition C.2 (Banach manifold). Let $\mathcal{M}$ be a Hausdorff topological space. A $\mathscr{C}^k$ *atlas* on $\mathcal{M}$ is a collection of triplets $\{(U_\alpha, \varphi_\alpha, \mathbf{E}_\alpha)\}_{\alpha\in A}$, where each U_α is an open set, $(\mathbf{E}_\alpha, ||\cdot||_\alpha)$ is a Banach space, and $\varphi_\alpha\colon U_\alpha \to \mathbf{E}_\alpha$ is a continuous map satisfying:

1. $\{U_\alpha\}_{\alpha\in A}$ is an open covering of $\mathcal{M}$
2. Each $\varphi_\alpha\colon U_\alpha \to \varphi_\alpha(U_\alpha) \subset \mathbf{E}_\alpha$ is a homeomorphism onto its image, which is open
3. If $U_\alpha \cap U_\beta \neq \varnothing$, then

$$\varphi_\alpha \circ \varphi_\beta^{-1}\colon \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

is a Fréchet $\mathscr{C}^k$ map.

Observe that we do not require the Banach spaces that model the manifold locally to be the same. However, whenever $U_\alpha \cap U_\beta \neq \varnothing$, it follows from the definition that $\mathbf{E}_\alpha$ and $\mathbf{E}_\beta$ are isomorphic (indeed, $D(\varphi_\alpha \circ \varphi_\beta^{-1})$ is a linear isomorphism between them). When all the spaces are the same (call it $\mathbf{E}$), we say that $\mathcal{M}$ is an $\mathbf{E}$-*manifold*. Each connected component of a Banach manifold is an $\mathbf{E}$-manifold, and usual finite-dimensional smooth manifolds are $\mathbb{R}^n$-manifolds. As in the finite-dimensional case, every atlas is contained within a maximal one. Obvious modifications to the previous definition allow us to define *Hilbert manifolds* and *locally convex manifolds* for manifolds that are locally modelled on those spaces. Observe that Hilbert manifolds are just a special case of Banach manifolds, but this is not the case for locally convex manifolds because we do not have norms to define Fréchet derivatives there. Smooth maps are defined analogously: if M and N are two Banach manifolds, and $f\colon \mathcal{M} \to \mathcal{N}$ is a map, we say that it is smooth if whenever (U, φ) and (V, ψ) are smooth charts satisfying $f(U) \subset V$, then the *localization of* f defined by $\psi \circ f \circ \varphi^{-1}\colon \varphi(U) \to \psi(V)$ is a $\mathscr{C}^\infty$ map (in the Fréchet sense).

C.1.1 The tangent space to a Banach manifold

In the category of finite-dimensional smooth manifolds, the tangent space to a manifold at a point can be defined as the space of derivations acting on germs of functions at that point, and as the set of equivalence classes of curves passing through that point. In the infinite-dimensional setting, these two definitions are no longer equivalent, and the definition chosen is the second one. With this idea in mind, let $I \subset \mathbb{R}$ be an interval, and $\mathcal{M}$ be a Banach manifold, a smooth curve is a smooth map $\alpha\colon I \to \mathcal{M}$. Let $\alpha\colon I \to \mathcal{M}$ and $\beta\colon I' \to \mathcal{M}$ be two smooth curves satisfying $\alpha(0) = \beta(0) = m \in \mathcal{M}$. We say that they are *equivalent* if there is a chart $(U, \psi, \mathbf{E})$ such that $(\psi \circ \alpha)'(0) = (\psi \circ \beta)'(0)$, where we defined

$$(\psi \circ \alpha)'(t) = [D(\psi \circ \alpha)(t)](1). \tag{C.1}$$

In that case, (C.1) holds for every chart. Observe that the right-hand-side makes sense; $D(\psi \circ \alpha)(t) \in \mathcal{L}(\mathbb{R}, \mathbf{E})$, so when evaluating at 1, the result is a vector in $\mathbf{E}$. This defines an equivalence relation on the set of all smooth curves passing through m at $t = 0$. A *tangent vector* is thus an equivalence relation of curves. We denote it by $[\alpha, m]$, or simply $[\alpha]$ if the point is clear from the context. The set of all tangent vectors is called the *tangent space* of $\mathcal{M}$ at m, and denoted $T_m\mathcal{M}$ as usual. This is indeed a vector space. To endow it with a linear structure, we can use any chart $(U, \phi, \mathbf{E})$, which defines a bijection $\bar{\phi}\colon T_m\mathcal{M} \to \mathbf{E}, [\alpha] \mapsto (\phi \circ \alpha)'(0)$. That is the content of the following lemma.

Lemma C.3. Let $\mathcal{M}$ be a Banach manifold.

1. Let $\phi\colon \mathcal{M} \to (\mathbf{E}, ||\cdot||_{\mathbf{E}})$ be a chart around $m \in \mathcal{M}$, and set $p_m := \phi(m) \in \mathbf{E}$. Then, the following maps are inverses

$$\begin{array}{rccc} \bar{\phi}\colon & T_m\mathcal{M} & \longrightarrow & \mathbf{E} \\ & [\alpha] & \mapsto & (\phi\circ\alpha)'(0) \end{array} \qquad \begin{array}{rccc} \bar{\phi}^{-1}\colon & \mathbf{E} & \longrightarrow & T_m\mathcal{M} \\ & v & \mapsto & [t \mapsto \phi^{-1}(p_m + t\cdot v)]. \end{array}$$

2. For all charts ϕ, ψ around m, we have $\bar{\psi}\circ\bar{\phi}^{-1} = D(\psi\circ\phi^{-1})(p_m)$.

3. $T_m\mathcal{M}$ admits a unique vector space structure and topology making each $\bar{\phi}$ into a linear isomorphism.

Proof. The proof on [51, Lemma 1.41] can be adapted to this situation. □

Remark C.4. As happens in the finite-dimensional case, if $\mathbf{E}$ is a Banach space, then $\mathrm{id}_{\mathbf{E}}\colon \mathbf{E} \to \mathbf{E}$ defines a chart, and $\overline{\mathrm{id}_{\mathbf{E}}}$ allows us to identify $\mathbf{E}$ and $T_e\mathbf{E}$ for every $e \in \mathbf{E}$. In particular, if $\mathbf{E} = \mathbb{R}$ we define $\frac{\mathrm{d}}{\mathrm{d}t}|_t := (\overline{\mathrm{id}_{\mathbb{R}}})^{-1}(1) \in T_t\mathbb{R}$.

We can then define the differential of a smooth map; if $f\colon \mathcal{M} \to \mathcal{N}$ is a smooth map between smooth Banach manifolds, then we define $\mathrm{d}_m f\colon T_m\mathcal{M} \to T_{f(m)}\mathcal{N}, [\alpha, m] \mapsto [f\circ\alpha, f(m)]$. This map is linear. Using the identification above, the differential of a map whose codomain is a Banach space is a linear map of the tangent space taking values in the same Banach space.

Many of the usual constructions of finite-dimensional manifold theory carry to the Banach manifold setting. Firstly, the tangent bundle associated to a Banach manifold is defined as usual:

$$T\mathcal{M} := \bigsqcup_{m\in\mathcal{M}} T_m\mathcal{M},$$

and it is a Banach manifold, endowed with a smooth projection $\pi([\alpha, m]) = m$, whose Banach manifold structure is obtained by requiring that for every chart $(U, \phi, \mathbf{E})$, the map $(\pi^{-1}(U), \Phi)$ is a diffeomorphism, where $\Phi\colon \pi^{-1}(U) \to \phi(U)\times\mathbf{E}, u \mapsto (\pi(u), \bar{\phi}(u))$. These maps define an atlas (see [51, Sec. 1.1.5]). Now, vector fields are defined as smooth sections of $T\mathcal{M}$, and the vector space of all vector fields in $\mathcal{M}$ is denoted $\mathfrak{X}(\mathcal{M})$.

Remark C.5. We still want to have vector fields acting on smooth real-valued functions to provide a Lie algebra structure on $\mathfrak{X}(\mathcal{M})$. This requires a few comments.

1. Let $f\colon \mathcal{M} \to \mathbb{R}$ be a smooth map, then, $\mathrm{d}_m f([\alpha, m]) = [f\circ\alpha, f(m)] \in T_{f(m)}\mathbb{R} \cong \mathbb{R}$ where the last isomorphism is given by $\overline{\mathrm{id}_{\mathbb{R}}}$. Thus, to see it as a real number, we understand $\mathrm{d}_m f([\alpha, m]) := \overline{\mathrm{id}_{\mathbb{R}}}([f\circ\alpha, f(m)]) = (f\circ\alpha)'(0)$, seeing the latter as the derivative at 0 of a smooth real function. That is, we essentially recover the standard finite-dimensional result: the differential of a smooth map acting on a vector is the derivative at 0 of $f\circ\alpha$ where α is a smooth curve satisfying $\alpha(0) = 0$ and $\alpha'(0)$ is the vector.

2. A vector field does induce a derivation on the algebra of smooth real-valued functions, which is given by $X(f)(m) := \mathrm{d}_m f(X_m)$.

As in the finite-dimensional setting, given two vector fields $X, Y \in \mathfrak{X}(\mathcal{M})$, there is a unique vector $[X, Y]$ whose action on a smooth function is given by $[X, Y](f) := X(Y(f)) - Y(X(f))$ [30, Prop. V-1.1 and Prop V-1.3]. The vector field $[X, Y]$ is called the *Lie bracket* of X and Y. It is bilinear, antisymmetric, and satisfies the Jacobi identity, and hence defines a Lie algebra structure on $\mathfrak{X}(\mathcal{M})$ (see [30, Cor. 1.4]).

C.1.2 Immersions, submersions and embeddings

The notions of immersion, submersion and embedding have generalizations to Banach manifolds. First, let $\mathcal{M}$ be a Banach manifold, and $\mathcal{N} \subset \mathcal{M}$. We say that $\mathcal{N}$ is a *Banach-submanifold* if for every $m \in \mathcal{N}$, there is a chart $(U, \varphi, \mathbf{E})$ of $\mathcal{M}$, such that $m \in \mathcal{M}$, and a closed subspace $\mathbf{F} \subset \mathbf{E}$ such that

$$\varphi(U \cap \mathcal{N}) = \varphi(U) \cap \mathbf{F}.$$

Moreover, given a Banach-space $(\mathbf{E}, ||\cdot||_{\mathbf{E}})$, and a closed vector subspace $\mathbf{F} \subset \mathbf{E}$, we say that $\mathbf{F}$ *splits* in $\mathbf{E}$, if there is another closed vector subspace $\mathbf{G} \subset \mathbf{E}$ such that $\mathbf{E} = \mathbf{F} \oplus \mathbf{G}$. We say that $\mathbf{G}$ is a *topological complement* of $\mathbf{F}$ in $\mathbf{E}$. Observe that every closed subspace of a Hilbert space splits (just take its orthogonal complement).

Definition C.6. Let $\mathcal{M}$ and $\mathcal{N}$ be two Banach manifolds. Let $f\colon \mathcal{M} \to \mathcal{N}$ be smooth.

1. We say that f is a *(Banach)-immersion* if $\mathrm{d}_m f$ is injective and its image is a closed subspace of $T_{f(m)}\mathcal{N}$ that splits for each $m \in \mathcal{M}$.
2. We say that f is a *(Banach)-submersion* if $\mathrm{d}_m f$ is surjective and $\ker \mathrm{d}_m f$ splits for every $m \in \mathcal{M}$. We say that $f(m)$ is a *regular value* of f.
3. We say that f is a *(Banach)-embedding* if it is an immersion and a homeomorphism into its image.

If f is an embedding, then $f(\mathcal{M}) \subset \mathcal{N}$ is a submanifold (in the sense defined above). It is sometimes called an *embedded submanifold*. As we see, the main difference with respect to finite-dimensional manifolds is the split condition (which is always satisfied in finite-dimensional spaces). The split condition is needed to recover the local immersion and submersion theorems, that is, to see immersions as inclusions and submersions as projections in suitable coordinates.

C.1.3 Banach Lie groups

Unsurprisingly, a *Banach-Lie group* is defined as a group that has the structure of a Banach manifold, and such that the product and inversion maps are smooth, and a *Hilbert-Lie group* is defined analogously. In this setting, right and left-translations L_g and R_g are diffeomorphisms, just as in the finite-dimensional case. The Lie algebra is also defined quite similarly to the finite-dimensional case. Firstly, two vector fields $X \in \mathfrak{X}(\mathcal{M}), Y \in \mathfrak{X}(\mathcal{N})$ are *f-related* (where $f\colon \mathcal{M} \to \mathcal{N}$ is a smooth map) if $\mathrm{d}_m f(X_m) = Y_{f(m)}$ for all $m \in \mathcal{M}$. We thus say that a vector field $X \in \mathfrak{X}(G)$ is *left-invariant* if it is L_g-related to itself. The *Lie algebra* of G, $\mathrm{Lie}(G)$ is the set of all left invariant vector fields on G, which can be identified with T_eG through the map $X_p := \mathrm{d}_e L_g(X_e)$ (we will reserve the notation $\mathfrak{g}$ for the Lie algebra of finite-dimensional Lie groups).

Lemma C.7. Let G be a Banach-Lie group. Then, $\mathrm{Lie}(G)$ is a Lie subalgebra of $\mathfrak{X}(G)$.

Proof. It is obvious that $\mathrm{Lie}(G)$ is a vector space, so we need only check that it is closed under the Lie bracket operation, but this follows from the fact that if X_i and Y_i are f-related, $i = 1, 2$, then $[X_1, X_2]$ and $[Y_1, Y_2]$ are f-related as well (see [**30**, Page 117]). □

Finally, Banach-Lie groups have a well-defined exponential map. This is not true for Lie groups modelled on arbitrary locally convex spaces. The definition is almost identical

to the finite-dimensional one. If $\alpha\colon I \to M$ is a smooth curve, we define

$$\dot{\alpha}(t) := \mathrm{d}_t\alpha\left(\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_t\right) \in T_{\alpha(t)}\mathcal{M}. \tag{C.2}$$

We say that a curve α is an *integral curve* of a vector field X if $X_{\alpha(t)} = \dot{\alpha}(t)$ for every $t \in I$. Given a Banach manifold, every vector field admits an integral curve defined on a maximal interval. The exponential map is just defined as

$$\begin{array}{rccc} \exp\colon & \mathrm{Lie}(G) & \longrightarrow & G \\ & X & \mapsto & \alpha(1) \end{array}$$

where α is the unique integral curve of X satisfying $\alpha(0) = e$. See [13] for more details. Finally, the adjoint and coadjoint actions of G on $\mathfrak{g}$ and $\mathfrak{g}^*$ admit direct generalizations to this context. To define Ad, simply observe that the map $C_g := L_{g^{-1}} \circ R_g$ is a diffeomorphism, hence we may just define $\mathrm{Ad}_g(X) := \mathrm{d}_e C_g(X)$ for $X \in \mathfrak{g}$ and $g \in G$. Similarly, we can just define Ad^* by setting $(\mathrm{Ad}^*_g\xi, X) := (\xi, \mathrm{Ad}_{g^{-1}}(X))$.

C.1.4 Differential forms in Banach manifolds.

Banach manifolds are sufficiently well-behaved that we can still see differential forms as sections of adequate vector bundles (which is not the case for general infinite-dimensional manifolds, see for example [51, Appendix E] for details). The definition of vector bundle admits a straightforward generalization to Banach manifolds by simply asking them to be locally isomorphic to the product of the base manifold and the Banach space, and asking that, for every pair of trivializations $(U, \Phi), (V, \Psi)$, the map $U \cap V \to \mathrm{Aut}(\mathbf{E}), m \mapsto \Phi_m \circ \Psi_m^{-1}$ is smooth.

Let $\mathrm{L}^r_{\mathrm{Alt}}(T\mathcal{M})$ be the vector bundle over $\mathcal{M}$ whose fibers are the Banach spaces of multilinear alternating maps from $T_m\mathcal{M}$ to $\mathbb{R}$. A differential form on $\mathcal{M}$ is defined as a section of this bundle. We keep the usual notation $\Omega^p_{\mathbb{R}}(\mathcal{M})$ for p-forms. As in the finite-dimensional case, $\Omega^p_{\mathbb{R}}(\mathcal{M})$ is a real vector space, and a $\mathscr{C}^\infty(\mathcal{M};\mathbb{R})$-module.

To define the exterior differential, one must observe that the usual definition in terms of local coordinates no longer makes sense in Banach manifolds. Instead, the exterior differential is defined using a coordinate-free relation that holds in the finite-dimensional setting as well (See [51, Prop. E.3] or [30, Prop. 3.2]).

Proposition C.8 (Exterior differential). Let $\mathcal{M}$ be a Banach manifold, and $\omega \in \Omega^p_{\mathbb{R}}(\mathcal{M})$. Then, there is a unique $(p+1)$-form on $\mathcal{M}$ such that for all smooth vector fields $X_1, \dots X_{p+1} \in \mathfrak{X}(\mathcal{M})$,

$$\begin{aligned} \mathrm{d}\omega(X_1, \dots, X_{p+1}) = &\sum_{i=1}^{p+1} (-1)^{i-1} X_i\big(\omega(X_1, \dots, \hat{X}_i, \dots, X_{p+1})\big) \\ &+ \sum_{1 \le i < j \le p+1} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots, \hat{X}_i, \dots, \hat{X}_j, \dots, X_{p+1}). \end{aligned} \tag{C.3}$$

Again, Banach manifolds are well-behaved enough so that most properties of the usual exterior differential are preserved. The exterior or wedge product is again defined analogously to the finite-dimensional case (see [30, Chapter V, section 3] for details).

C.2 Basics of locally convex spaces

First, recall that a *topological vector space*, TVS for short, is a vector space endowed with a topology such that addition and scalar products are continuous. Morphisms in the

category **TVS** are continuous linear maps.

Definition C.9 (Seminorm). Let V be a TVS, a *seminorm* is a map $p\colon V \to [0,\infty)$ satisfying

1. $p(\lambda v) = |\lambda| \cdot p(v)$ for every $\lambda \in \mathbb{R}, v \in V$.
2. $p(v+w) \leq p(v) + p(w)$ for all $v, w \in V$.

Observe that a seminorm is just a norm for which the condition $p(v) = 0$ if and only if $v = 0$ is omitted. If we consider $p_i^{-1}(0)$, which is a vector subspace of V, then the quotient $V/p_i^{-1}(0)$ has a natural norm $||x + p_i^{-1}(0)|| := p(x)$ for any representative x of the equivalence class. Hence, the quotient has a natural topology as a normed vector space. We can now define locally convex spaces.

Definition C.10 (Locally convex TVS). Let V be a TVS, we say that it is *locally convex* if there is a family of continuous seminorms $\{p_i\}_{i\in I}$ such that:

1. The topology of V is the initial topology with respect to the canonical projections $q_i\colon V \to V/p_i^{-1}(0)$.
2. If $v \in V$ and $p_i(v) = 0$ for all $i \in I$, then $v = 0$.

The second property of the definition implies that V is Hausdorff. [**51**, Prop. A.10] gives explicitly a subbase of the TVS topology on V and gives a proof of the Hausdorff condition. Observe that asking for V to be a TVS is superfluous. Indeed, if V is a vector space endowed with the initial topology with respect to the natural quotient maps then sum and product by scalars are continuous. Let us see it for sums. Let τ be the initial topology on V, and τ_i be the metric topology on $V_i := V/p_i^{-1}(0)$ (which turns each V_i into a TVS). Consider the map $s\colon V \times V \to V, (x,y) \mapsto x + y$.

By the universal property of the initial topology, s is continuous if and only if each $q_i \circ s$ is continuous. However, observe that the diagram commutes. s_i is the sum in V_i (which is continuous). Hence, $q_i \circ s = s_i \circ (q_i \times q_i)$, the right-hand side being continuous. This means that s is continuous. The same reasoning works for scalar multiplication.

$$\begin{array}{ccc} (V,\tau) & \xrightarrow{\quad q_i \quad} & (V_i,\tau_i) \\ \uparrow{\scriptstyle s} & & \uparrow{\scriptstyle s_i} \\ (V\times V, \tau\times\tau) & \xrightarrow{\quad q_i\times q_i \quad} & (V_i\times V_i, \tau_i\times\tau_i) \end{array}$$

Locally convex TVS are important because they retain one of the most celebrated and useful results of functional analysis: the Hahn-Banach theorem.

Theorem C.11 (Hahn-Banach). Let V be a locally convex TVS. Then, for every pair of distinct points $x, y \in V$, there is a continuous linear map $\alpha\colon V \to \mathbb{R}$ such that $\alpha(x) \neq \alpha(y)$.

References

[1] Ralph Abraham, Jerrold E. Marsden, and Tudor S. Ratiu. *Manifolds, tensor analysis, and applications*. English. 2nd ed. Applied mathematical sciences; v. 75. New York: Springer-Verlag, 1988. ISBN: 0387967907.

[2] Luis Alvarez-Consul, Mario Garcia-Fernandez, and Oscar Garcia-Prada. "Coupled equations for Kähler metrics and Yang–Mills connections". In: *Geometry & Topology* 17.5 (2013), pp. 2731–2812. DOI: 10.2140/gt.2013.17.2731.

[3] Boudjemaa Anchouche and Indranil Biswas. "Einstein-Hermitian connections on polystable principal bundles over a compact Kähler manifold". In: *American Journal of Mathematics* 123.2 (2001), pp. 207–228. DOI: 10.1353/ajm.2001.0007.

[4] Michael Francis Atiyah and Raoul Bott. "The yang-mills equations over riemann surfaces". In: *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* 308.1505 (1983), pp. 523–615.

[5] G Barajas et al. "Non-connected Lie groups, twisted equivariant bundles and coverings". In: *Geometriae Dedicata* 217.2 (2023), p. 27. DOI: https://doi.org/10.1007/s10711-022-00764-w.

[6] Pilar Benito, Javier Rández-Ibáñez, and Jorge Roldán-López. "Skew-Adjoint Maps and Quadratic Lie Algebras". In: *Mediterranean Journal of Mathematics* 21.3 (May 2024), p. 115. ISSN: 1660-5454. DOI: 10.1007/s00009-024-02656-7.

[7] D. Bleecker. *Gauge Theory and Variational Principles*. Dover Books on Mathematics. Dover Publications, Incorporated, 2013. ISBN: 9780486151878.

[8] Steven B Bradlow, Oscar García-Prada, and Ignasi Mundet I Riera. "Higgs pairs, twisted equivariant structures and non-connected groups". To be published. 2026.

[9] Steven B Bradlow, Oscar García-Prada, and Ignasi Mundet i Riera. "Relative Hitchin–Kobayashi correspondences for principal pairs". In: *Quarterly Journal of Mathematics* 54.2 (2003), pp. 171–208. DOI: 10.1093/qmath/hag013.

[10] T. Bröcker and T. Dieck. *Representations of Compact Lie Groups*. Graduate Texts in Mathematics. Springer Berlin Heidelberg, 2003. ISBN: 9783540136781.

[11] Bram Brongers. *Homological Mirror Symmetry and the Thomas-Yau Conjecture*. Master's Thesis at the University of Groningen supervised by N. Martynchuk and M. Kool. Last accesed on 06/06/2026. 2023. URL: https://fse.studenttheses.ub.rug.nl/30506/17/mMATH_2023_BrongersBA.pdf.

[12] Daniel Bump. *Lie Groups*. 2nd ed. Graduate Texts in Mathematics. Springer New York, 2013. ISBN: 978-1-4614-8024-2.

[13] C. T. J. Dodson, George Galanis, and Efstathios Vassiliou. *Geometry in a Fréchet Context: A Projective Limit Approach*. London Mathematical Society Lecture Note Series. Cambridge University Press, 2015.

[14] S. K. Donaldson. "A new proof of a theorem of Narasimhan and Seshadri". In: *Journal of Differential Geometry* 18.2 (1983), pp. 269–277. DOI: 10.4310/jdg/1214437664.

[15] O. Garcia-Prada and R.O. Wells. *Differential Analysis on Complex Manifolds*. Graduate Texts in Mathematics. Springer New York, 2007. ISBN: 9780387738925.

[16] Oscar Garcia-Prada, Peter B. Gothen, and Ignasi Mundet i Riera. *The Hitchin-Kobayashi correspondence, Higgs pairs and surface group representations*. 2012. arXiv: 0909.4487 [math.DG]. URL: https://arxiv.org/abs/0909.4487.

[17] Oscar García-Prada and André Oliveira. *Connectedness of Higgs bundle moduli for complex reductive Lie groups*. 2016. arXiv: 1408.4778 [math.AG]. URL: https://arxiv.org/abs/1408.4778.

[18] Daniele Giannetto. *Infinite dimensional symplectic reduction and the dynamics of a rigid body moving in a perfect fluid*. Master's Thesis at Università degli Studi di Padova. Last accesed on 27/05/2026. 2024. URL: https://thesis.unipd.it/handle/20.500.12608/68351.

[19] M.J.D. Hamilton. *Mathematical Gauge Theory: With Applications to the Standard Model of Particle Physics*. Universitext. Springer International Publishing, 2018. ISBN: 9783319684383.

[20] Peter Heinzner and Alan Huckleberry. "Analytic Hilbert Quotients". In: *Several Complex Variables*. Ed. by Michael Schneider and Yum-Tong Siu. Mathematical Sciences Research Institute Publications. Cambridge University Press, 2000, pp. 309–350.

[21] J. Hilgert and K.H. Neeb. *Structure and Geometry of Lie Groups*. Springer Monographs in Mathematics. Springer New York, 2011. ISBN: 9780387847948.

[22] Victoria Hoskins. *Geometric invariant theory and symplectic quotients*. Lecture notes for a course given at Freie Universität Berlin. Last accesed on 05/06/2026. URL: https://userpage.fu-berlin.de/hoskins/GITnotes.pdf.

[23] James E Humphreys. *Linear Algebraic Groups*. Vol. 21. Graduate Texts in Mathematics. New York: Springer-Verlag, 1975. ISBN: 978-0-387-90108-4.

[24] D. Huybrechts. *Complex Geometry: An Introduction*. Universitext. Springer Berlin Heidelberg, 2005. ISBN: 9783540266877.

[25] A.W. Knapp. *Lie Groups Beyond an Introduction*. Progress in Mathematics. Birkhäuser Boston, 2013. ISBN: 9781475724530.

[26] Shoshichi Kobayashi. *Differential Geometry of Complex Vector Bundles*. Vol. 15. Publications of the Mathematical Society of Japan. Princeton, N.J. and Tokyo: Princeton University Press and Iwanami Shoten, 1987. ISBN: 978-0-691-08467-1.

[27] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry, Vol. I*. Wiley Classics Library. Reprint of the 1963 original. MR 97c:53001a. John Wiley & Sons, 1996. ISBN: 978-0-471-15733-5.

[28] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry, Vol. II*. Vol. 15. Wiley Classics Library. Reprint of the 1969 original. MR 97c:53001b. New York-London-Sydney: John Wiley & Sons, 1969. ISBN: 978-0-471-15732-8.

[29] Andreas Kriegl. *Advanced Functional Analysis: Locally Convex Spaces and Spectral Theory*. Lecture notes for a course given at Wien University. Last accesed on 05/06/2026. July, 2019. URL: https://www.mat.univie.ac.at/~kriegl/Skripten/2019SSe.pdf.

[30] Serge Lang. *Differential and Riemannian manifolds*. 3rd ed. Graduate texts in mathematics; New York : Springer-Verlag, 1995.

[31] Dong Hoon Lee. *The Structure of complex Lie groups*. Boca Raton etc: Chapma & Hall/CRC, 2002. ISBN: 1584882611.

[32] John Lee. *Introduction to Riemannian manifolds*. New York, NY: Springer Berlin Heidelberg, 2018. ISBN: 978-3-319-91754-2.

[33] John M. Lee. *Introduction to complex manifolds*. Graduate studies in mathematics, volume 244. Providence, Rhode Island: American Mathematical Society. ISBN: 9781470476953.

[34] John M. Lee. *Introduction to Smooth Manifolds*. 2nd ed. New York: Springer, 2012. ISBN: 978-1-4419-9982-5.

[35] John M. Lee. *Introduction to Topological Manifolds*. 2nd ed. Vol. 202. Graduate Texts in Mathematics. New York: Springer, 2011. ISBN: 978-1-4419-7939-1.

[36] Jerrold Marsden and Alan Weinstein. "Reduction of symplectic manifolds with symmetry". In: *Reports on Mathematical Physics* 5.1 (1974), pp. 121–130. ISSN: 0034-4877. DOI: https://doi.org/10.1016/0034-4877(74)90021-4. URL: https://www.sciencedirect.com/science/article/pii/0034487774900214.

[37] P.K. Mitter and C. M. Viallet. “On the bundle of connections and the gauge orbit manifold in Yang-Mills theory”. In: *Communications in Mathematical Physics* 79.4 (1981), pp. 457–472. DOI: https://doi.org/10.1007/BF01209307.

[38] G. D. Mostow. “Self-Adjoint Groups”. In: *Annals of Mathematics* 62.1 (1955), pp. 44–55. ISSN: 0003486X, 19398980. DOI: https://doi.org/10.2307/2007099. URL: http://www.jstor.org/stable/2007099 (visited on 06/05/2026).

[39] D. Mumford, J. Fogarty, and F. Kirwan. *Geometric Invariant Theory*. Ergebnisse der Mathematik und Ihrer Grenzgebiete, 3 Folge/A Series of Modern Surveys in Mathematics Series. Springer Berlin Heidelberg, 1994. ISBN: 9783540569633.

[40] J.R. Munkres. *Topology*. Featured Titles for Topology. Prentice Hall, Incorporated, 2000. ISBN: 9780131816299.

[41] Vicente Muñoz, Ángel González-Prieto, and Juan Ángel Rojo. *Geometry and Topology of Manifolds: Surfaces and Beyond*. Vol. 208. Graduate Studies in Mathematics. American Mathematical Society, 2020. ISBN: 978-1-4704-6132-4.

[42] Masayoshi Nagata. “Invariants of group in an affine ring”. In: *Journal of Mathematics of Kyoto University* 3.3 (1963), pp. 369–378.

[43] M. S. Narasimhan and C. S. Seshadri. “Stable and Unitary Vector Bundles on a Compact Riemann Surface”. In: *Annals of Mathematics* 82.3 (1965), pp. 540–567. ISSN: 0003486X, 19398980. URL: http://www.jstor.org/stable/1970710 (visited on 05/27/2026).

[44] A. Ramanathan. “Moduli for principal bundles over algebraic curves: I”. In: *Proceedings of the Indian Academy of Sciences - Mathematical Sciences* 106.3 (Aug. 1996), pp. 301–328. ISSN: 0973-7685. DOI: 10.1007/BF02867438.

[45] A. Ramanathan. “Stable principal bundles on a compact Riemann surface”. In: *Mathematische Annalen* 213.2 (1975), pp. 129–152. DOI: https://doi.org/10.1007/BF01343949.

[46] Ignasi Mundet i Riera. *A Hitchin-Kobayashi correspondence for Kaehler fibrations*. 1999. arXiv: math/9901076 [math.DG]. URL: https://arxiv.org/abs/math/9901076.

[47] G. Rudolph and M. Schmidt. *Differential Geometry and Mathematical Physics: Part I. Manifolds, Lie Groups and Hamiltonian Systems*. Theoretical and Mathematical Physics. Springer Netherlands, 2012. ISBN: 9789400753457.

[48] Gerd Rudolph and Matthias Schmidt. *Differential Geometry and Mathematical Physics. Part II. Fibre Bundles, Topology and Gauge Fields*. Theoretical and Mathematical Physics. Springer, 2017. ISBN: 978-94-024-0959-8. DOI: 10.1007/978-94-024-0959-8.

[49] Robin Rüßmann. *Tensor Products of Hilbert Spaces*. Master’s Thesis at Technische Universität Kaiserslautern Supervised by Klaus Ritter. Last accesed on 05/06/2026. 2020. URL: https://kluedo.ub.rptu.de/frontdoor/deliver/index/docId/7459/file/Tensor+Products+of+Hilbert+Spaces.pdf.

[50] Sheila Sandon. *On the Chern correspondence for principal fibre bundles with complex reductive structure group*. Master’s Thesis at Leiden University. Supervised by Martin Lubke. Last accesed on 13/05/2026. 2005. URL: https://hdl.handle.net/1887/3597572.

[51] Alexander Schmeding. *An Introduction to Infinite-Dimensional Differential Geometry*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2022.

[52] Lars Martin Sektnan. *Canonical metrics and stability on holomorphic vector bundles*. Lecture notes for a course given at Aarhus University. Last accesed on 12/05/2026. 2021. URL: https://sites.google.com/cirget.ca/lars-sektnan/bundles.

[53] Jean de Siebenthal. “Sur les groupes de Lie compacts non connexes”. In: *Commentarii Mathematici Helvetici* 31.1 (Dec. 1956), pp. 41–89. ISSN: 1420-8946. DOI: 10.1007/BF02564352.

[54] A.C. da Silva. *Lectures on Symplectic Geometry*. Lecture Notes in Mathematics n.º 1764. Springer, 2001. ISBN: 9783540421955.

[55] S.B. Sontz. *Principal Bundles: The Classical Case*. Universitext. Springer International Publishing, 2015. ISBN: 9783319147642.

[56] S. Subramanian and A. Ramanathan. “Einstein-Hermitian connections on principal bundles and stability.” In: *Journal für die reine und angewandte Mathematik* 1988.390 (1988), pp. 21–31. DOI: doi:10.1515/crll.1988.390.21.

[57] Gábor Székelyhidi. *An Introduction to Extremal Kähler Metrics*. Vol. 152. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 2014, pp. xvi+191. ISBN: 978-1-4704-1047-6.

[58] Andrei Teleman. "The Newlander-Nirenberg Theorem for Principal Bundles". In: *Mathematische Zeitschrift* 306.2 (Jan. 2024), p. 20. ISSN: 1432-1823. DOI: 10.1007/s00209-023-03413-4.

[59] Samuel Trautwein. *A survey of the GIT picture for the Yang-Mills equation over Riemann surfaces*. 2015. arXiv: 1511.08122 [math.DG]. URL: https://arxiv.org/abs/1511.08122.

[60] Loring W Tu. *Differential Geometry: Connections, Curvature, and Characteristic Classes*. Cham: Springer, 2017. ISBN: 978-3-319-55082-4.

[61] K. Wehrheim. *Uhlenbeck Compactness*. EMS series of lectures in mathematics. European Mathematical Society, 2004. ISBN: 9783037190043.